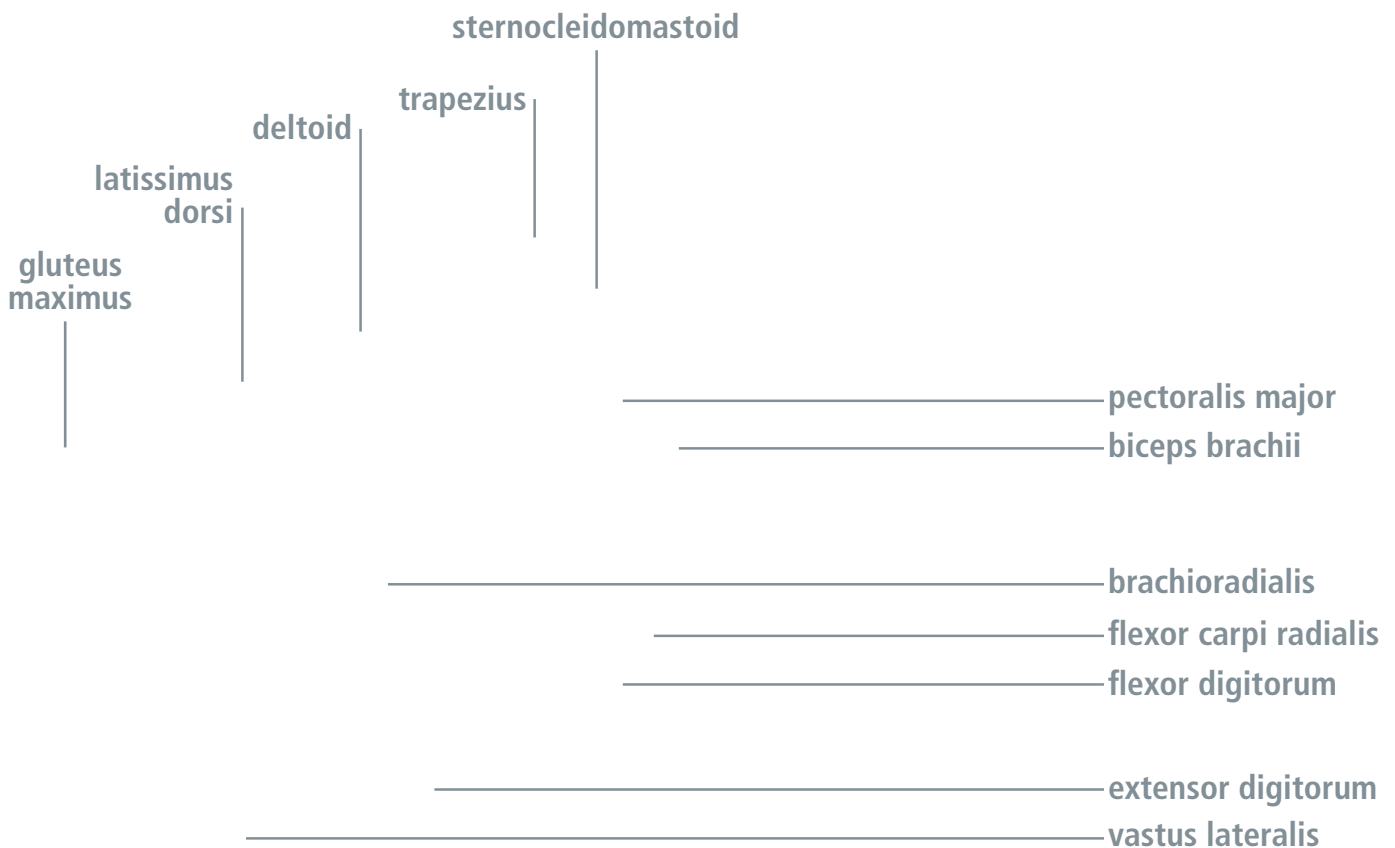


ANATOMY of EXERCISE

A TRAINER'S INSIDE GUIDE TO YOUR WORKOUT



PAT
MANOCCHIA

anatomy of

EXERCISE

anatomy of **EXERCISE**

PAT MANOCCHIA

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INTRODUCTION

This is not the first book ever written that concerns itself with the anatomical structures that are involved in exercise, nor is it intended to be the final word. It is not meant to be an exhaustive exploration of exercise vocabulary either. This book takes a unique and comprehensive stance on the anatomy of exercise, useful to laymen and professionals alike.

Books that have included exercises with anatomical representations, whether or not they included instructions for performing the exercises, are predominately targeted at two groups of readers: bodybuilders and scientists. What I've tried to do with this book is to make this kind of information accessible and useful to anyone who exercises, including bodybuilders and scientists. To that end, this book includes exercise types typically overlooked in similar works, such as aerobic activities, stretches, and stability work. Moreover, this book takes into consideration the ways in which the body's systems depend upon one another to generate movement.

In other texts of this nature, exercises have for the most part been depicted as utilizing a specific muscle group because the book was intended to show how to develop size or strength in that particular muscle or group. Unfortunately, what often doesn't get mentioned is how the adjacent muscles and structures, as well as some that are not directly or obviously involved, contribute to the exercises and subsequent improvement. Hip and spine position, for instance, contribute to almost every major exercise and are integral not only to the proper biomechanics of a given movement, but also to the subsequent improvement of the targeted muscle.

For each exercise, the muscles indicated in the illustrations are identified as the ones that are primarily involved in the movement, whether they are active or stabilizing. Active or primary muscles are defined as those that contract to move a structure, while stabilizing muscles are defined as those that either co-contract or, by their activation, stabilize either the primary structure or a secondary structure to allow movement. In a push-up, for instance, the primary active muscles act to extend the elbow and adduct the humerus (upper arm) at the shoulder joint. Primary stabilizers act to ensure that the elbow and shoulder joints remain steady and track properly; however, without the contraction of the deep spinal and pelvic musculature, as well as anterior leg musculature that contracts to keep those joints stable and allow the ankle joint to act as a fulcrum, the movement is not possible.

The contribution of the secondary stabilizers varies in degree, depending upon the movement. For example, in a barbell curl, because the weight is in front of the body and is translated in a curvilinear fashion that creates a greater forward lever and subsequent need for stabilization as it moves upward, the back and hip muscles contribute more to the movement. If the movement could simply not be performed without the contribution of these muscles, they are included.

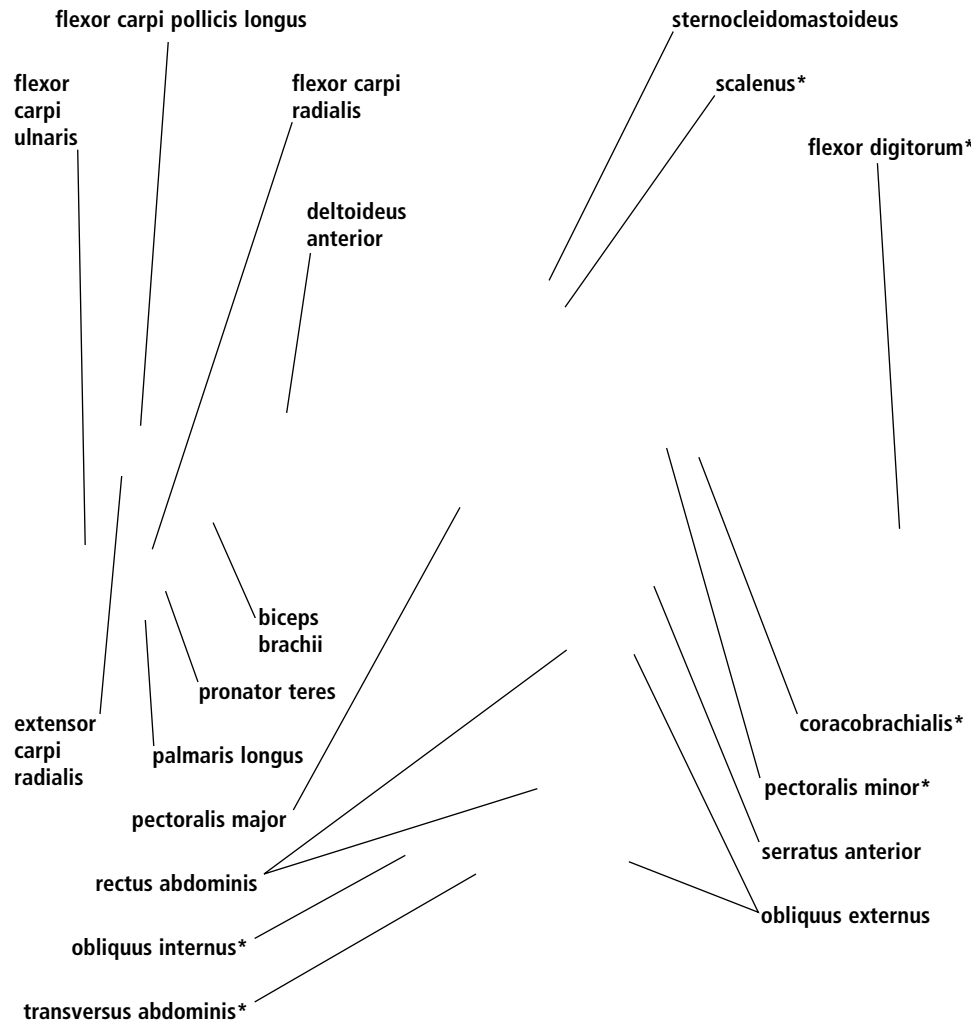
The point here is to make the reader aware that during any given movement, some muscles that may not play a major role in the actual execution may still be necessary contributors for proper biomechanics and form while the exercise is being performed. The basic method I used to determine this was to ask whether or not the movement could be performed if the secondary stabilizers were injured, but readers should be aware that the specifics are open to some debate.

There is an enormous amount of potential variation to these exercises because for any one single exercise there are perhaps four or five different ways to alter the stimulus (by changing the grip, foot position, altering the speed of the movement, and so forth). I have included some of these variations for many of the exercises.

This book contains the basic exercise vocabulary that any program can, and should, be built around, whether you are an elite athlete, a raw beginner, or are suffering from an injury. The specific exercises to use as well as intensity (the weight used, when relevant), volume (number of sets and repetitions), duration (time per session), and frequency (sessions per week) will all be determined by your own specific capacities and goals. The best and most effective way to determine these things is to consult a professional in the fitness/wellness/strength-training profession for a program and prescription which suits your unique abilities and objectives.

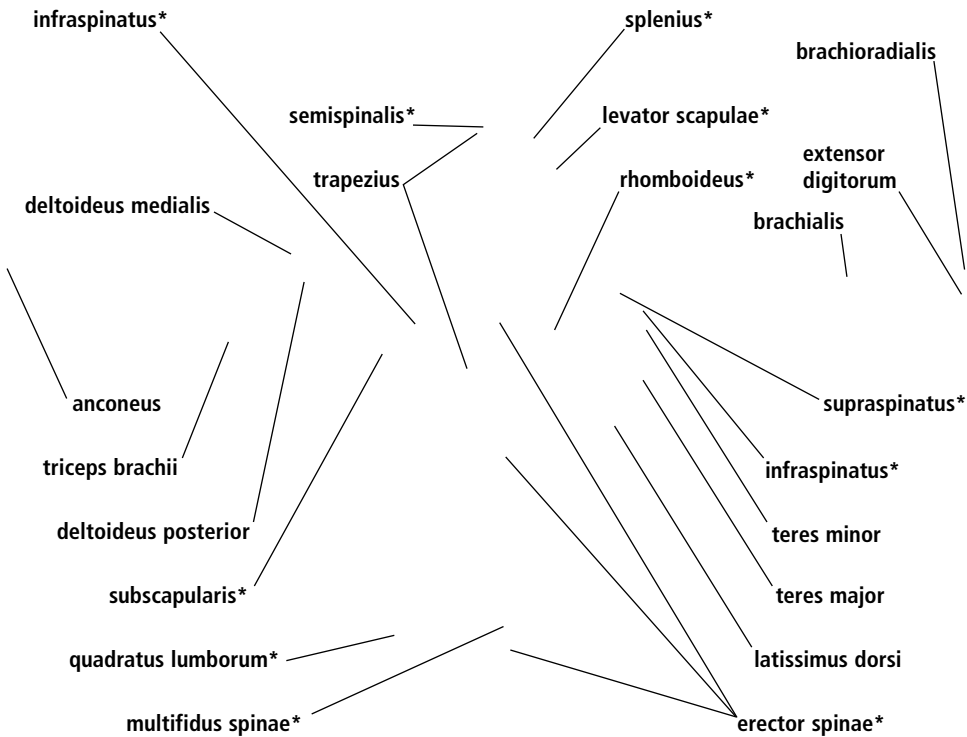
The text is laid out in a structure that mimics the progression of a typical workout. The text encompasses all of the elements necessary for a comprehensive workout, but it is not intended to be prescriptive in any way. The best use of this book is as a reference manual for understanding both positioning and muscular involvement for the included exercises. It should stimulate some thought when performing a given exercise about how the rest of your body plays a part in any particular movement.

UPPER BODY (FRONT)



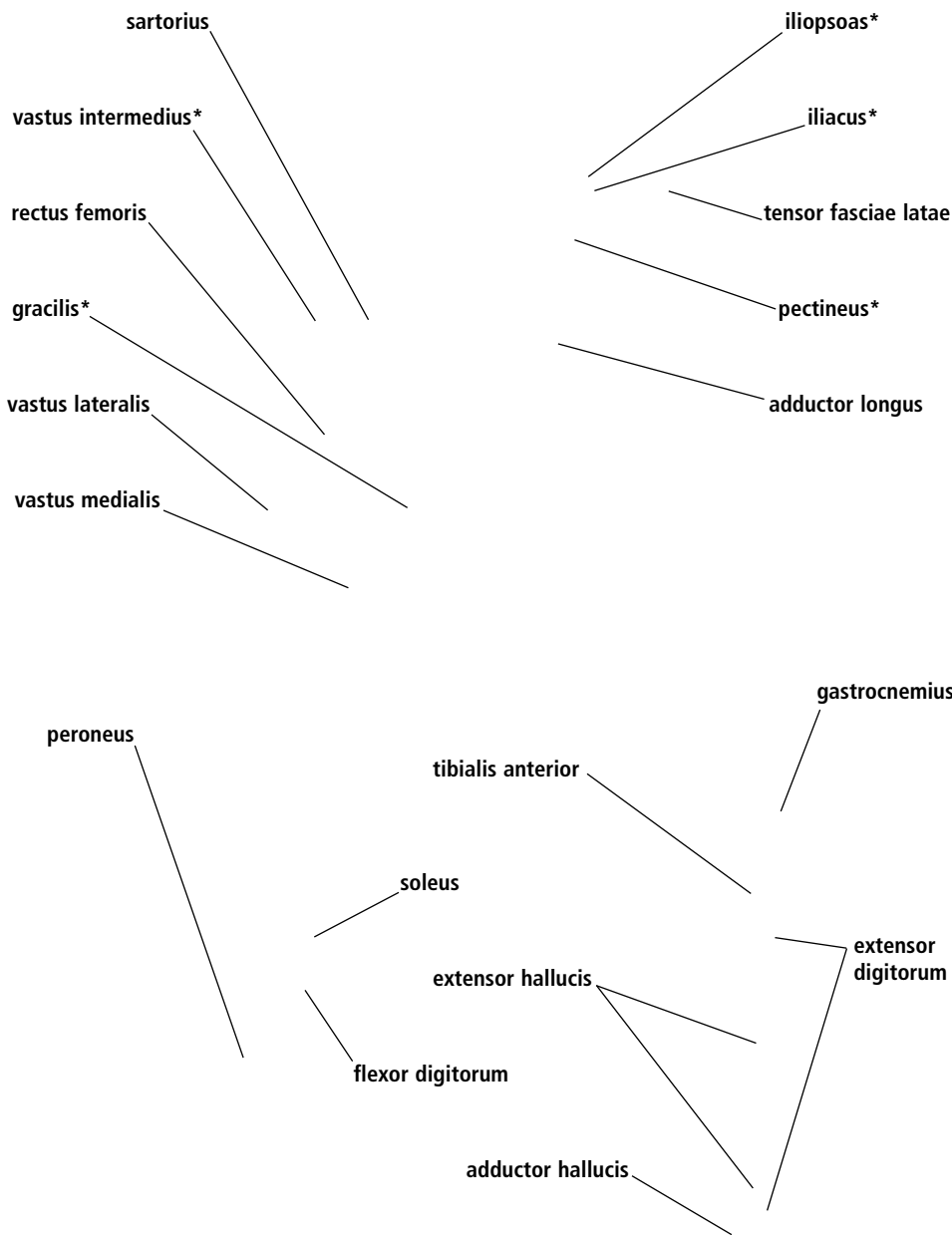
ANNOTATION KEY
* indicates deep muscles

UPPER BODY (BACK)



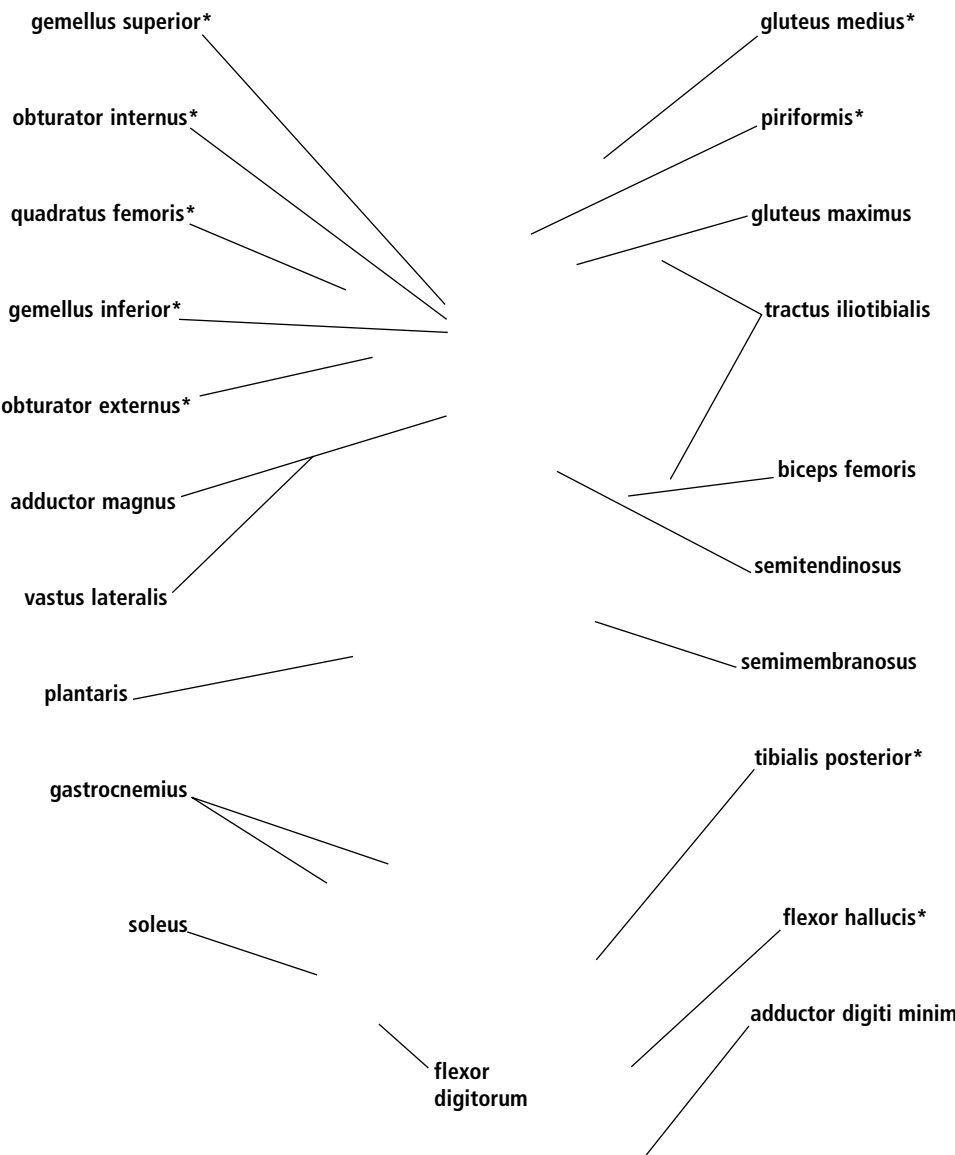
ANNOTATION KEY
* indicates deep muscles

LOWER BODY (FRONT)



ANNOTATION KEY
* indicates deep muscles

LOWER BODY (BACK)



ANNOTATION KEY
* indicates deep muscles

WARM-UPS

The exercises included in the warm-up section increase the heart rate due to a demand for blood delivery. As a result of this activity, not only does oxygen and nutrient-rich blood become available to working muscles, but joint activity also becomes easier, and body temperature rises.

There are many effective and acceptable methods of increasing circulation and raising temperature, both systemically (as depicted in these exercises) and locally in specific areas of the body. The one factor to keep in mind is that warm-ups should be performed *any* time exercise is done, in order to maximize benefit and minimize the potential for injury.

RUNNING

LOOK FOR

- Each foot to strike the ground with the forward portion (your forefoot) directly beneath your center of mass.
- An S-like shape when viewed in profile.
- A soundless transition from foot to foot.
- An even cadence of forefoot strikes.

AVOID

- Reaching forward by extending your knee and landing in front of your center of mass.
- Striking the ground with your heel—ever.
- Vertical movement of your center of mass.

Starting Position: Stand with your knees slightly bent and your feet close together, directly under your hips. Your spine should be in a neutral position, with your head and chest up. Look directly forward.

Action: Correct form, often overlooked, is extremely important in order to derive the greatest benefit from running, and is described in detail here. Begin by rocking your body forward from your hips and torso. Allow your weight to roll forward onto the balls of your feet, allowing both heels to rise. Once your body begins to fall forward, remove one foot from the ground. As your body continues the forward motion, allow your forward foot to fall on the ball of the foot, not allowing your heel to hit the ground. Your foot should land directly under your center of mass while at the same time removing the supporting foot in the same manner.

Movement Path: Your center of mass (located in the region of your hips) is translated along a horizontal path directly forward.

The lower extremities move vertically, alternating in repetitive actions, directly beneath the center of mass.

STABILIZE BY

- Keeping your spine in a neutral position and your shoulders down and relaxed.
- Lifting up your head and chin.
- Keeping your arms bent and comfortable.

JUMP ROPE

Starting Position: Stand with your knees slightly bent and your feet close together, directly under your hips. Your spine should be in a neutral position, with your head and chest up. Look directly forward. With your elbows slightly bent and your palms facing forward, lightly grasp the handles of the jump rope with your elbows at your sides. The rope should be behind your heels.

Action: Bend your knees slightly, and rotate your wrists backward and upward, bringing the rope up behind your body and over your head. As the rope begins the downward phase in front of your body, your wrists pull down and your palms rotate forward slightly, bringing the rope forcefully toward the ground slightly in front of your toes. Jump just enough to allow the rope to pass beneath your feet, and repeat.

Movement Path: There is a vertical translation of your entire body simultaneously, while your wrists rotate in a forward direction.

STABILIZE BY

- Keeping your rib cage up and your spine in a neutral position.
- Keeping your ankles, knees, and hips taut but relaxed.

LOOK FOR

- Your ankles, knees, and hips to bend and extend with the same amplitude.
- A bouncing movement.
- A short foot-strike duration, always landing on your forefeet.
- The rope to have continuous tension.
- Jumping to be relatively quiet.

AVOID

- Allowing your heels to touch the ground.
- Excessive arm movement.
- Excessive knee or ankle movement.
- Landing and pushing

BIKE

Starting Position: Sit on a bike, with your torso bent forward, resting your weight either on your hands or your elbows. Place your feet on the pedals with one leg flexed and bent toward your torso and the other leg extended so that the pedal at the bottom range of the cycle allows your knee to extend fully, without locking it or requiring a hip shift.

Action: Extend one knee and hip, pushing down while simultaneously drawing the other knee and hip up. Repeat.

Movement Path: Your ankles and feet perform a repetitive, alternating circular motion beneath and slightly forward of your center of mass.
Your knees and hips perform a pumping movement by flexing up and in toward the torso and then extending down and away.

- STABILIZE BY**
- Keeping your spine in a neutral position and your shoulders down.

- LOOK FOR**
- An even movement between your two legs.
 - Your head and spine to remain stationary.

- AVOID**
- Rounding your back or shoulders.
 - Focusing only on the extension of your knee and hip.
 - An improper seat height, i.e., either not allowing your knee and hip to fully extend, or requiring a change in your hip position to provide tension in the pedal through the bottom of the arc.

ROWING

Starting Position: Sit in the rower so that your spine is vertical and your pelvis is tilted up. Your feet should be securely fastened. Bend your knees, sliding your torso forward, and grasp the handles.

1

- STABILIZE BY**
- Keeping your spine from head to tailbone rigid throughout the movement.
 - Maintaining a vertical alignment of your knees, ankles, and hips.
 - Keeping your shoulders down and away from your ears throughout the movement.

Action: Push your feet into the foot stand by extending your knees, hips, and back. As your knees reach the midpoint of the range, your spine should be perpendicular to the ground and your elbows should be slightly bent as they pull the handle toward the midline of your torso. Finish the movement when your knees are completely extended, your spine is 5 to 10 degrees past perpendicular, and the handle is contacting your torso just below your chest,

with your elbows retracted and your shoulders pulled down and back. Return by bending your knees and allowing your torso and arms to extend forward so that your torso is flexed forward, 5 to 10 degrees past perpendicular, and your arms are fully extended, so that your hands are above your ankles. Inhale on the downstroke, and exhale as the stroke is executed.

2

Movement Path: Your torso bends forward with your arms extended forward from the hips and your knees bent.
Your center of mass is translated horizontally as your torso flexes and extends along with your hips and knees.

- LOOK FOR**
- Your ankles, knees, hips, shoulders, and elbows to all move simultaneously.
 - Your forearms to remain horizontal at all times.
 - Your head and chest to remain high and forward.

- AVOID**
- Rounding your back.
 - Excessive movement of your spine in either direction.
 - An unsynchronized movement pattern.
 - Allowing your knees to migrate either inward or outward.

STRETCHING

There are several different methods of stretching, only some of which are included here. The objective in performing these stretches is primarily to help improve the range of motion of both the joints and muscles.

They may be performed before exercising as a preparatory activity to stimulate neurological awareness, during an exercise session to provide blood to working muscles, after exercising as a method of cooling down and “reminding” joints and muscles of their movement patterns, or as a workout in themselves, as a method of recovery and regeneration from a prior bout of activity.

HIP FLEXOR/ HAMSTRING STRETCH

Starting Position: Kneel on one knee and place the opposite foot forward so that your front knee and hip are bent at 90 degrees. Form a direct line through your torso and spine to the knee on the floor. Let your hands rest by your sides, and keep your shoulders down and relaxed and your chest up. Make sure your hips, knees, and ankles are in alignment.

Action: Raise your arms and move your hips forward simultaneously, until your arms are directly over your head and your front knee is over the toes of your front foot. Your torso remains vertical, and your bottom leg is opened at the hip joint. Continue by bringing your arms down and flexing your torso forward. Move your hips backward, straightening the front leg, while reaching your hands to the ground around your knee while your back knee remains on the ground.

1

LOOK FOR

- All joints to move at the same time.

AVOID

- Allowing your front foot to leave the floor.
- Any hip rotation.
- Elevating your shoulders toward your ears.

Movement Path: During the forward movement, your center of mass translates forward horizontally and slightly downward.

During the backward movement, the same occurs in the opposite direction, with the exception of your torso flexing forward.



2

STABILIZE BY

- Keeping your hips parallel during all phases of the movement.
- Keeping your front foot flat.
- Maintaining the alignment of your hip, knee, and ankle throughout the movement.

MODIFICATIONS

More Difficult: The forward movement is the same, but during the backward movement, let your back knee leave the ground and straighten your back leg as your hands reach directly down to the floor beneath your hips.



STRADDLE ABDUCTOR STRETCH

Starting Position: Spread your feet widely apart and bend your knees. Keep your hands on your knees and your weight evenly distributed throughout your feet. Your spine should be in a neutral position and your torso should bend forward at a 45-degree angle to the floor, so that your hips are behind your heels.

Action: Alternate leaning from side to side, bending the appropriate knee, while extending and straightening the opposite knee. Keep your hips back and your shoulders and chest facing forward.

1

LOOK FOR

- Your torso to remain stationary as it moves from side to side.

AVOID

- Letting any part of your feet leave the floor.
- Allowing your knees to extend beyond your toes.
- Any rotation of your hips.

Movement Path: Your hips remain behind your heels, and your knees remain over your toes on the bent leg, as your center shifts from side to side, with a small arc in the movement (the highest point being the center of the arc and the lowest the widest).

2

STABILIZE BY

- Keeping your spine in a neutral position.
- Keeping your feet flat and your weight evenly distributed.
- Bearing some weight on your knees with your hands.

CROSS-LEG GLUTEAL STRETCH

Starting Position:

Standing upright, cross your ankle to just above the opposite knee joint. Your lower body should appear as a figure 4. Let your arms hang at your sides (the position should be repeated on the opposite side after completing the stretch).

**STABILIZE BY**

- Fixing and slightly arching your spine. If you are raising your arms to the front, stabilize your scapula and make sure your crossed leg is positioned firmly above your knee.

Action: Bend your standing leg by flexing your hip and knee to 90 degrees; essentially the stretch is a one-legged squat. Maintain the crossed leg in a stationary position. (To increase the intensity of the stretch apply a slight downward pressure to your crossed knee and a slight upward pressure on the same ankle). Hinge at your hips, not your spine.

Movement Path: To descend into the stretch position, your hips should move backward first. Lower until the femur of your standing leg is parallel to the floor (your knee should bend to 90 degrees). Raise your arms from your sides 90 degrees in front of you to your chest level, with your palms facing inward and your scapula stabilized.

2

LOOK FOR

- Your hips to move backward, prior to flexion at your knee.
- Your lower leg to be perpendicular to the floor with your toes pointed forward.
- A fixed spine and pelvic position with no anterior rotation of pelvis.
- Your head to remain in an upright position.

AVOID

- Rounding your spine or tilting your pelvis.
- Your front knee coming too far forward over your toes.

PRESS-UPS

Starting Position: Lie flat on your stomach, with your palms down and next to your shoulders. Place your elbows by your sides, so that your forearms rest on the ground.

Action: Squeeze and depress your shoulder blades and squeeze your gluteals. Contract your spinal muscles and lift your chest and head upward and backward. As your chest rises, assist them by pressing your forearms and palms into the floor. Extend your elbows as far as your torso will allow without raising your hips off the floor, forcing your shoulders to shrug, or causing pain in your lower back or legs. Exhale as you rise, and inhale as you return to the starting position.

1

Movement Path: Your spine moves in a curvilinear motion, both backward and upward, while your arms extend forward and downward.

2

STABILIZE BY

- Contracting your gluteals and pressing your hips to the floor.
- Keeping your elbows at your sides.
- Keeping your chest and head high.

LOOK FOR

- Your spine to curl on the way up.
- Your arms to do no more than 50 percent of the work in lifting your torso.

AVOID

- Elevating your hips off the ground.
- Forcing or pushing your torso upward with arms.
- Raising or pinching your shoulders.

HIP/LOW-BACK STRETCH

Starting Position: In a seated position with your torso vertical, extend one leg straight in front of you. Bend the other leg and cross it over the outstretched leg, so that the foot is flat on the floor and the ankle of the bent leg lies adjacent to the outside of the knee of the straight leg.

Action: Wrap the opposite arm around your bent leg. Pull your chest up, and rotate your torso toward the bent knee by pulling around your leg. Reach directly behind your back with the hand on the bent-knee side, placing it on the ground and using it to keep your spine high and long while twisting. Hold the twisted position for 20 seconds. Repeat on the opposite side.

Movement Path: Your upper torso rotates with your spine vertical and perpendicular to your hips and the straightened leg, while your lower body remains stable.

STABILIZE BY

- Using both arms to keep your torso in a vertical position.
- Pushing the foot on the bent-knee side flat into the ground and keeping your hips stationary.

- LOOK FOR**
- Your opposite arm to wrap around the bent knee with the crease in your elbow aligned just below your kneecap.
 - Your shoulders to remain in an even plane.
 - Your chest to stay high.

- AVOID**
- Rounding your spine
 - Letting any part of the foot on the bent-leg side rise off the ground.

QUAD STRETCH

Starting Position: Lie on your side. With your bottom hand, pull your bottom knee toward your chest. Grasp the shoelaces of your top shoe, keeping your knees parallel.

Action: Contract your gluteals and extend your hip and heel backward. Gently assist the movement by pulling your upper leg back and upward, pausing at the farthest point in your range of the motion. Return by flexing your hip joint and bringing the top knee forward until it is parallel with your bottom knee. Repeat on the opposite side.

Movement Path: Your torso and bottom leg are stationary as your upper leg and hip move from in front of your chest down and backward (your heel reaches toward your buttocks).

STABILIZE BY

- Pulling your abdomen in and keeping your hips parallel.
- Keeping your chest and rib cage up and your shoulders back and down.

- LOOK FOR**
- Your top knee (the moving leg) to be pointing directly down from your hips at the bottom of the movement.
 - Your knee to be fully flexed during the movement.
 - Your leg and hip to be actively engaged during the entire stretch.

- AVOID**
- Using your arm to generate the stretch (your arm should only assist with the movement, not create it).

ILIOTIBIAL BAND STRETCH

Starting Position: Standing up straight, cross one leg over the other, keeping your feet flat and slightly apart, with your back knee straight and your front knee only slightly bent.

Action: Keeping your legs crossed at the ankle, bend over completely to the ground while reaching your hands toward your toes. Pause, then return to a standing position and uncross your legs. Cross your legs in the opposite direction and repeat the movement.

Movement Path: Your torso flexes directly forward.

STABILIZE BY

- Keeping your feet flat and your weight distributed slightly toward the back of the foot.
- Keeping a slight bend in the front leg.

LOOK FOR

- Your hands to remain fairly close to your body on the way down.

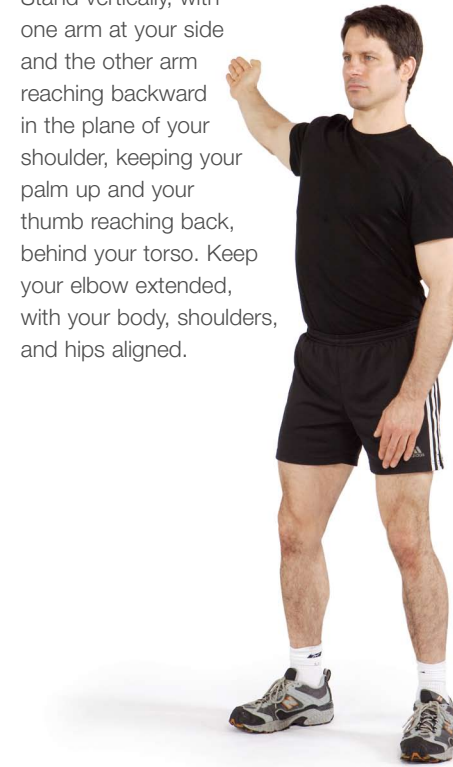
AVOID

- Bouncing or forcing the extension of your hands toward the ground.

UPPER-BACK/ SHOULDER STRETCH

Starting Position:

Stand vertically, with one arm at your side and the other arm reaching backward in the plane of your shoulder, keeping your palm up and your thumb reaching back, behind your torso. Keep your elbow extended, with your body, shoulders, and hips aligned.



STABILIZE BY

- Keeping your back straight, your chest up, and your hips facing forward.

2

Action: Contract your chest and bring your arm forward in a horizontal plane. As your arm reaches the front of your body and is pointing forward, place the opposite hand and wrist under and around the moving arm and continue the movement by gently assisting the pressure of the arm across your chest. Repeat on the opposite side.

Movement Path: Your straight arm moves in a horizontal plane and a circular pattern around your body from front to back. Your legs, hips, and torso are stationary.

LOOK FOR

- Your shoulders to remain down throughout the movement.
- Your head to face forward.

AVOID

- Bending the elbow on the moving arm.
- Letting the shoulder of the moving arm shrug upward.
- Rotating any other part of your body.

ACTIVE HAMSTRING STRETCH

LOOK FOR

- Your knee to be fully extended at the top of the movement (this may require changing your upper-leg position to accommodate a full extension).
- An achievement of the end range of motion, or the farthest point at which the knee can straighten out.

AVOID

- Flattening out your lower back during the knee extension.
- Pulling on the extended leg with any appreciable force.

Starting Position: Lie flat on your back with one knee bent and your foot flat on the floor. Clasp your fingers behind and below the knee on the opposite leg, with the knee joint on your upper leg pointing straight at the ceiling while keeping the leg relaxed.

1

Action: Extend the foot of the supported leg toward the ceiling by contracting your quadriceps (the front upper thigh), until your leg is fully extended. When fully extended, continue to contract your upper leg, bringing it up toward your chest. Use your hands to exert a *very* minimal, gentle pull upward in the same direction for no more than two seconds. Release the pull and allow your leg to drop slowly to its original position. Keep your foot flexed and your toes pulled up toward your shin.

Movement Path: Your lower leg moves in an upward arc toward the ceiling.

STABILIZE BY

- Keeping your spine in a neutral position.
- Keeping the opposite foot flat on floor.
- Making sure your hips remain stationary.

2

ACTIVE HAMSTRING STRETCH • STRETCHING

MODIFICATION

More difficult: Keep one leg straight, bringing it up until you can grasp it behind the knee joint and gently assist it, while continuing to bring your leg toward your chest.

1

2

LEGS AND HIPS

The muscles of the legs span three joints: the hip, knee, and ankle. The muscles of the upper leg operate the humerus at the hip and the tibia and fibula at the knee. Lower-leg muscles move the foot. Multiple ligaments at each joint stabilize movement in all directions. The most critical function of the legs is interaction with hip, core, and back muscles to generate force, provide locomotion, and decelerate movement.

The muscles of the hips either originate or traverse the pelvis and are responsible primarily for moving the legs. These muscles very often bear the brunt of improper training routines that tend to focus on movement in only one plane or direction and consequently develop strength imbalances that result both in poor performance and/or injury. In order for any force to be translated from the lower to upper body (particularly the spine), hip muscles must be addressed and trained properly.

LEG AND HIP ANATOMY

LEGS AND HIPS

gluteus maximus

tensor fasciae latae

iliopsoas*

pectineus*

adductor longus

adductor magnus

sartorius

semimembranosus

gracilis*

ANNOTATION KEY

* indicates deep muscles

LEGS AND HIPS

gluteus medius*

piriformis*

tensor fasciae latae

rectus femoris

vastus medialis

gluteus maximus

vastus lateralis

biceps femoris

soleus

peroneus longus

peroneus brevis

tibialis posterior*

gastrocnemius

tibialis anterior

extensor digitorum longus

extensor hallucis longus

ANNOTATION KEY

* indicates deep muscles

BARBELL SQUAT

1

Starting Position: Stand with your feet slightly wider than shoulder-width apart, resting a barbell across the top of your shoulder blades. Your hands should be wider than hip width. Keep your chest up and your spine in a neutral position. Keep your eyes looking up, 20 degrees above horizontal. Your weight should be evenly distributed across your feet.

Action: Retract your hips, pulling them backward. Bend your knees, keeping your spine in a neutral position, allowing your body to hinge at the hip and knee joints, until your upper legs are parallel with the floor and your spine is at a 45-degree angle to the floor. Your knees should extend slightly forward, with your chest and head up. Your knees should be directly over the front of your feet.

Movement Path:

Your hips and spine slide down and backward, while your knees slide forward until the bottom of the movement; then your knees, hips, and back move straight up simultaneously.

STABILIZE BY

- Keeping your spine in a neutral position and your shoulder blades pinched down and back.
- Keeping your abdominal muscles taut and your hands on the bar, with some tension in your grip.

2

LOOK FOR

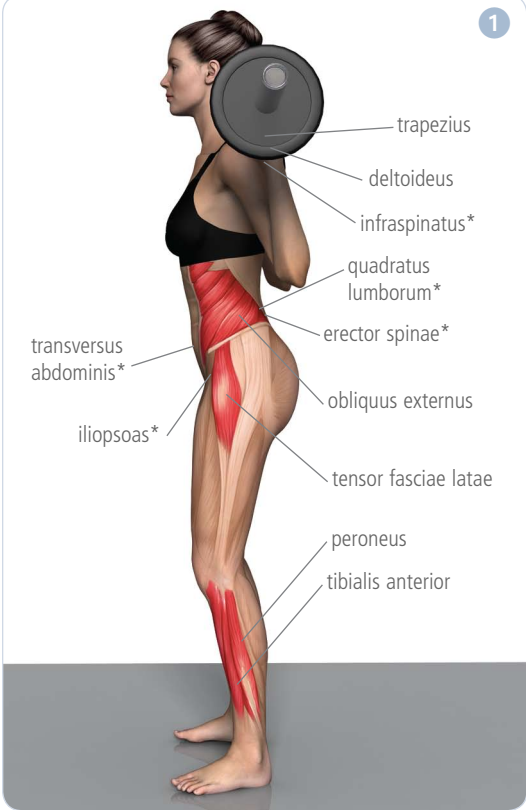
- The barbell to drop down in a vertical line, directly above the mid-foot to heel.
- All joints to move at the same time.
- Balance throughout the movement.

AVOID

- Sliding your knees forward, beyond your toes.
- Rounding your back.
- Allowing the bar to roll up on your neck, toward your head.
- Letting your knees slide until they are either wider or narrower than your feet width.

BARBELL SQUAT • LEGS AND HIPS

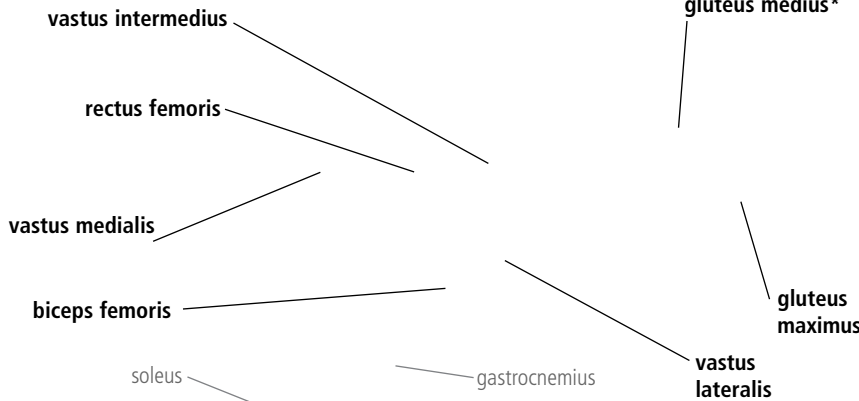
1



ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

2



BEST FOR

- biceps femoris
- gluteus maximus
- gluteus medius
- rectus femoris
- vastus intermedius
- vastus lateralis
- vastus medialis

FRONT SQUAT

Starting Position: Stand with your chest and head up, your arms extended forward, and a barbell resting on the top of your shoulder muscles. Keep your spine neutral and your feet slightly wider than shoulder width.

Action: Retract your hips and bend your knees, keeping your head and chest up and your spine in a neutral position. Lower your hips until your upper legs and thighs are at least parallel with the ground. Your head and spine should be at a 45-degree angle to your hips and to the ground. Your hands can either be extended forward or crossed to grip the bar so that it does not slip.



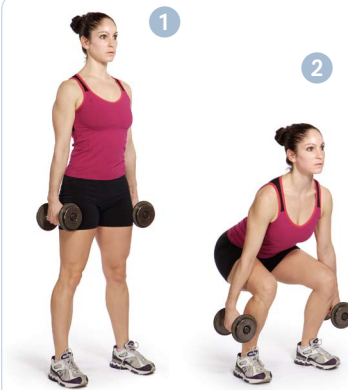
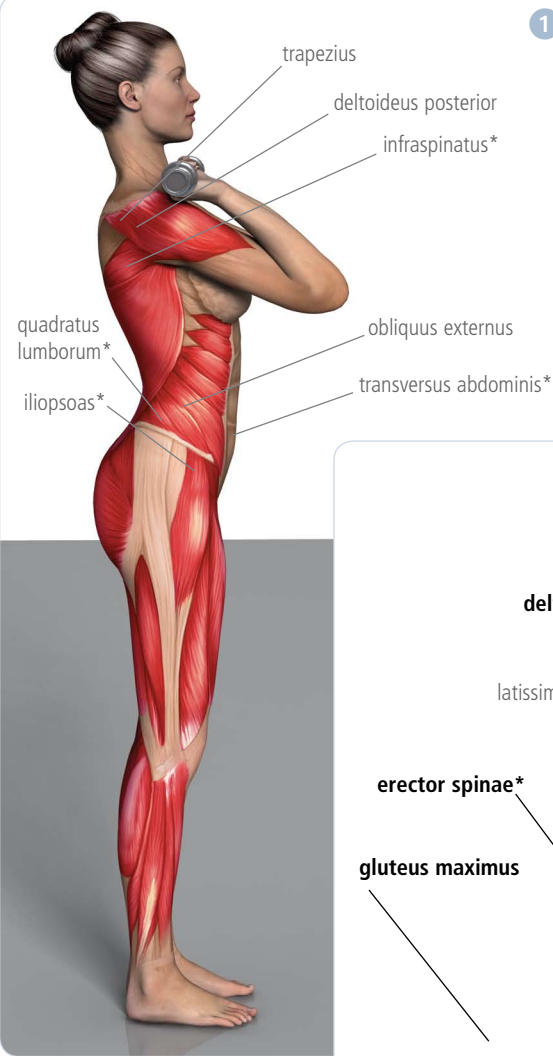
- STABILIZE BY**
- Retracting your shoulder blades.
 - Keeping your abdomen pulled up and in.
 - Keeping your knees parallel and over your feet.

- LOOK FOR**
- A slow, controlled descent and ascent.
 - No movement in the vertical line of the bar as it drops.
 - No foot movement or heel lift.

- AVOID**
- Extending your knees beyond your toes.
 - Dropping your elbows; keep them and your upper arms parallel to the ground at all times.
 - Extending your head forward or elevating your scapula and shoulders.

Movement Path: Drop straight down and push straight up by extending your arms on the way down and extending your legs on the way up. Your hips, shoulders, and ankles move up simultaneously as you push through your knees.

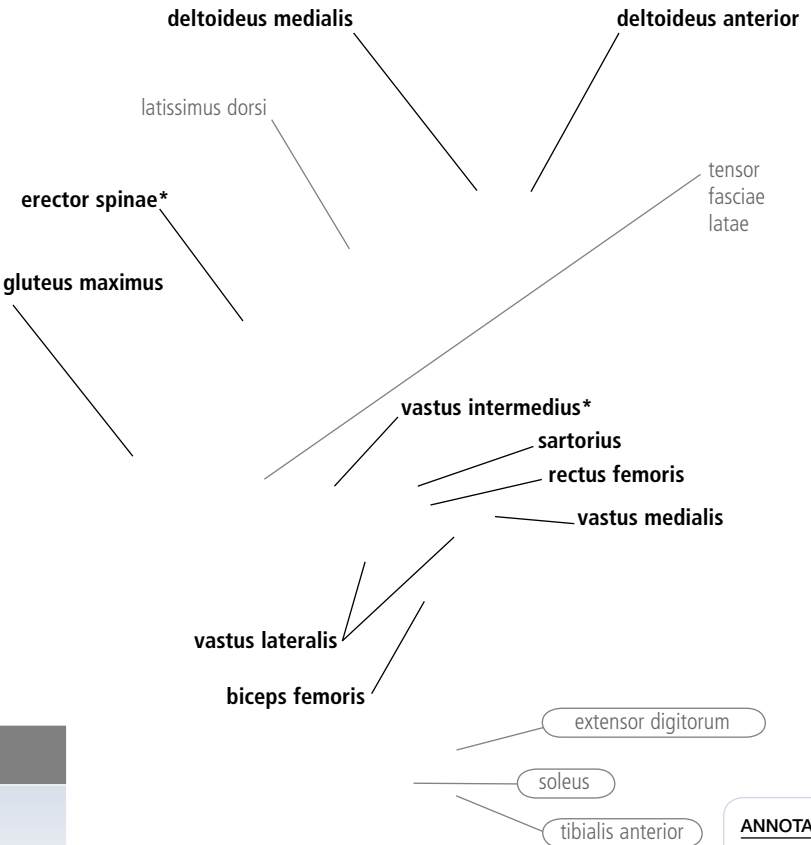
FRONT SQUAT • LEGS AND HIPS



MODIFICATION
Similar difficulty: Instead of a barbell, hold dumbbells at your side. Maintain the same activation pattern and movement sequence.

BEST FOR

- biceps femoris
- erector spinae
- gluteus maximus
- rectus femoris
- vastus intermedius
- vastus lateralis
- vastus medialis



ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

LUNGE

Starting Position: Stand with your feet close together and your hands on your hips.

Action: Keeping your head up, your spine in a neutral position, and your hands on your hips, take a step forward, bending your front knee to a 90-degree angle and dropping your front thigh until it is parallel with the ground. Your back knee drops straight down behind you, so that you are balancing on the toe of your foot to create a 90-degree angle in your knee joint and a straight line from your spine through your bottom knee. Return to the starting position by pushing on your front foot and elevating with your back leg until you are again standing.

Movement Path: The general motion is forward and descending. Your spine stays in a vertical position and is translated forward and downward by the step and the descent.

- STABILIZE BY**
- Keeping your chest high, stomach up, and spine neutral.
 - Evenly distributing your weight across your front foot, from front to back.
 - Keeping your back foot on the toe and your weight in the back of the stepping leg.

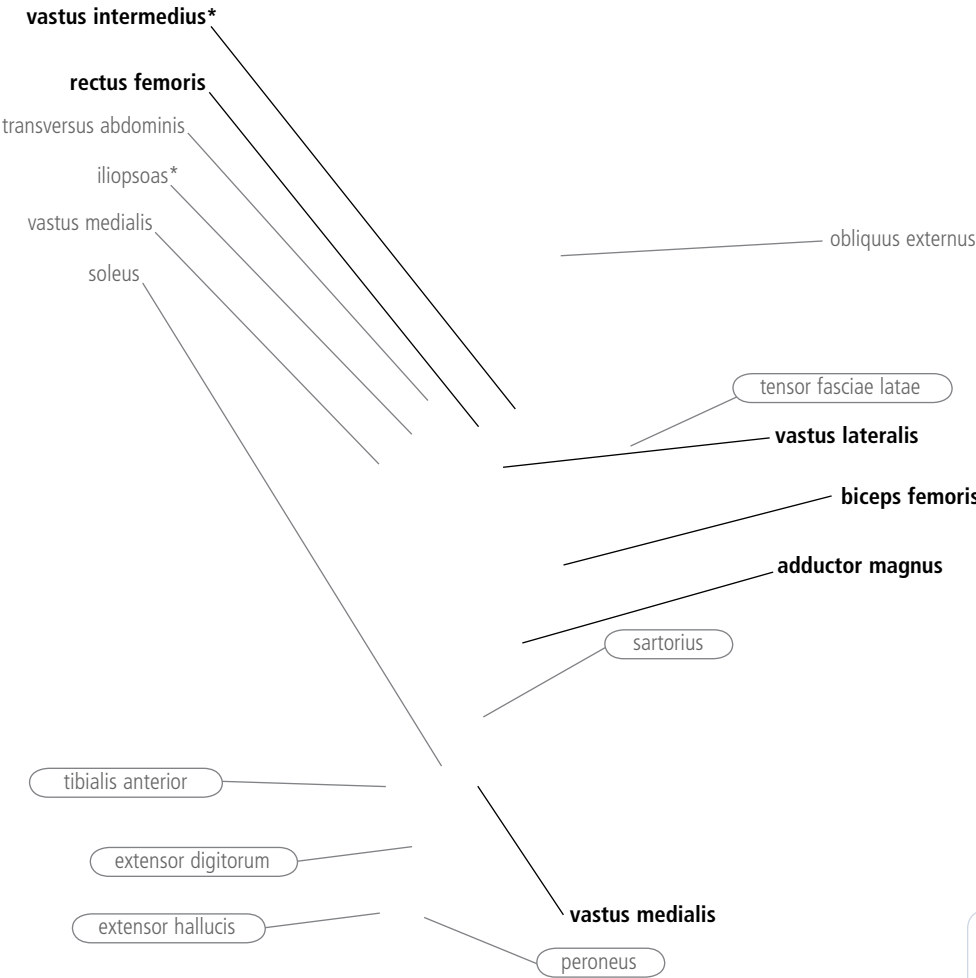
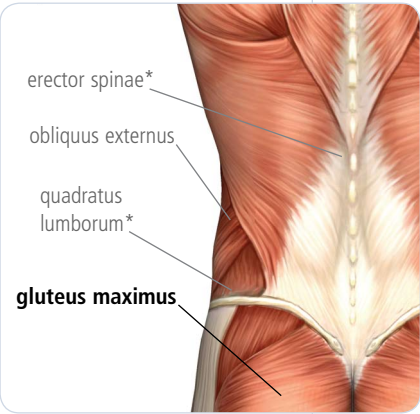
- LOOK FOR**
- No translation forward from the hips.
 - Your spine to remain in the same position as it moves down and up.
 - No lateral movement of your leg as you step, either landing or pushing.

- AVOID**
- Raising the heel of your stepping foot off the ground or rotating your hips or torso.

LUNGE • LEGS AND HIPS

BEST FOR

- adductor magnus
- biceps femoris
- gluteus maximus
- rectus femoris
- vastus intermedius
- vastus lateralis
- vastus medialis



ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

LATERAL LUNGE

LOOK FOR

- A simultaneous movement of your arms and hips.
- Your chest to remain up and your shoulders to remain down.

AVOID

- Any part of the stepping foot losing contact with the ground or your knee extending forward beyond your toe.
- An excessive drop in torso angle beyond or below 45 degrees.

Action: Step directly out to the side at 180 degrees, retracting your hips and keeping your spine neutral. As your chest moves forward and your hips retract, extend your arms in order to ensure balance. Stop at the bottom of the movement when the upper thigh of your stepping leg is parallel to the ground. The opposite knee should be extended, your hips should be behind the stepping foot, and your knee should not exceed the toe line and should be directly over the foot. Your upper arms should be parallel with the ground. Pushing back off the stepping leg, return to the starting position.

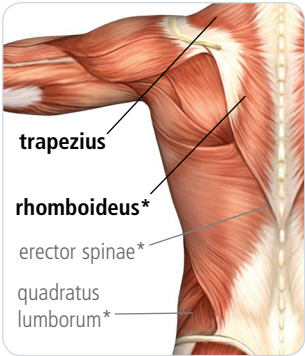
Starting Position: Stand vertically with your feet directly below your hips and your hands on your hips.

Movement Path: You move laterally to the side; your arms go forward and your hips go back.
Your torso drops as your hips retract. Use one foot as both the decelerator and accelerator. Use the standing or stationary foot as a balance lever.

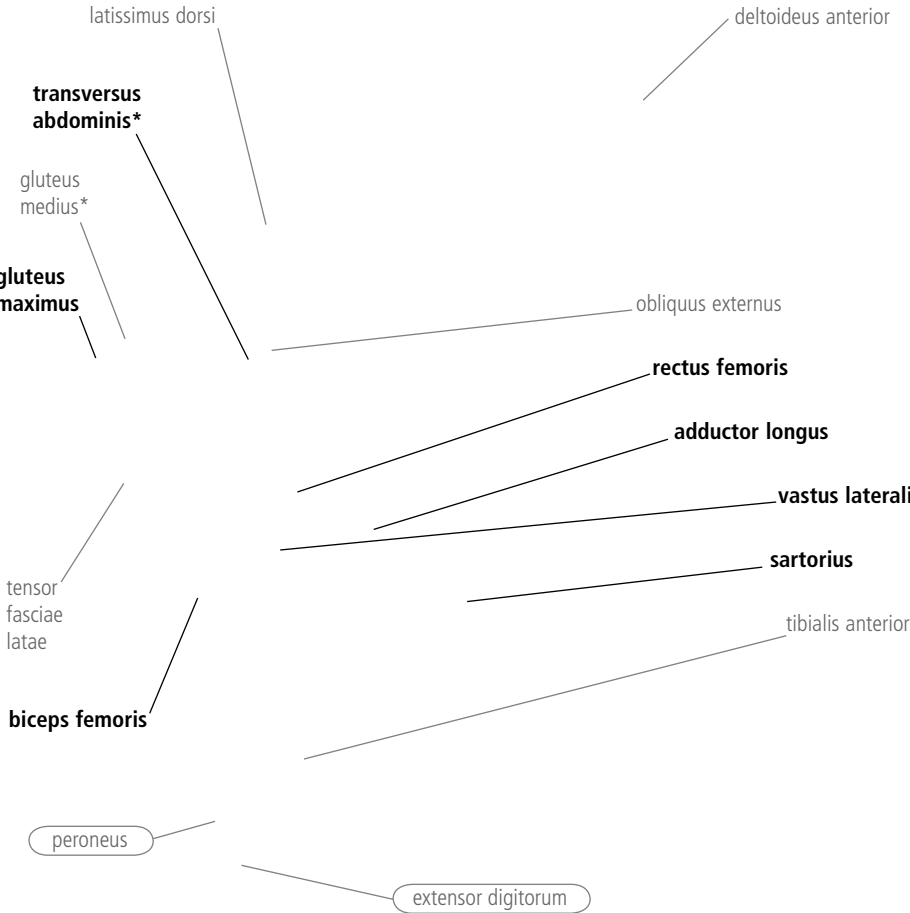
STABILIZE BY

- Keeping your hips retracted, your chest up, and using your arms as a counter-balance to the retraction of your hips.
- Keeping the opposite leg in contact with the floor to maintain tension on your quadriceps and hamstrings, so that your knee is locked and extended.

LATERAL LUNGE • LEGS AND HIPS



BEST FOR	
• adductor longus	• gluteus maximus
• adductor magnus	• rectus femoris
• biceps femoris	• sartorius
	• vastus lateralis



ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

STEP-UP

1



Starting Position: Place one foot on a block in front of you, preferably with the raised knee at close to a 90-degree angle. Make sure your body is vertical, your chest is up, and your knee is directly over your foot. Your raised knee should not exceed the toe line and your foot should be flat on the surface of the step. Grasp the dumbbells by the side of each hip.

2

LOOK FOR

- A slight forward translation and directly upward movement of your spine.

AVOID

- Bending or extending your back knee.
- Allowing your front knee to slip forward beyond the toe line or any part of your front foot to lift off the step.
- Moving your knee either laterally or medially; keep it directly over the stepping foot.

STABILIZE BY

- Keeping your upper back muscles and shoulders down and back.
- Not allowing your momentum to bring forward either the weights or your torso.
- Keeping your hip, shoulder, and ankle in a line from the bottom weight.

Action: Begin by leaning slightly forward. Allow the weights to swing forward so that they are close to the plane of your ankle. Keeping your back leg straight, push through your top leg, extending your knee and hips simultaneously to drive your body up and over the step. Do not allow your back leg to push off the floor.

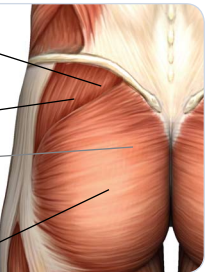
Movement Path: The movement is slightly forward and directly upward.
 Your head should begin behind the elevated foot and end directly over it.
 Allow your arms to simply stabilize the weight and hang as pendulums, following the body's natural path.
 Descend in the same fashion.

STEP-UP • LEGS AND HIPS

BEST FOR

- biceps femoris
- gastrocnemius
- gluteus maximus
- sartorius
- semimembranosus
- semitendinosus
- soleus
- vastus medialis

gluteus medius*
 gluteus minimus*
 piriformis*
 gluteus maximus



1

pectineus*
 semimembranosus
 gastrocnemius
 biceps femoris
 soleus

2

trapezius
 deltoideus posterior
 triceps brachii
 extensor carpi radialis
 extensor digitorum
 gluteus medius*
 flexor digitorum
 deltoideus anterior
 biceps brachii
 brachioradialis
 flexor carpi radialis
 adductor longus
 adductor magnus
 rectus femoris
 sartorius
 gracilis*
 semitendinosus
 tibialis posterior*
 tibialis anterior

ANNOTATION KEY
 Black text indicates active muscles
 Gray text indicates stabilizing muscles
 * indicates deep muscles

STEP-DOWN

Starting Position: Begin by standing in a vertical position on a block, with one foot unsupported off to the side.



1

STABILIZE BY

- Keeping your spinal muscles active, your shoulders retracted and depressed, the opposite leg extended and taut, and your opposite foot flexed.

Action: Drop your torso by retracting the hip over the standing foot and bending your knee, allowing your torso to ride forward. Extend your arms, dropping the non-weight-bearing leg to below the level of the step. Drop to just above 90 degrees, or to a point at which balance can be achieved. Return by extending your hips, straightening your knee, and lifting your chest up and away.

Movement Path: While descending directly down, your arms extend and your hips retract. Use your hands and arms for balance.

2

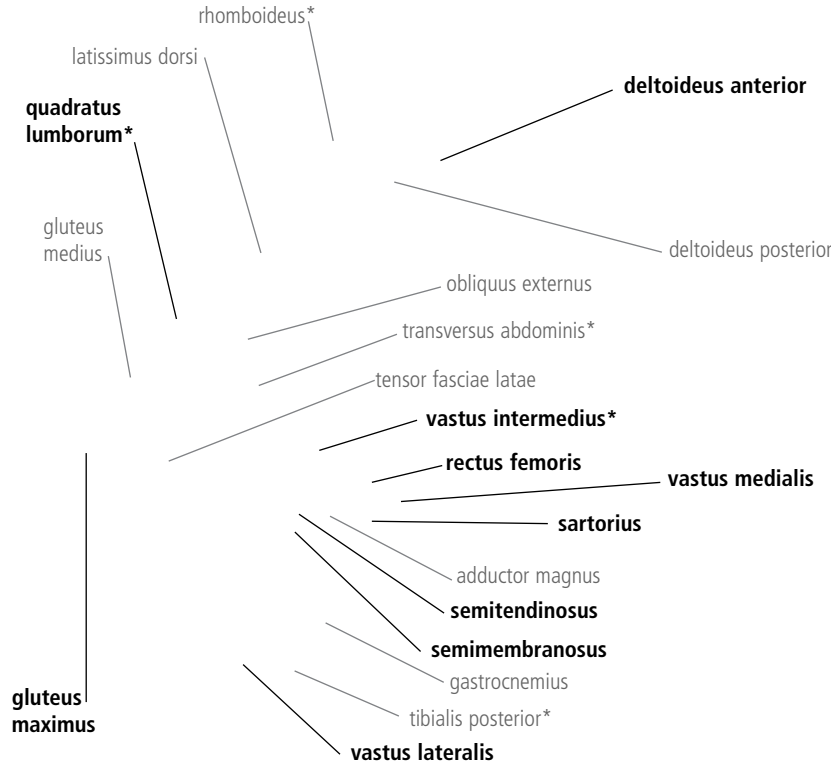
LOOK FOR

- Your head to remain directly above your ankle.
- Your knee and hips to move simultaneously.

AVOID

- Extending your knee beyond your toe line.
- Any rotation in your hips or torso
- Any deviation of the standing knee from above the weight-bearing foot.

STEP-DOWN • LEGS AND HIPS



BEST FOR

- gluteus maximus
- rectus femoris
- semimembranosus
- semitendinosus
- vastus intermedius
- vastus lateralis
- vastus medialis

ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

MODIFICATION

More difficult: Grasp a weight, such as a medicine ball, at your chest in the initial standing position. As you descend by retracting your hips and bending your knee, keeping your spine taut and your chest up, extend the weight forward so that at the bottom of the movement the weight and your arms are fully extended from your torso and balance is achieved. Return to the starting position by extending your hip and knee and elevating your torso, while at the same time drawing the weight back toward your chest to the starting position.



1

2

CALF RAISE

Starting Position: With your body vertical, place the front part of one or both feet (from the ball joint to the toe) on a step. Either one or both feet should be extended, so that the heel and arch of each foot is beyond the edge of the step. Your hips, ankle(s), and shoulders should be aligned; your spine should be in a neutral position and your head should be up. Grasp a dumbbell in one hand if using only a single leg.

Action: Allow your heels to descend in a vertical line, dropping toward the floor. At the bottom of the motion, push down on your toes and the arch of the foot, keeping your spine straight and elevating your body directly upward.

Movement Path: Your body moves in a vertical descent and ascent. Use the front part of your foot as the activator.



STABILIZE BY

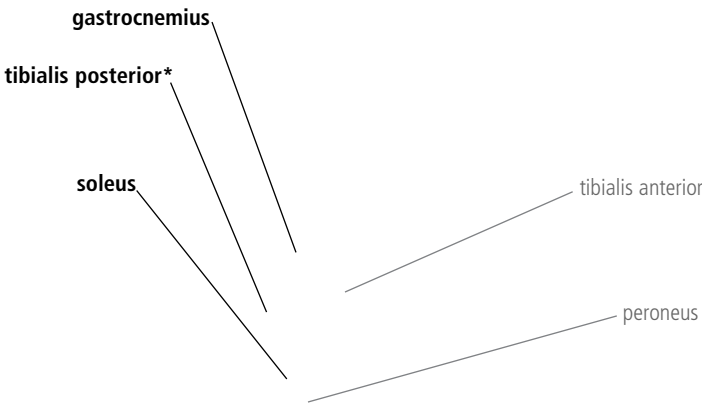
- Keeping your spine in a neutral position.
- Keeping your feet evenly placed on the step and your upper legs locked.
- Placing the fingertips of one or both hands on supporting surfaces. Do not allow your hands to influence movement up or down; simply allow them to help stabilize your spinal position and torso angle.



CALF RAISE • LEGS AND HIPS

BEST FOR

- gastrocnemius
- soleus
- tibialis posterior



ANNOTATION KEY

Black text indicates active muscles
 Gray text indicates stabilizing muscles
 * indicates deep muscles

CABLE ABDUCTION

Starting Position: Stand so that you are facing perpendicularly to the resistance. Your outside leg should bear the weight, attached directly at the ankle bone. Hold onto the column with the hand closest to it.

Action: Allow the hip attached to the cable to slightly elevate, so that the foot that is attached to the cable does not bear any of your weight. Keeping your hips solid and on a level plane, push your leg away from the midline of your body as far as possible, until your hip position can no longer be maintained or is beginning to be compromised. Allow the weight and your leg to return slowly to the starting position.

Movement Path: Your foot should move directly away from your body, in an arc to the side and up. Extend by pushing your hip up and leading with outside edge of your foot.

STABILIZE BY

- Using your hand to keep your torso in a vertical position, directly over the standing leg.
- Maintaining a neutral spinal position.

LOOK FOR

- Your hips to remain even.
- The movement to come from your leg.
- Your feet to remain parallel during the movement.

AVOID

- Any excessive knee bending.
- Any rotation of your hip or shoulder.
- Swinging or creating momentum in the cable.

1

2

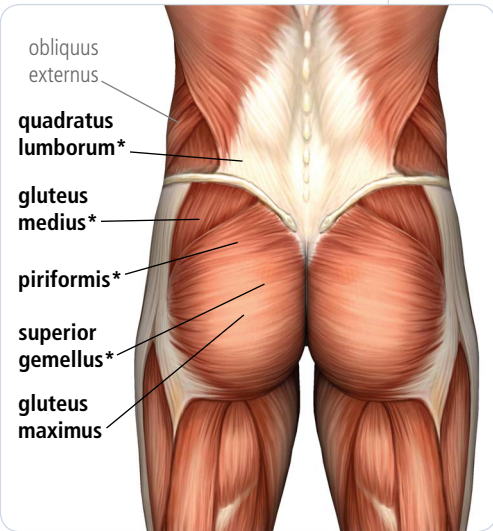
MODIFICATION
More difficult:
 Don't hold on to the column—place your hands on your hips instead. Maintain the same activation pattern and movement sequence.



BEST FOR

- gluteus maximus
- gluteus medius
- piriformis
- gemellus superior
- tensor fasciae latae

ANNOTATION KEY
 Black text indicates active muscles
 Gray text indicates stabilizing muscles
 * indicates deep muscles



CLAMSHELLS

Starting Position: Lie on your side with your knees bent so that your shoulders, hips, and ankles are in a straight line. Make sure that your spine is in a neutral position and that your bottom arm is bent with your head resting on it to allow it to remain in line with your spine. Hold a weight firmly against your thigh.

1

Action: Raise your upper knee by rotating your hip toward the ceiling, contracting your gluteals. The toes of your upper foot rise, too, as the ankle on the active leg remains in a fixed position relative to the leg. Return to the starting position by lowering the knee joint.

Movement Path: Both your upper and lower leg move in a curvilinear path away from the body's midline, from horizontal toward vertical and back.

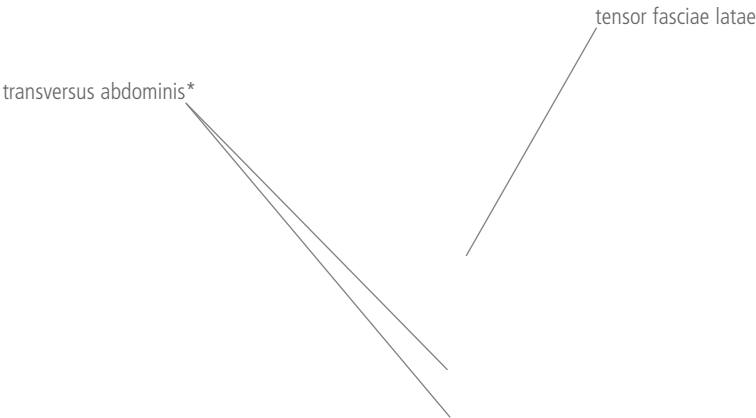
2

- STABILIZE BY**
- Keeping your abdominal muscles pulled in and up throughout the entire movement.
 - Maintaining a neutral spinal position throughout the movement.
 - Keeping the ankle on the active leg locked in a fixed position.
 - Firmly controlling the hip joint on the active side.

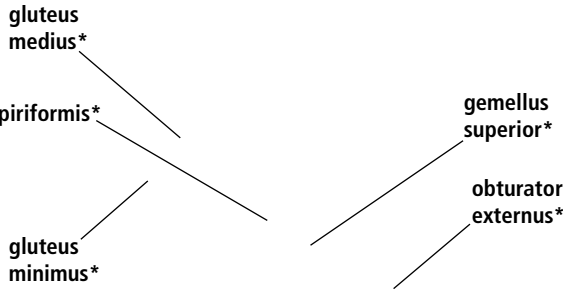
CLAMSHELLS • LEGS AND HIPS

BEST FOR

- gluteus medius
- gluteus minimus
- obturator externus
- piriformis
- superior gemellus



ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles



SINGLE-LEG DEADLIFT



1

Starting Position: Stand on your right leg and bend your left leg to a 45-degree angle. Keep your torso upright, and squeeze your shoulder blades together. Hold dumbbells in both hands.

STABILIZE BY

- Maintaining a focus on balance—it is key! Look at and focus on a spot in front of you as you bend over, balancing on one leg.
- Contracting your quadriceps on the eccentric movement and your hamstrings and gluteals on the concentric movement.

Action: Bend your right leg slightly as you bend over from your hip and reach the dumbbells toward the floor. Make sure you keep your chest up and your back slightly arched. Your left leg remains bent at 45 degrees throughout the exercise. Once you've touched the floor or gone as deep as you can, squeeze your gluteals and shoulder blades as you stand up again.

Movement Path: Your center of mass descends vertically and your torso moves in an arc as though rotating around the center of a circle.

LOOK FOR

- Your spine to remain in a constant position throughout the movement
- Your torso to flex forward from the hip joint.
- Your hamstrings to stretch; allow your pelvis to rotate forward from below the belt.

AVOID

- Rounding your back.
- Allowing your shoulder blades to slip forward.
- Improper form. Correct posture is extremely important: make sure your chest is up and your back is slightly arched. If you can't bend over very far in this position, that is okay. It is better to have proper posture than greater range of motion.

2

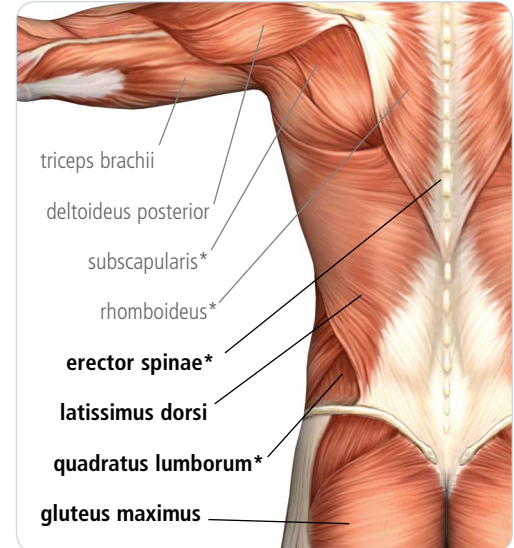
SINGLE-LEG DEADLIFT • LEGS AND HIPS

1

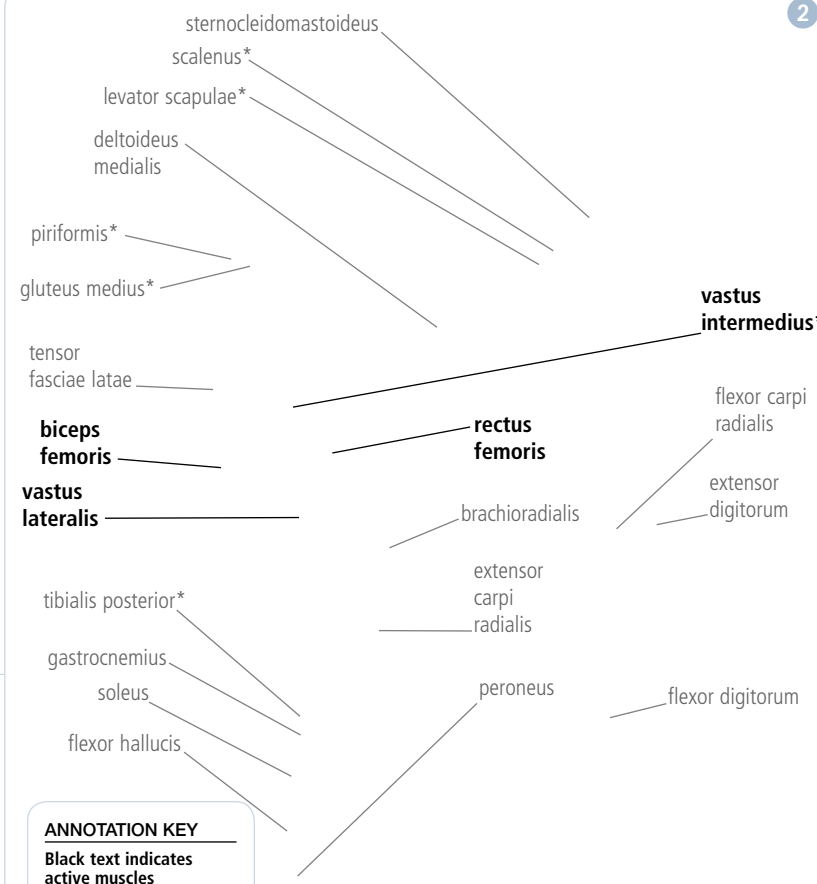


BEST FOR

- biceps femoris
- erector spinae
- gluteus maximus
- quadratus lumborum
- rectus femoris
- latissimus dorsi
- semimembranosus
- semitendinosus
- vastus lateralis
- vastus medialis



2



ANNOTATION KEY

Black text indicates active muscles
 Gray text indicates stabilizing muscles
 * indicates deep muscles

LEG PRESS PLIÉ

1

Starting Position: Sit with your feet placed slightly wider than shoulder-width apart. Ensure that they are flat, directly under your hips.

Grasp the handles firmly, keeping your chest and rib cage high. Make sure that your knees are directly above your toes and pointing in the same direction.

Action: Using the entirety of each foot, push firmly into your foot base, extending both knees and hips simultaneously while exhaling until your knees are fully extended. Inhale, allow your knees, hips, and feet to return to the starting position.

2

Movement Path: Your torso remains in a constant position as your legs and hips are extended, so that your center of mass moves away from your feet in a straight line horizontally. The return path is the same but opposite action.

STABILIZE BY

- Pulling your hips down firmly into the seat.
- Keeping your feet flat.
- Keeping your knees parallel and over your feet at all times.

LOOK FOR

- A simultaneous extension of both your knee and hip joints as well as a slight flexion of your ankles.
- Your gluteals to remain in firm contact with the seat.

AVOID

- Allowing your knees to migrate either toward or away from each other.
- Rotating your toes outward.
- Allowing your hips or gluteals to roll up off the seat.

LEG PRESS PLIÉ • LEGS AND HIPS

BEST FOR

- | | |
|-------------------|-------------------|
| • adductor longus | • rectus femoris |
| • adductor magnus | • sartorius |
| • gluteus maximus | • semimembranosus |
| • gluteus medius | • semitendinosus |
| • gracilis | • soleus |
| | • vastus medialis |

gluteus medius*
 piriformis*
 adductor magnus
 erector spinae*
 gluteus maximus
 semitendinosus

sartorius
 gracilis*
 adductor longus
 gastrocnemius
 rectus femoris
 vastus medialis
 tibialis posterior*
 tibialis anterior
 extensor digitorum longus
 soleus
 semimembranosus
 biceps femoris
 vastus lateralis
 tensor fasciae latae

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

MODIFICATION

More difficult: Use only one leg, starting with the active leg bent at a 90-degree angle and your foot at chest height. Keep your opposite leg straight and avoid rotating your hips or pelvis. Maintain the same activation pattern and movement sequence.



LEG EXTENSION

Starting Position: In a seated position, ensure that the backs of your knees rest directly against the edge of the seat. Your knee joints should be directly adjacent and bent to at least 90 degrees to the machine's pivot arm. The resistance pad should lie just above ankle height. Grasp the handles firmly, with your torso upright and your hips pulled down securely into the seat.



LOOK FOR

- Only your lower legs to move; the rest of your body should remain stationary.
- Control of the movement in both directions.

AVOID

- Allowing any space between the back of your knees and the seat edge.
- Any spinal movement.
- Elevating your shoulders.
- Allowing your hips or gluteals to elevate off the seat.

STABILIZE BY

- Keeping your hips firmly down into the seat throughout the movement.
- Maintaining an upright torso, right against the seat back.
- Pulling your toes up slightly so that your ankle joints are slightly dorsi-flexed.

Action: Exhale while contracting your quadriceps and extending your legs, kicking your feet outward and up until your knee joints are completely extended. Allow your lower legs to drop while inhaling to return to the starting position.

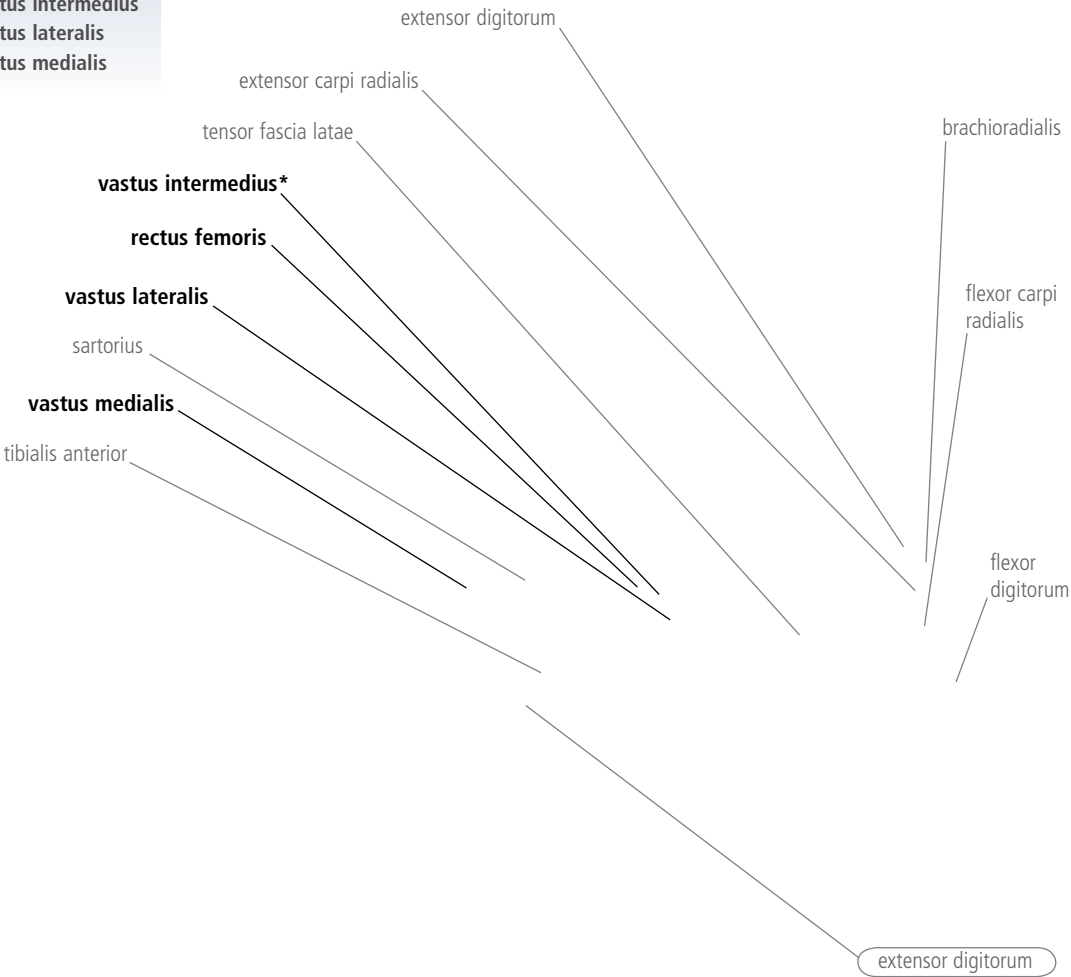
Movement Path: Your feet are kicked out and away from under the upper leg though a curvilinear arc.

2

LEG EXTENSION • LEGS AND HIPS

BEST FOR

- rectus femoris
- vastus intermedius
- vastus lateralis
- vastus medialis



ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

LOOK FOR

- The only movement to occur from the knee joint to the ankle joint as your lower leg is drawn up toward your hips.

AVOID

- Separating your toes.
- Arching your back.
- Allowing your hips to elevate off the bench.
- Letting the resistance pad slide on the lower leg as your muscles are contracted and your knee is flexed.

LEG CURL

Starting Position: Lie in a prone position, with your hips directly over the apex of the bend in the bench. Adjust your lower legs so that the resistance arm pad rests on your Achilles tendon, just above the top of your shoe. Your knee joint should be directly adjacent to the pivot joint of the machine.

Action: Exhaling, contract your hamstrings and pull your heels up and in until the bar comes into contact with your lower gluteals (maximum range) or until your knee joint reaches 90 degrees of flexion. Inhale as the weight is lowered in a controlled movement to the starting position.

Movement Path: From a 180-degree flat plane with your knees completely extended, your lower leg swings up in an arc toward your hips until there is less than 90 degrees of knee flexion and your ankles are close to your hips. Return in the same curvilinear path.

STABILIZE BY

- Keeping your hips in contact with the bench at all times.
- Firmly grasping the handles.
- Keeping your feet and ankles at a 90-degree angle.

LEG CURL • LEGS AND HIPS

BEST FOR

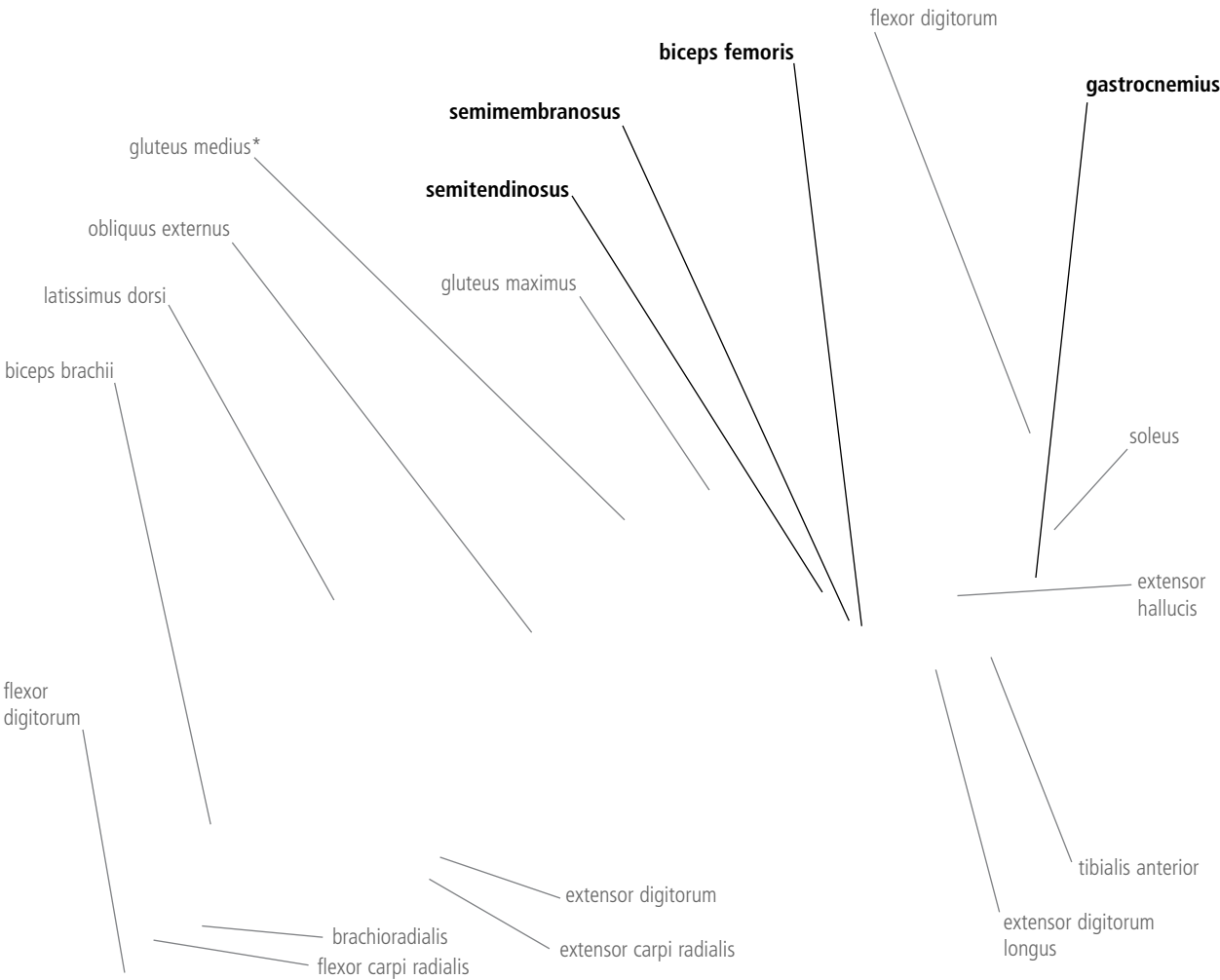
- biceps femoris
- gastrocnemius
- semimembranosus
- semitendinosus

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



WALL SIT

Starting Position: Place your back against a wall in a standing position, with your feet directly under your hips and your hands placed against the wall for balance. Walk your feet forward and slowly slide your torso down the wall until your hips, knees, and ankles are all at 90-degree angles. Your knees should be directly over your ankles. Your legs should be parallel. Make sure that your feet are flat on the floor. Keep your spine in a neutral position, your hands on your hips, and your head up.

Action: None (isometric muscle contraction). The position should be maintained until fatigue makes your muscles burn and the form is difficult to maintain.

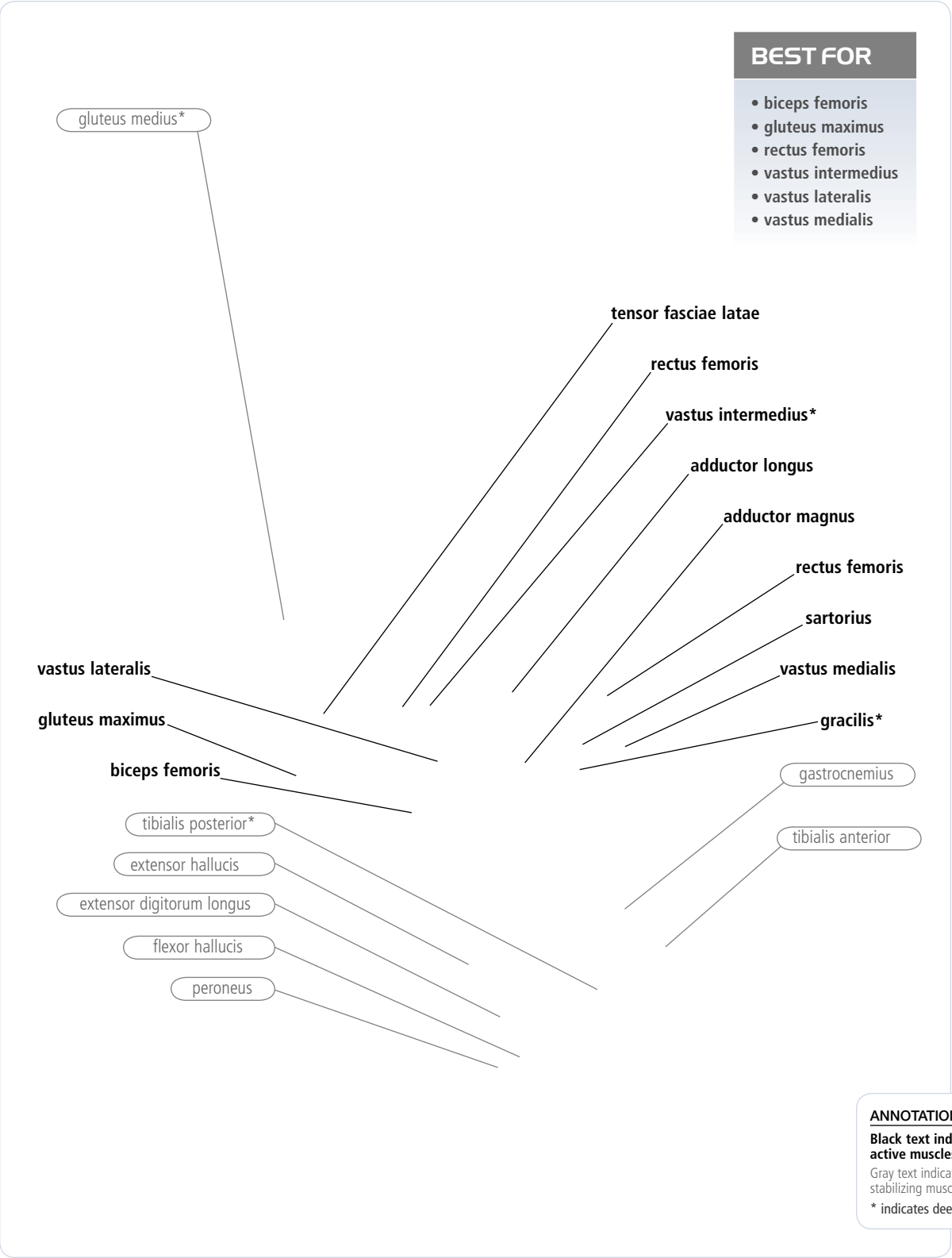
Movement Path: None.

- LOOK FOR**
- No movement whatsoever.
 - Firm 90-degree angles.
 - Your muscles to shake when fatigued.

- AVOID**
- Any movement.
 - Reaching a level of fatigue that requires you to sit rather than stand to remove yourself from the wall.

- STABILIZE BY**
- Keeping your shoulders, hips, and upper back pressed firmly against the wall.
 - Keeping your feet flat and the pressure evenly distributed.
 - Maintaining tension on the transversus abdominis by pulling it up and in with your ribs high.

WALL SIT • LEGS AND HIPS



ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

THE SKATER

Starting Position: Bend your knees and hips so that your torso is at a 45-degree angle. Place your feet slightly closer than hip-width apart (the active side hip should be flexed to a slightly greater degree). Grasp the cable frame in front of you with your palms facing down. Stand on one foot, and to the opposite leg, attach an ankle strap connected to a low pulley located directly in front of the involved leg. Raise the involved leg slightly.

1

STABILIZE BY

- Keeping your spine in a neutral position.
- Keeping the standing leg in a fixed position and your weight evenly distributed.

2

Action: Your knee extends as the active hip both extends and externally rotates, until your hip and knee are fully extended. Return by allowing the resistance to pull your leg back to the starting position.

Movement Path: Your foot moves away and out from both your midline and center of mass on the extension and up and into them upon return to the starting position.

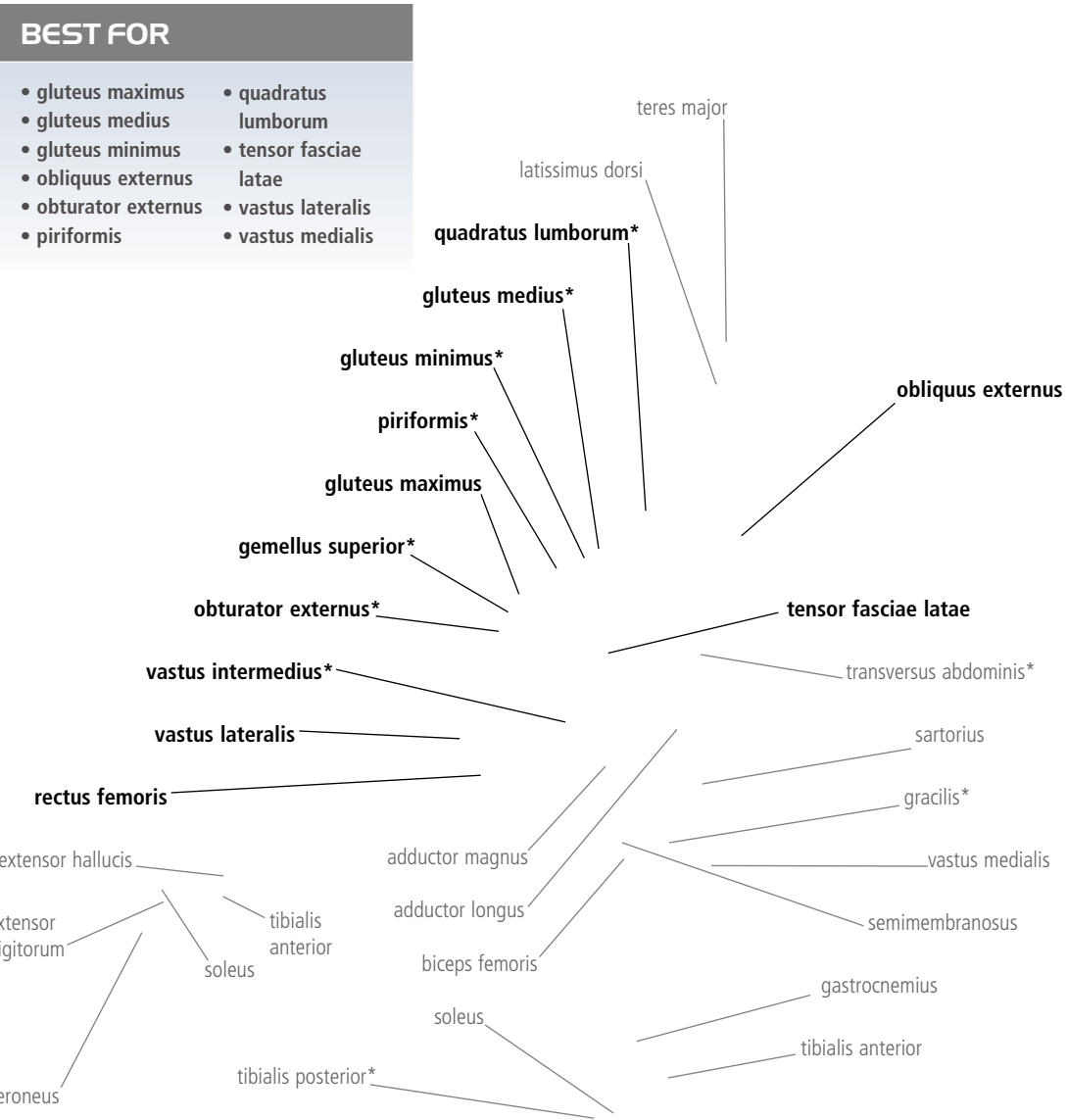
LOOK FOR

- The sole of your foot to rotate outward at the same time as the knee joint.
- The movement to appear balanced and timed so that it flows easily.

AVOID

- Rotating your upper body.
- Excessively arching your back.
- Gripping the column so that balance is transferred from your standing foot to your hands.

THE SKATER • LEGS AND HIPS



ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

PLOW

LEGS AND HIPS

LOOK FOR

- A simultaneous movement while your hips raise, so that your spine is at a 45-degree angle from your hip to your shoulder from the ground.

AVOID

- Dropping your knees toward the floor.
- Bending your elbows.
- Allowing your shoulders to either elevate toward your ears or round forward.

STABILIZE BY

- Keeping your chest high and contracted.
- Elongating your neck and extending your elbows throughout the movement.

Starting Position: Place your hands on the ground, with your legs extended so that the tops of your shoes are on top of a Swiss ball in a push-up position. Keep your spine neutral.

1

Action: Pull your knees up toward your chest while flexing your feet, balancing your toes on the ball, driving your hips toward the ceiling, and retracting your abdomen.

Movement Path: Your torso flexes in a straight line and a single plane. Your feet move up toward your midline in a horizontal plane.

2

PLOW • LEGS AND HIPS

BEST FOR

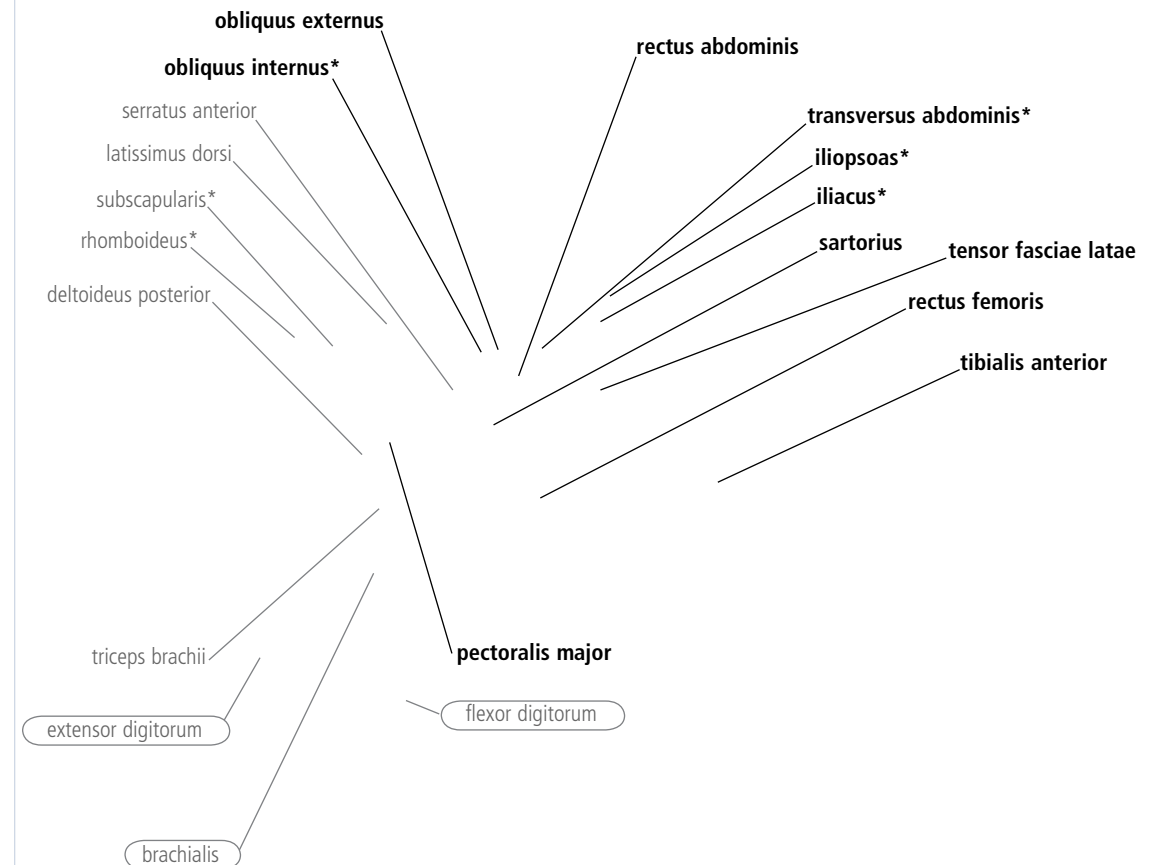
- iliacus
- iliopsoas
- obliquus externus
- obliquus internus
- rectus abdominis
- sartorius
- tibialis anterior
- transversus abdominis

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



MODIFICATION

More difficult: Push your hips toward the ceiling, keeping your knees straight. The soles of your feet should be on the ball, with your head and chest between your arms and your legs and torso at a 90 degree angle. Avoid rounding your back, bending your knees, or bending your elbows.



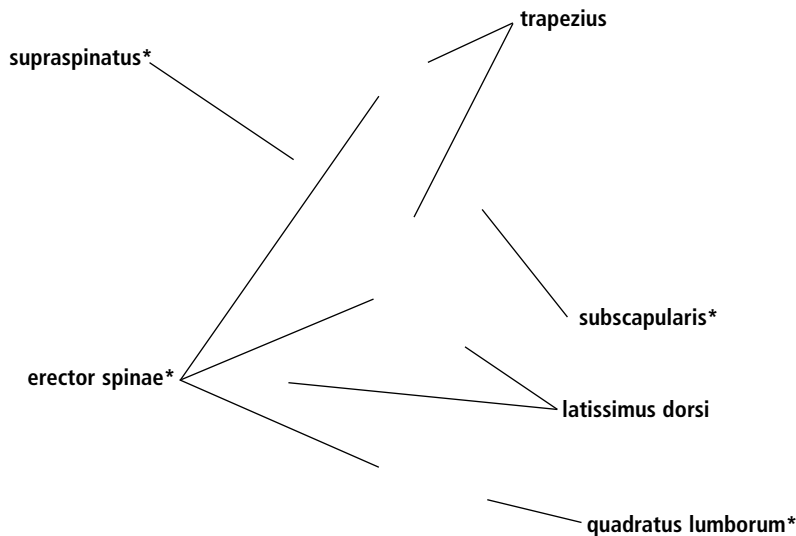
BACK

The muscles of the back are used to move not only the spine, but also the hips, head, arms, and pelvis.

There are three groups of back muscles: lower, upper, and deep spinal. Lower-back muscles work with hip muscles to tilt the pelvis backward and forward, as well as flex and extend the lower spine. This combination of muscles keeps the spine in a normal lordosis, without which neither the upper nor lower body could function properly or efficiently. The upper-back muscles depress, elevate, and rotate the scapula, and retract, rotate, adduct, and abduct the humerus. These muscles are primarily responsible for all “pulling” movements, along with the biceps. The deep spinal muscles both move and stabilize the vertebrae. There are no movements in this book, or in everyday life for that matter, which do not require some contribution from the muscles of the back.

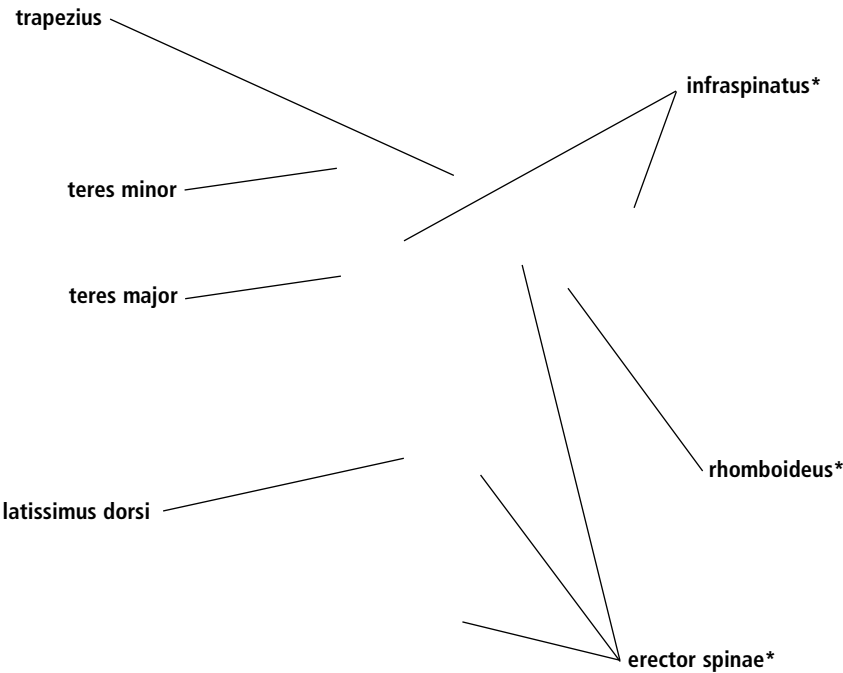
BACK ANATOMY

BACK



ANNOTATION KEY
* indicates deep muscles

BACK



ANNOTATION KEY
* indicates deep muscles

BACK EXTENSION

Starting Position: Rest your upper legs on pads so that your hip bones are slightly above the top of the pads. Your legs and knees should be straight and contracted, with the calves on your lower legs securely pushed back into the lower pad, with your feet flat. Your torso should form a straight line from head to toe, with your spine in a neutral position. Clasp your fingers behind your head, with your shoulders down, your chin up, and your elbows pointing directly out from your shoulders.

Action: Drop your entire torso forward from your hips, keeping your spine solid and your elbows back. Contract your hamstrings, gluteals, and back to return to the starting position. Inhale as you drop forward, and exhale as you return to the starting position.

Movement Path: Your entire upper body moves from a vertical position in a downward arc and returns.

STABILIZE BY

- Contracting all of your leg muscles.
- Keeping your chest high, your shoulders down, and your head neutral.
- Keeping your abdomen pulled up and in.

LOOK FOR

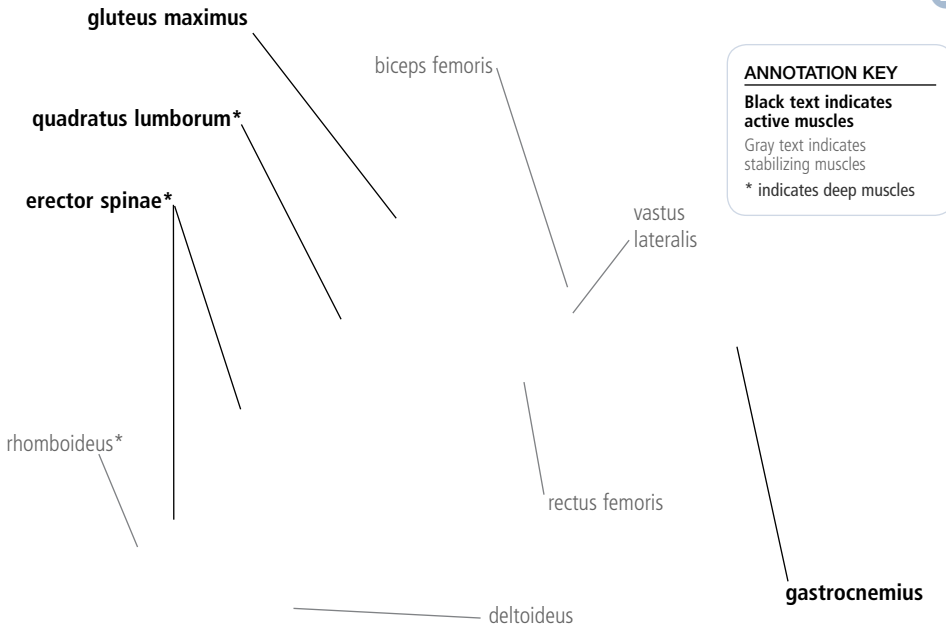
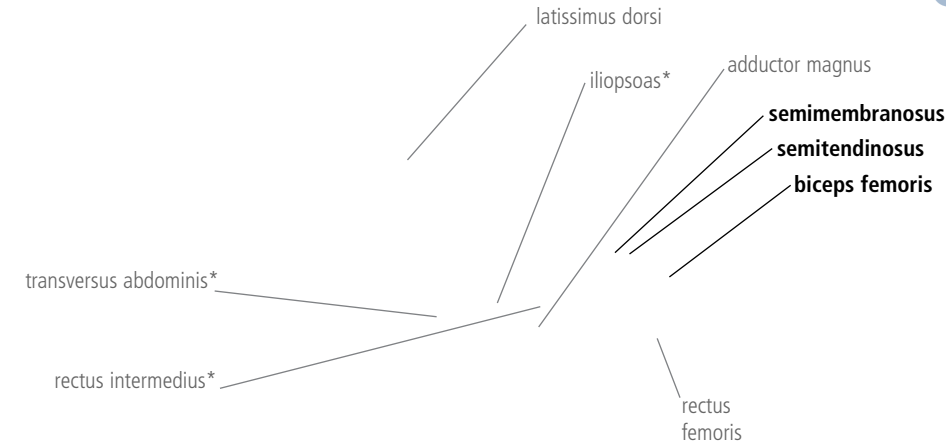
- Your body to move forward from the lower gluteal, not your back or spine.
- Your lower body to remain taut throughout the movement.
- Your lower legs to be in continuous contact with the support pads.

AVOID

- Elevating your shoulders.
- Bending your knees.
- Improper positioning (your hip bones should be in direct contact with the pads).

1

2



ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

BEST FOR

- biceps femoris
- erector spinae
- gluteus maximus
- quadratus lumborum
- semimembranosus
- semitendinosus

MODIFICATIONS



More difficult: As you rise, twist your torso so that your upper body faces either left or right. Untwist as you descend.



More difficult: Lie on your stomach on a Swiss ball with your arms behind your head and your legs spread. Twist as you rise as described above.



More difficult: Hold dumbbells below your chest to begin; raise them to 180 degrees. This is also a modification of the Rear Raise; see pages 120–121.

ROMANIAN DEADLIFT

BACK

LOOK FOR

- All movement to happen at the same time.
- Your spine to remain completely stable from hips to head.
- Your head to be up, with your eyes forward and looking upward.

AVOID

- Allowing your spine to round (by flexing forward) or change position in segments as it moves.
- Bending so that your hips are above your shoulders during the movement.
- Bending your elbows or shrugging your shoulders.
- Allowing your weight to rest in the front part of the foot or the bar to be forward of the toe line.

1

Starting Position: With the barbell on the ground and your feet shoulder-width apart, stand so that your shins contact the bar. Grasp the bar with an alternating grip (with one palm facing toward you and the other away). Bend your knees slightly, keeping your spine in a neutral position and your hips elevated, so that your head, shoulders, and hips are in a straight line and parallel to the floor.

Action: Create tension from your hands through the back of your body, all the way to your heels. Drive your back up and your hips forward, drawing the bar in a straight line vertically adjacent to your shins, and continue until you are in a full standing position.

2

Movement Path:
Your center of mass moves vertically upward as the line of your torso rotates in an arc.



3

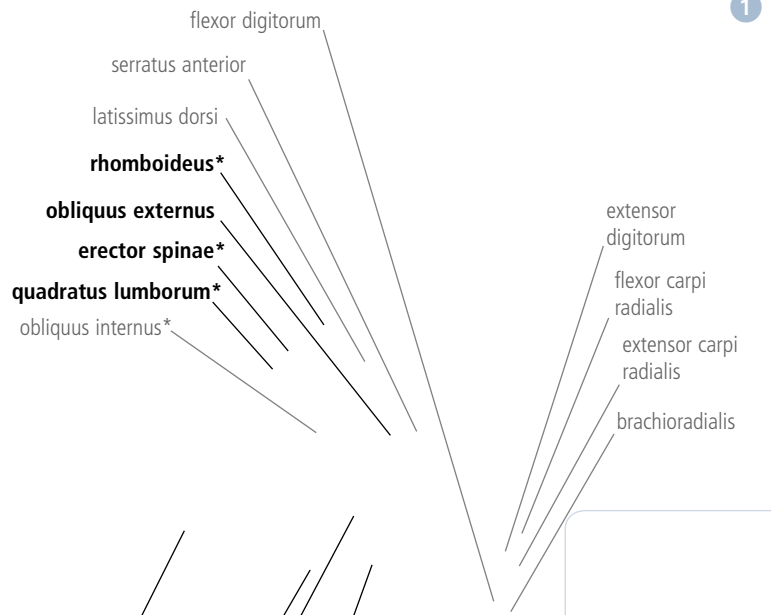


STABILIZE BY

- Keeping your rib cage high, your head up, and your shoulders down and back, with your shoulder blades flat on your rib cage.

ROMANIAN DEADLIFT • BACK

1



flexor digitorum
serratus anterior
latissimus dorsi
rhomboideus*
obliquus externus
erector spinae*
quadratus lumborum*
obliquus internus*

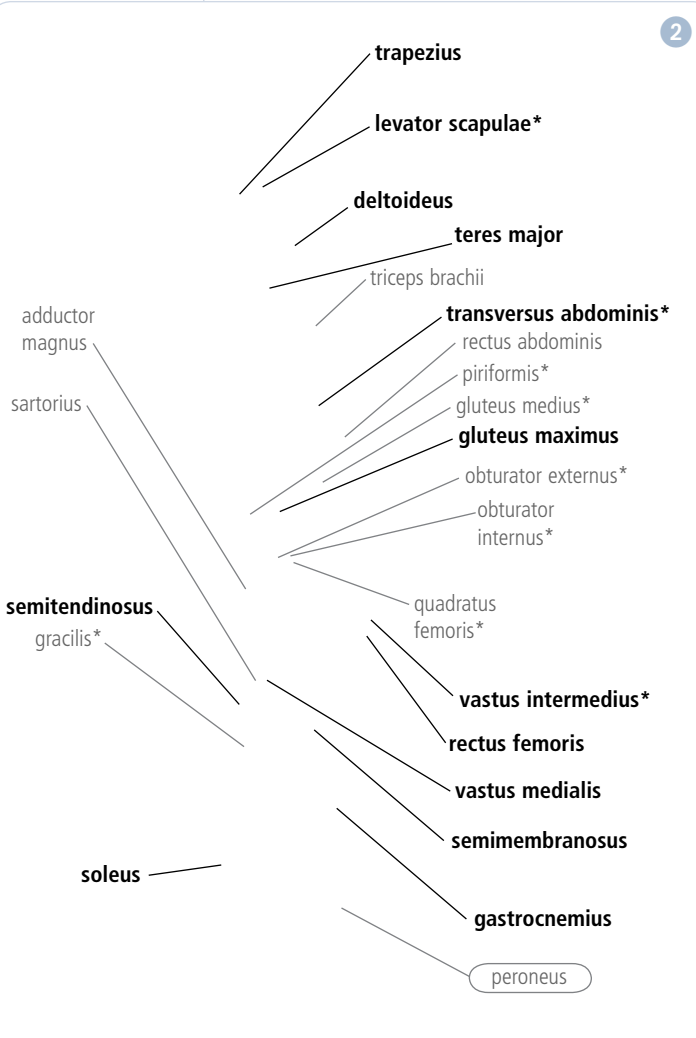
extensor digitorum
flexor carpi radialis
extensor carpi radialis
brachioradialis

adductor magnus
semitendinosus
vastus lateralis
biceps femoris

BEST FOR

- biceps femoris
- erector spinae
- gluteus maximus
- quadratus lumborum
- levator scapulae
- rhomboideus
- semimembranosus
- semitendinosus
- trapezius

2



trapezius
levator scapulae*
deltoideus
teres major
triceps brachii
transversus abdominis*
rectus abdominis
piriformis*
gluteus medius*
gluteus maximus
obturator externus*
obturator internus*
quadratus femoris*
vastus intermedius*
rectus femoris
vastus medialis
semimembranosus
gastrocnemius
peroneus

adductor magnus
sartorius
semitendinosus
gracilis*

soleus

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

ONE-ARM DUMBBELL ROW

Starting Position: Place one knee and one hand on a bench, with your spine horizontal and parallel to the bench. Your knee should be directly below your hip and your hand directly below your shoulder. Place your opposite foot on the ground, bending your knee slightly. Your feet should be slightly wider than shoulder width. Grasp a dumbbell in your free hand, and allow your arm to hang perpendicular to the floor.

1

- LOOK FOR**
- The dumbbell to move from your shoulder toward the mid-torso as it rises.
 - The shoulder on the working side to stay down and away from your ear.

- AVOID**
- Any spinal movement.
 - Any hip rotation.
 - Lifting the dumbbell in a straight line.
 - Allowing your elbow to migrate away from the side of your torso during any part of the movement.

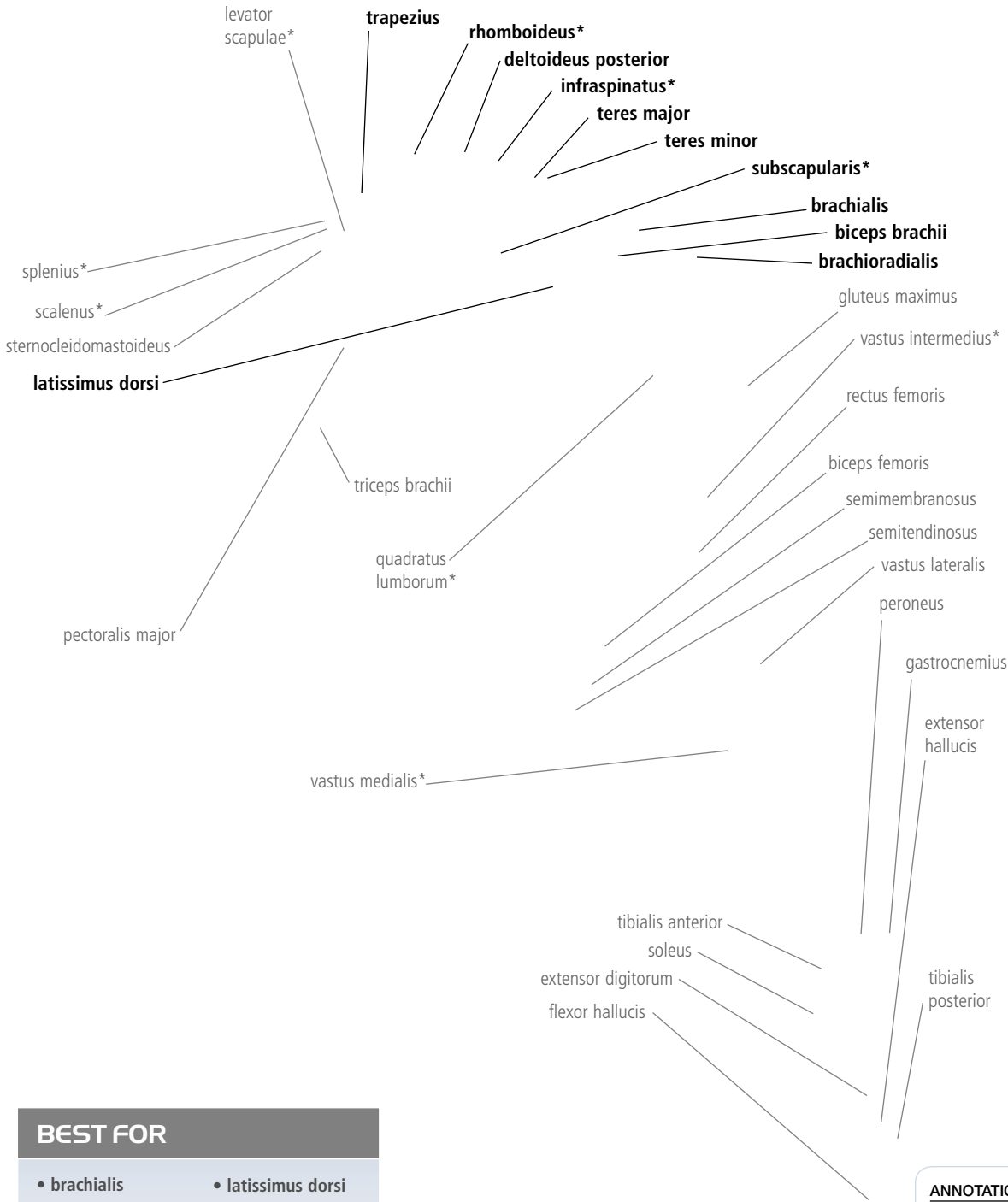
Action: Exhale, and drive your elbow directly up toward the ceiling by retracting your shoulder blade and flexing the elbow joint. Inhale, and lower your elbow, keeping the rest of your body motionless.

2

Movement Path: Your lower arm moves from an extended position, perpendicular to the torso and pointing directly down toward the floor, straight up, remaining perpendicular to the torso in the vertical plane throughout the movement. Your upper arm moves from the beginning position upward and backward in an arc, past the torso until your hand is adjacent to your rib cage.

- STABILIZE BY**
- Keeping your hips and shoulders even.
 - Bearing your weight evenly on all three contact points (the weight-bearing hand, the knee on the bench, and the foot on the floor).
 - Keeping your chest high and maintaining a neutral spinal position.

ONE-ARM DUMBBELL ROW • BACK



BEST FOR

- | | |
|------------------------|--------------------|
| • brachialis | • latissimus dorsi |
| • brachioradialis | • rhomboideus |
| • infraspinatus | • teres major |
| • deltoideus posterior | • teres minor |
| | • trapezius |

ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

BARBELL ROW

LOOK FOR

- The bar to travel in a straight vertical path.
- Your hips and torso to remain motionless, with your head up.

AVOID

- Any spinal movement.
- Rounding your shoulders forward.
- Allowing the bar to descend in a path that is in front of toe line.
- Changing either your knee or hip position during the movement.
- Holding your breath.

Starting Position: Bend over so that your back is almost parallel to the floor, with your feet hip-width apart. Hold a barbell with straight arms and your palms facing your body. Make sure your spine is in a neutral position. Your hips should be behind your feet, and your knees should be bent, with your torso angled forward so that your shoulders are slightly above your hips. The barbell should be above your shoelaces. Your chest should be up, your abdomen pulled in, and your chin slightly raised.

STABILIZE BY

- Pulling your abdomen up and in.
- Contracting your spinal muscles and maintaining your spinal position.
- Keeping your knees bent at a fixed angle.
- Keeping your weight toward the heel of your feet and the barbell close to your legs throughout the movement.
- Contracting your gluteals and hamstrings throughout the movement.
- Keeping your chest elevated.

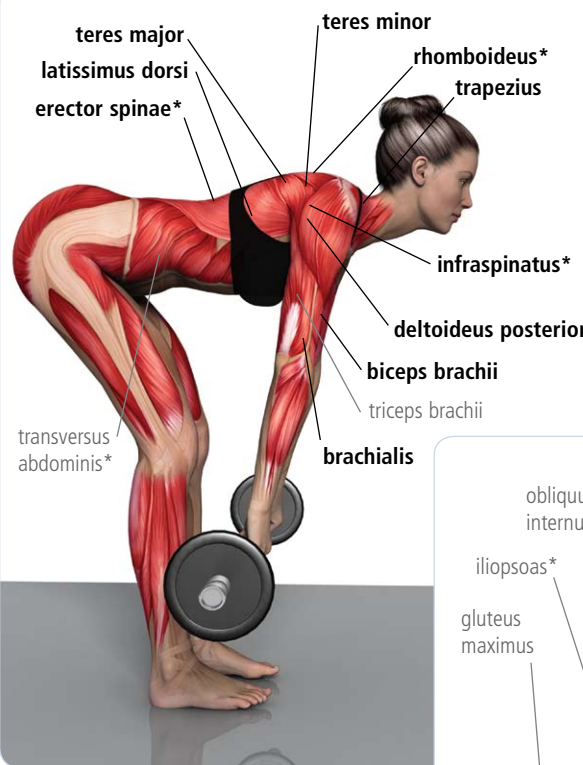
Action: Exhale, and pull the barbell directly upward to your mid-torso, just below your rib cage. Your forearms travel vertically and your elbows are pulled directly upward, above the plane of your back. Squeeze your shoulders down and back. Inhale, and lower the bar in the same path, close to and parallel with your lower legs.

Movement Path: Your center of mass remains stationary. Your upper arms move from a forward, downward-facing position backward and upward, while your lower arms remain forward and downward-facing.

BARBELL ROW • BACK

BEST FOR

- biceps brachii
- brachialis
- brachioradialis
- erector spinae
- infraspinatus
- latissimus dorsi
- deltoideus posterior
- rhomboideus
- teres major
- trapezius

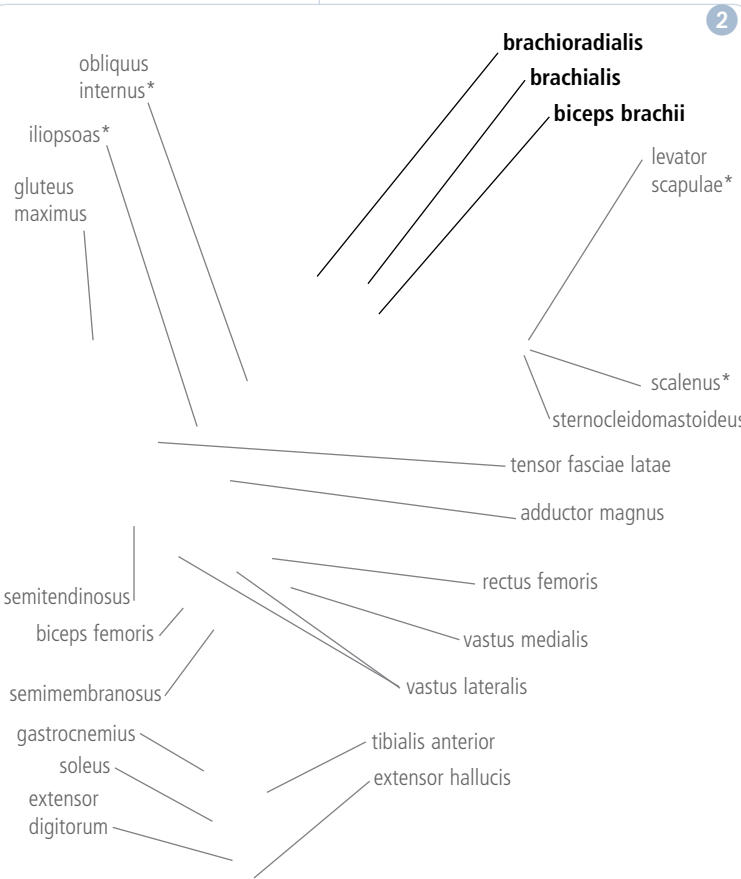


ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



DEADLIFT

Starting Position: With the barbell on the ground and your feet shoulder-width apart, stand so that your shins contact the bar. Grasp the bar with an alternating grip (with one palm facing toward you and the other away). Keep your spine neutral, positioned at a 45-degree angle to perpendicular. Drop and retract your hips so that your upper legs are parallel to the ground (or as close to parallel as your flexibility will allow). Position your shoulder joints directly over the bar. Make sure your feet are flat and your weight is evenly distributed. Pull your chest, head, and rib cage up and your abdominal muscles up and in. Inhale at the bottom of the position.



Action: Exhale, and drive your torso up and backward and your hips up and forward. Push your feet into the ground, extending your knees and pulling backward on the bar with your upper back and shoulders until you arrive at a vertical position.



Movement Path: Your hips move upward and forward while your spine and torso move upward and backward, your knees extend, and your entire body moves upward and away from the floor.

STABILIZE BY

- Keeping your rib cage high and your head up.
- Pushing your shoulders down and back, with your shoulder blades flat on your rib cage.
- Keeping your knees directly over your feet.



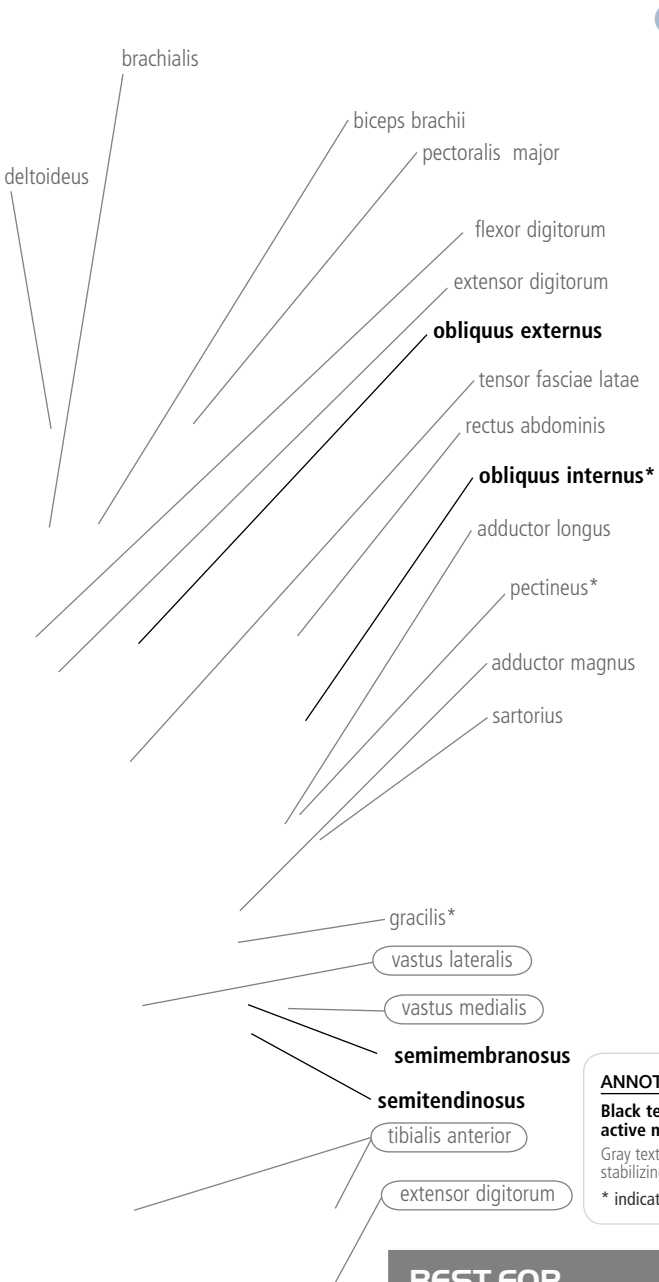
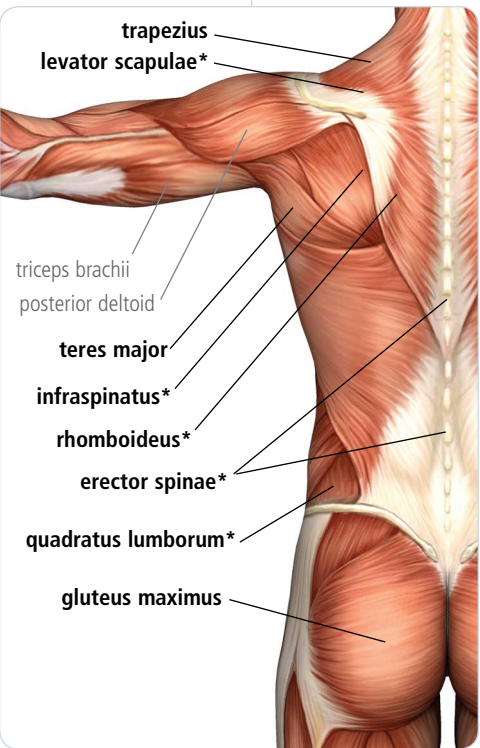
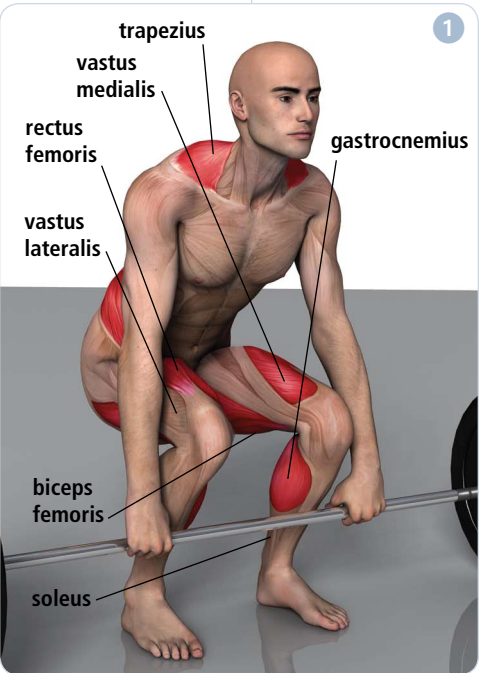
LOOK FOR

- The angle of your spine to never drop below 45 degrees during the movement.
- A slight arch in your back throughout the movement.
- All of your joints to move at the same time and at the same rate.

AVOID

- Straightening your knees prior to extending your back and hips.
- Rounding your back.
- Elevating your shoulders or lowering your head.
- Allowing your knees to migrate either inward or outward.

DEADLIFT • BACK



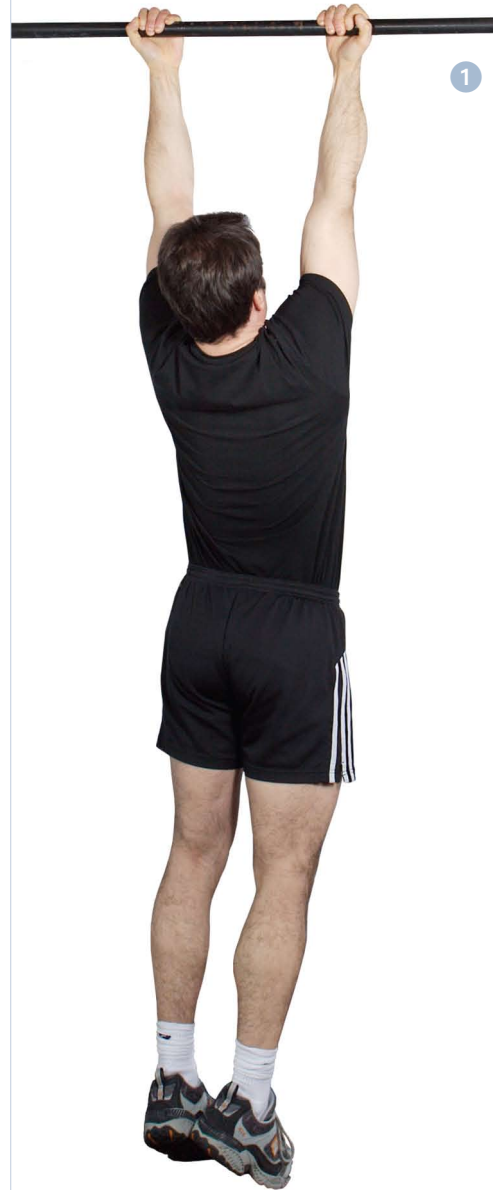
ANNOTATION KEY
Black text indicates active muscles
 Gray text indicates stabilizing muscles
 * indicates deep muscles

BEST FOR

- | | |
|----------------------|--------------------|
| • biceps femoris | • rectus femoris |
| • erector spinae | • rhomboideus |
| • gluteus maximus | • semimembranosus |
| • latissimus dorsi | • semitendinosus |
| • levator scapulae | • soleus |
| • obliquus externus | • trapezius |
| • quadratus lumborum | • vastus lateralis |
| | • vastus medialis |

CHIN-UP

Starting Position: Hanging from a bar, with your knees bent only very slightly, grip the bar with your palms in (i.e., facing your body). Keep your head in a neutral alignment. Your hands should be shoulder-width apart.



LOOK FOR

- Your arms to return to a full extension.
- You shoulder blades to draw together and downward at the beginning of the movement.

AVOID

- Swinging, jerking, chin “pecking,” or hyper-extension of elbows.

Action: Pull your body up vertically until your upper chest is aligned with the bar: this is the end of the concentric phase. Lower your body back down to the starting position with your elbows fully extended (the end of the eccentric phase).



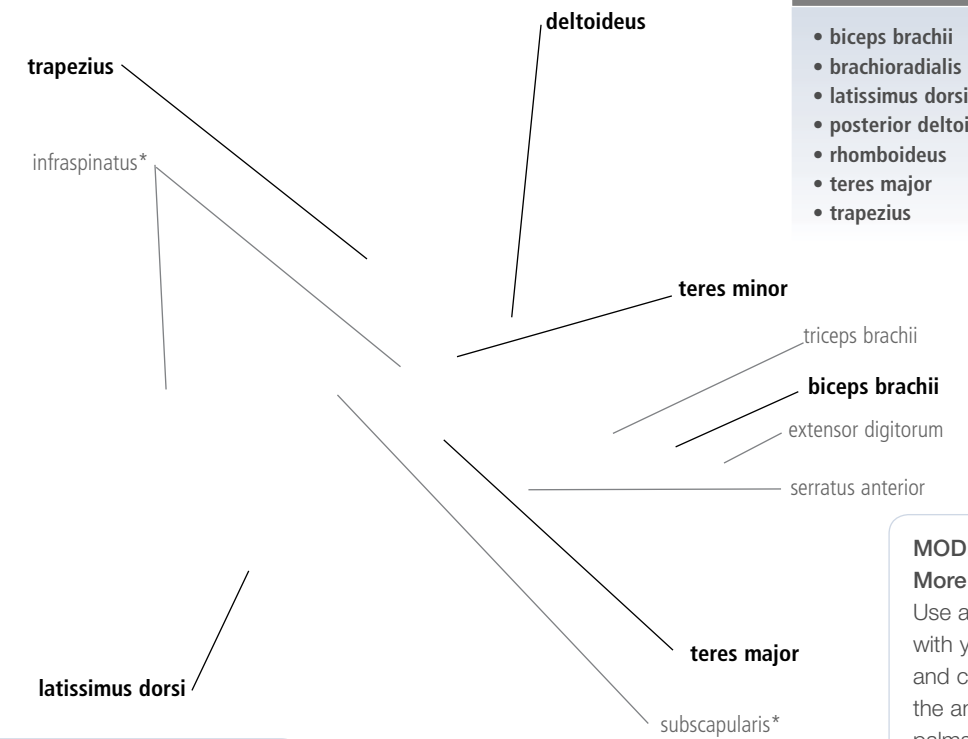
Movement Path: Your body moves vertically up. Your upper body tilts back slightly to allow your chin to smoothly pass the bar line.

STABILIZE BY

- Retracting your scapulae.
- Keeping your core tight to prevent swinging.

CHIN-UP • BACK

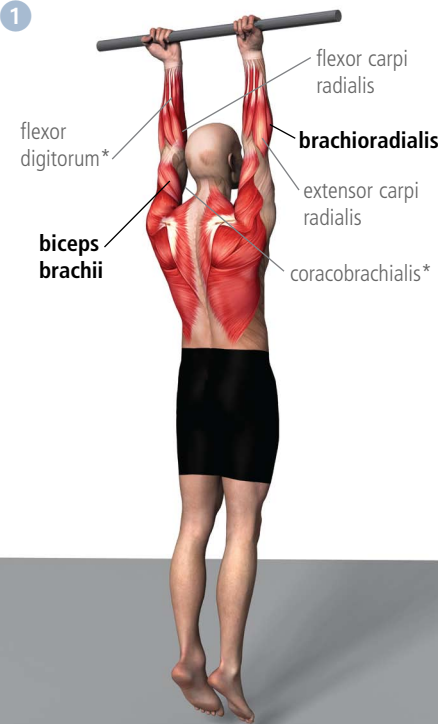
2



BEST FOR

- biceps brachii
- brachioradialis
- latissimus dorsi
- posterior deltoid
- rhomboideus
- teres major
- trapezius

1



MODIFICATION

More difficult:

Use a wide grip with your legs bent and crossed at the ankle and your palms facing away from your body. Let a partner assist you by spotting from the back, either from the bent crossed knees or the mid-back.



ANNOTATION KEY

Black text indicates active muscles

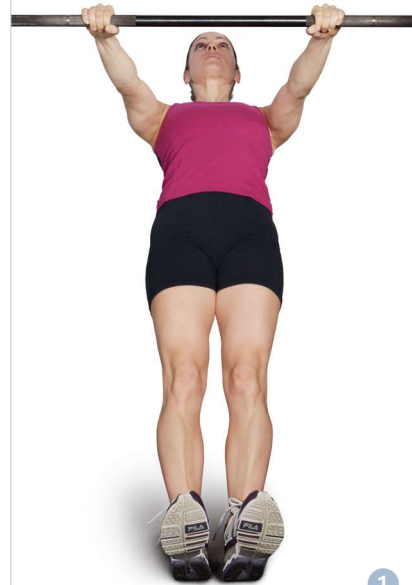
Gray text indicates stabilizing muscles

* indicates deep muscles

BODY ROW

BACK

Starting Position: Hang from a bar with your body in a flat plane. The line of your body should be at a 45-degree angle to the floor. Grasp the bar with both arms in supine or prone grips. Your elbows should be at 90-degree angles.



STABILIZE BY

- Fixing your shoulders in one position
- Locking your knees
- Keeping your ankles in a fixed position
- Keeping your hips, abdominal muscles, and lower back rigid

LOOK FOR

- A single plane of movement, maintaining a straight line from your head to your ankles.

AVOID

- A segmental elevation, such as your shoulders rising before your hips or vice versa.
- Elevating your shoulders towards your ears.
- Moving your head forward.

Action: Move your feet away from the bar until your arms are straight, keeping on your heels. Pull your body toward the bar until your chest touches it. Lower yourself slowly, and repeat. The bottom of your chest should always touch the bar at the end of the movement. Keep your body in a straight line on your heels.

Movement Path: Your entire body moves in a single arc with your feet as the fulcrum.

biceps brachii

extensor carpi radialis

pectoralis major

pectoralis minor*

coracobrachialis*

flexor digitorum

serratus anterior

flexor carpi radialis
transversus abdominis*

vastus intermedius*

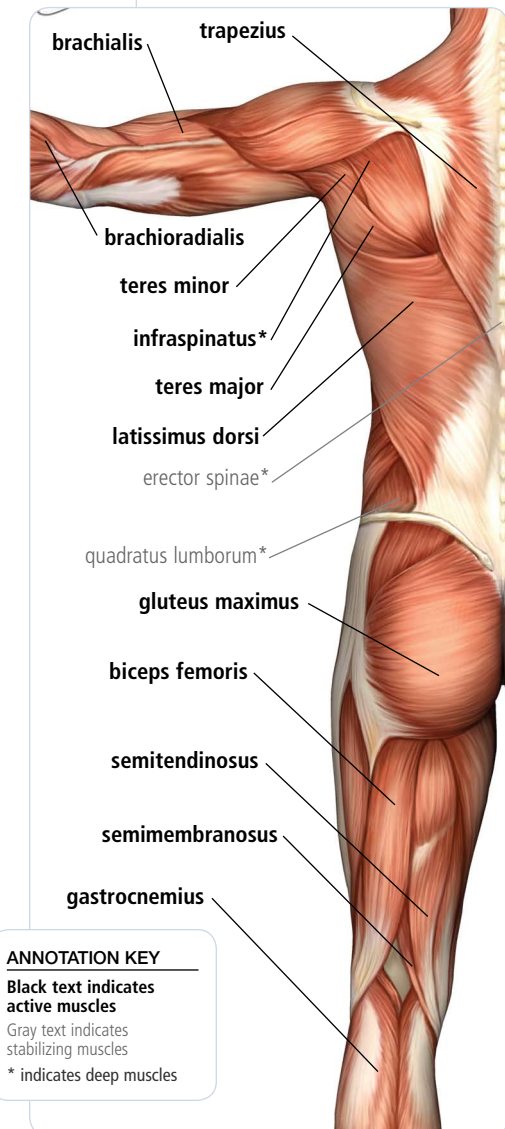
rectus femoris

vastus medialis

vastus lateralis

BEST FOR

- biceps brachii
- brachialis
- brachioradialis
- infraspinatus
- latissimus dorsi
- rhomboideus
- teres major
- teres minor
- trapezius



MODIFICATION

More difficult: Put both of your feet on a Swiss ball.



ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

PULL-OVER

Starting Position: Lying on your back on a bench, extend your arms over your chest, and bend your elbows slightly. Grasp a kettle bell so that it is positioned directly over your chin. Keep your feet flat and your spine in a neutral position.

1

Action: Inhale, and lower the weight toward the floor behind and over your head while expanding your chest and rib cage. Rotate backward from your shoulder joint, keeping your elbow movement to a minimum.

Movement Path: The weight travels from in front of your torso overhead and back. Your shoulders and arms are rotated upward and backward, then downward and forward. Your torso remains stationary.

2

- LOOK FOR**
- Your chest to remain up.
 - Your upper arms to be parallel with your ears at the bottom of the movement.
 - Your hips, middle and upper back, and feet to remain flat.

- AVOID**
- Excessive arching or flattening of your spine.
 - Bending your elbows as the weight is lowered or extending your elbows as the weight is pulled up.
 - Allowing your shoulders to shrug during the movement.

- STABILIZE BY**
- Contracting your chest and elevating your rib cage.
 - Keeping your abdominals pulled up and in.
 - Contracting your gluteals.
 - Keeping your feet flat.
 - Contracting your arms so that the elbow angle remains fixed throughout the movement.

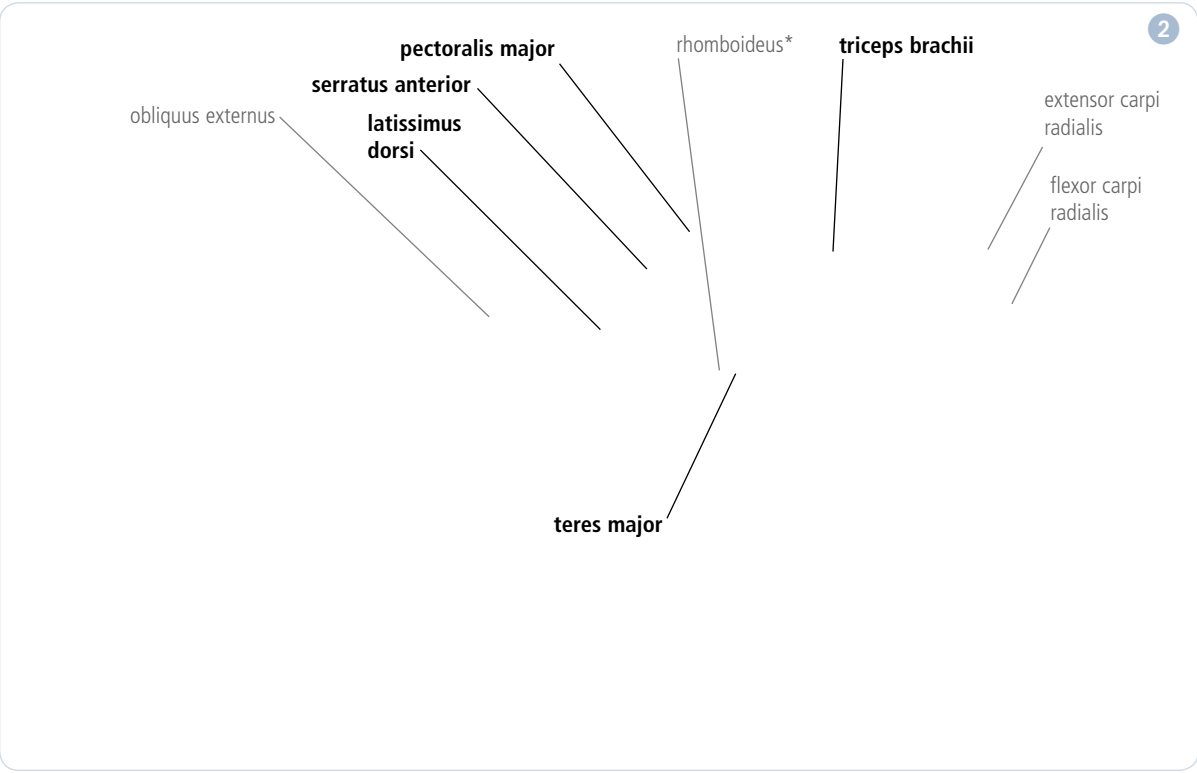
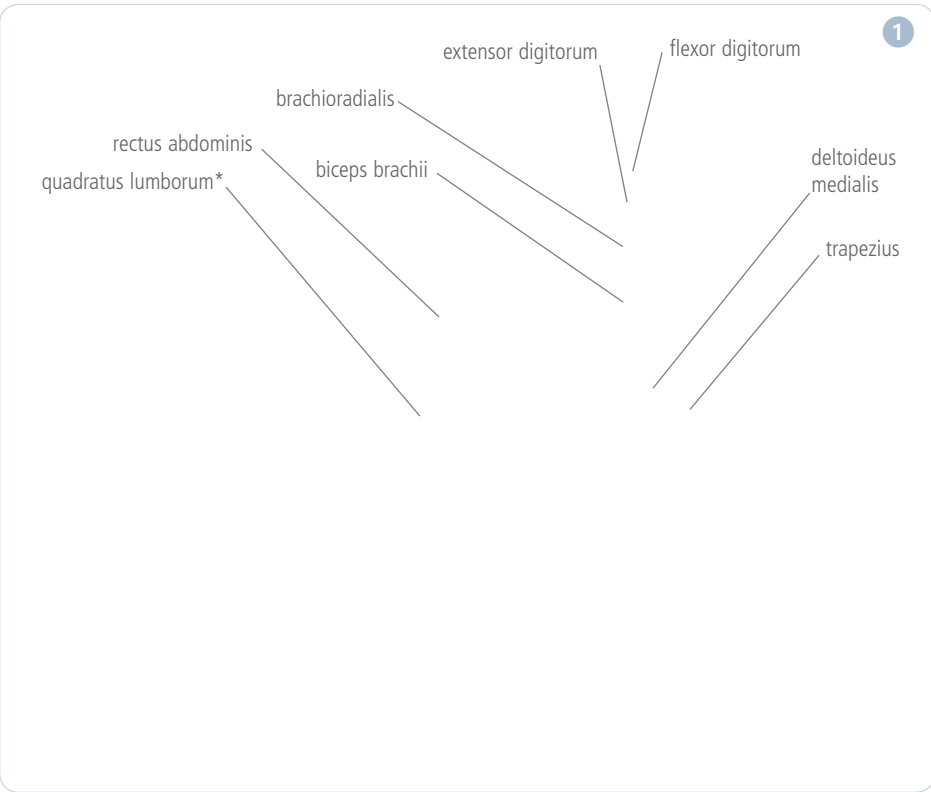
PULL-OVER • BACK

BEST FOR

- latissimus dorsi
- pectoralis major
- serratus anterior
- teres major
- triceps brachii

ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles



LAT PULL-DOWN

BACK

LOOK FOR

- A slow continuous movement in the vertical plane.

AVOID

- Leaning back too far or a swinging motion in the torso.
- Reaching your chin up higher to clear the bar.
- Rising out of your seat.
- Elevating your shoulders.

Starting Position: Sit facing the weight stack with your knees under the thigh pads and your feet flat on the floor. Hold the bar with your hands slightly wider than shoulder-width apart in an overhand grip (with your palms facing away from you). Let your arms be pulled upward until your elbows are fully extended. Your upper body should be straight with a slight (5-degree) backward lean, your head should be neutral with your eyes looking straight ahead, and your legs should be bent at 90 degrees, with the pad resting against your thighs to keep your legs grounded.



Action: Keeping your torso in a 5- to 10-degree backward lean, squeeze your shoulder blades back and down as you pull the bar in front of your head, below your chin almost to your collarbone. Slowly raise the bar back up to the starting point and repeat.

Movement Path: The movement of the bar is a vertical line.

STABILIZE BY

- Using your abdominal and back muscles to maintain an upright posture.



LAT PULL-DOWN • BACK

BEST FOR

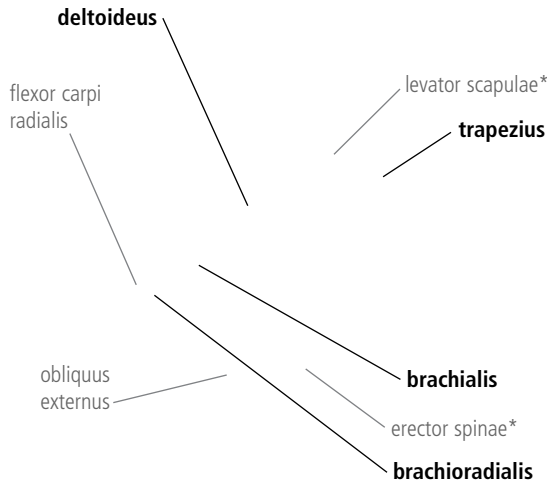
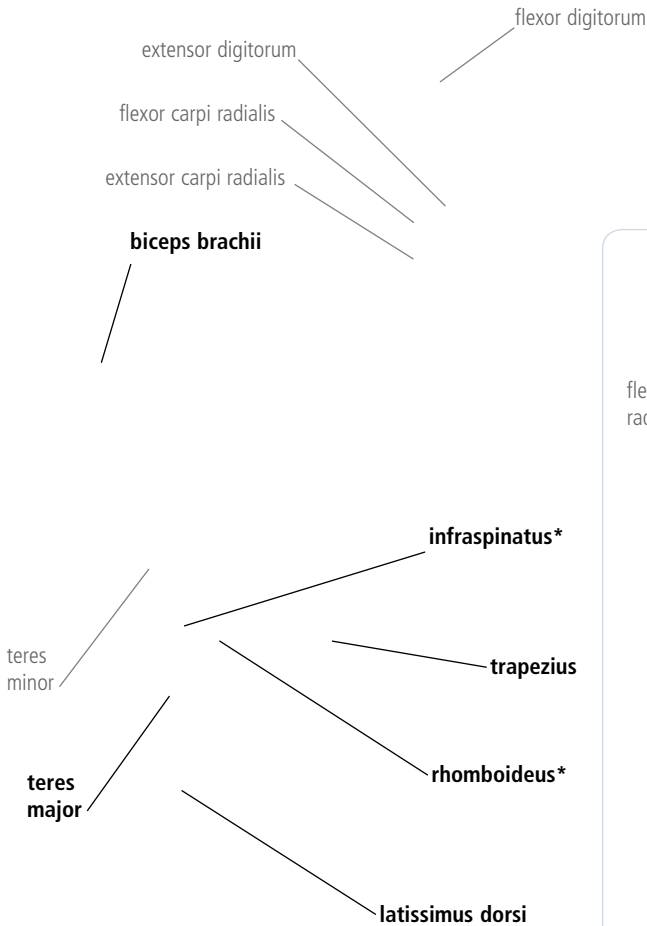
- latissimus dorsi
- rhomboideus
- lower trapezius
- teres major
- biceps brachii
- brachialis
- brachioradialis

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



MODIFICATION

Similar difficulty: Use parallel bar handles, and maintain the same activation pattern and movement sequence.



CABLE ROW

Starting Position: Sit on the bench with your legs extended in front of you. Keep your legs slightly bent and reach forward to grab the handle with both of your hands in a neutral grip (so that your palms face each other). Start with your hands over your ankles, your back arched, and your knees slightly bent.

STABILIZE BY

- Using your abdominal and back muscles to maintain an upright posture.

1

LOOK FOR

- A slow continuous movement in the horizontal plane.

AVOID

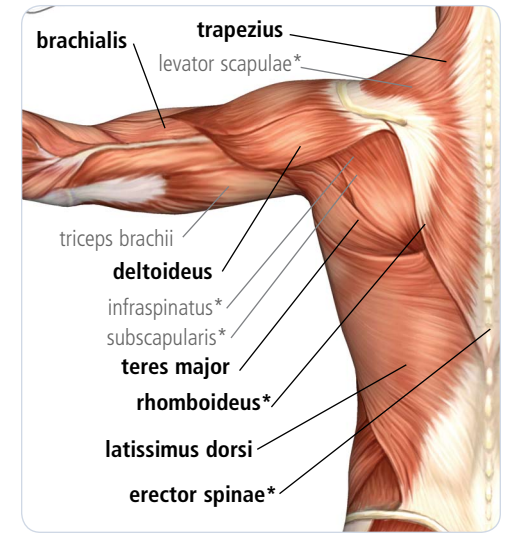
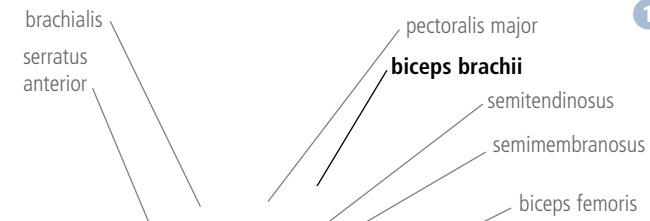
- Rounding your back.
- Too much movement in your torso (i.e., leaning forward or backward too much).
- Elevating your shoulders.

Action: As you pull the handle towards your midsection, squeezing your shoulder blades back and down, lean backward (keeping your back arched) to about 5 degrees past perpendicular. To repeat, lean forward to about 5 degrees less than perpendicular and slowly straighten your arms, then pull the handle towards your torso again as you lean back to 5 degrees past perpendicular. The set should be done within this 10-degree range. Make sure to keep your legs slightly bent, your back arched, and your elbows shoulder-width apart.

Movement Path: The movement of the bar is horizontal; your torso oscillates in a narrow 10-degree arc.

2

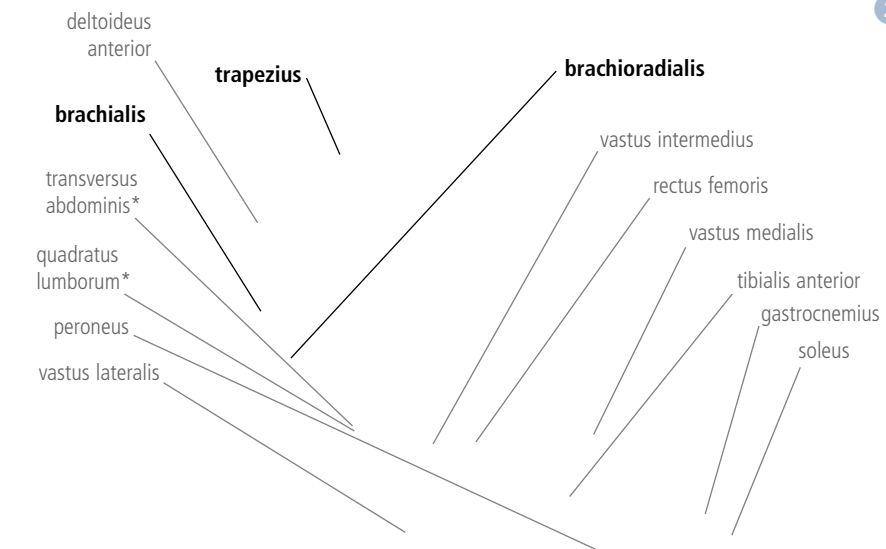
CABLE ROW • BACK



2

BEST FOR

- latissimus dorsi
- trapezius
- rhomboideus
- erector spinae
- teres major
- deltoides anterior
- brachialis
- brachioradialis
- biceps brachii

**MODIFICATION****Similar difficulty:**

Change your grip position to either supinated (with your palms up) or pronated (with your palms down).

**ANNOTATION KEY**

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

STRAIGHT-ARM PULL-DOWN

Starting Position: Stand, facing the high pulley, with your legs shoulder-width apart and your spine in a neutral position. Grasp the bar in an overhand grip with your palms facing down and your arms extended.

1

Action: Pull the bar straight down toward your lap, bringing your shoulders down and back, so that your palms face your thighs at the bottom of the movement. Return the attachment and repeat.

Movement Path: Your arms should extend slightly beyond your body in one plane of movement. Your torso rotates around your center of mass at the same rate as you pull the cable.

STABILIZE BY

- Drawing in your navel.
- Retracting your shoulder blades.
- Keeping your shoulders in one position and your upper arms alongside your body.

2

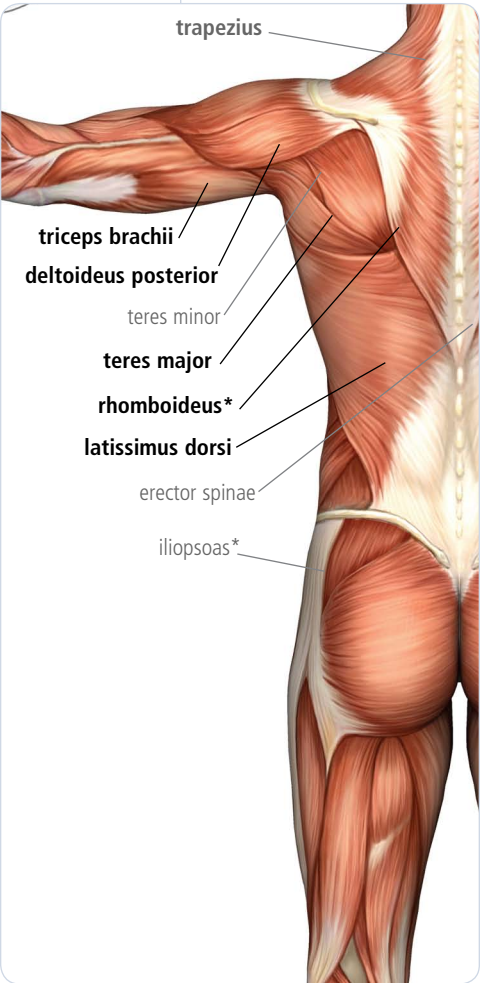
- LOOK FOR**
- A neutral spine.
 - Retracted scapulae.

- AVOID**
- Elevating your shoulders toward your ears.
 - Arching your back.
 - Bending your arms.

STRAIGHT-ARM PULL-DOWN • BACK

BEST FOR

- latissimus dorsi
- deltoideus posterior
- rhomboideus
- teres major
- triceps brachii



ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

CABLE REAR RAISE

Starting Position:

Standing equidistant and directly between two high cables, grasp the handles with the opposing hands so that your palms face each other. Take two full steps backward, and extend your arms so that your hands land directly in front of your shoulder joints with your arms completely straight, parallel to each other and the ground. The cable should be crossed between your hands at approximately chin height.



1

STABILIZE BY

- Keeping your chest high and your abdomen pulled up and in.
- Keeping your shoulders down and back throughout the movement.
- Keeping your shoulders, hips, knees, and feet in alignment.

LOOK FOR

- Your arms to be fully extended throughout the movement.

AVOID

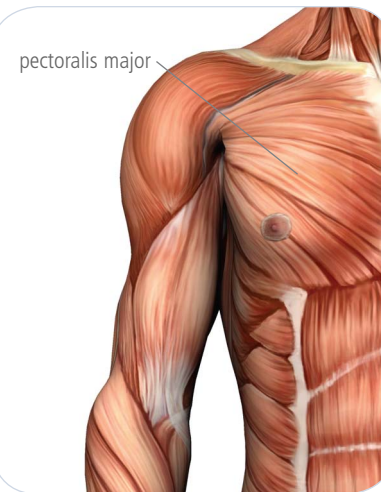
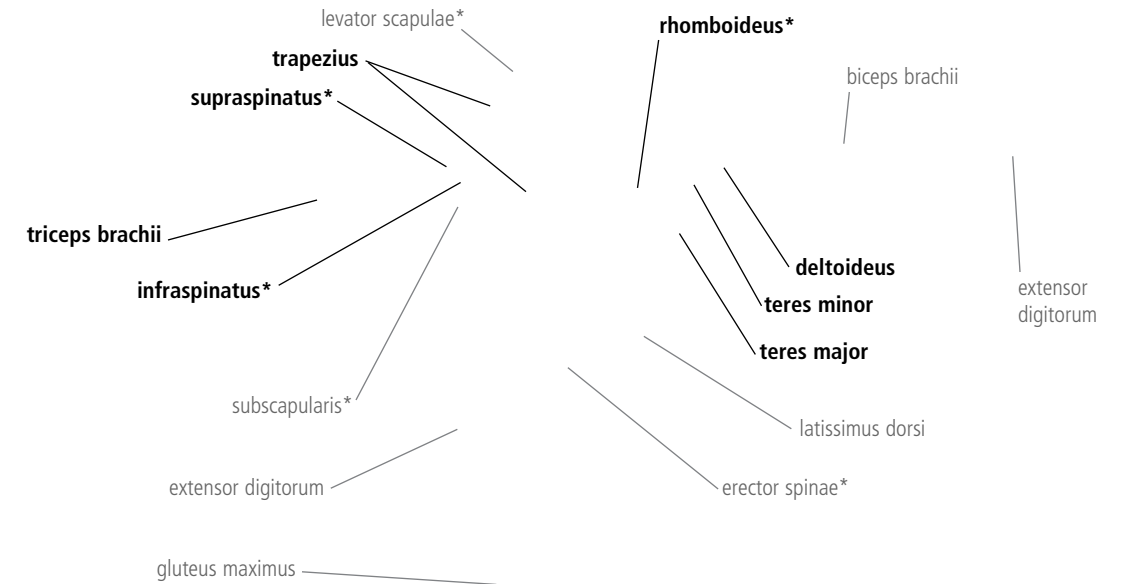
- Bending your elbows.
- Shrugging your shoulders.
- Changing the plane of your arm movement.

Action: Exhale, and retract your shoulders and arms backward and out to the sides until your arms are pointing directly out. Make sure your arms remain in a horizontal plane. Inhale, and allow your arms to return in the same but opposite movement.

Movement Path: Your torso, hips, and legs remain stationary while your arms move in a horizontal, 90-degree arc from the front of your body out to both sides.

2

CABLE REAR RAISE • BACK



BEST FOR

- infraspinatus
- deltoideus
- rhomboideus
- supraspinatus
- teres major
- teres minor
- trapezius
- triceps brachii

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

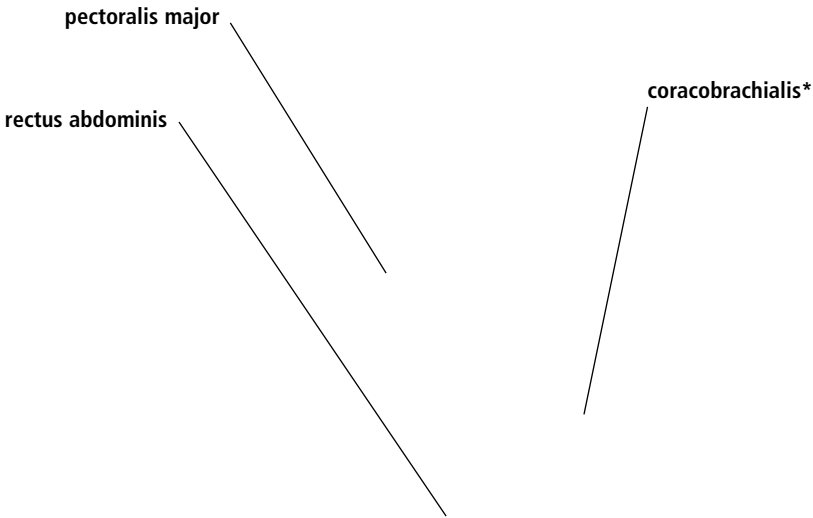
* indicates deep muscles

CHEST

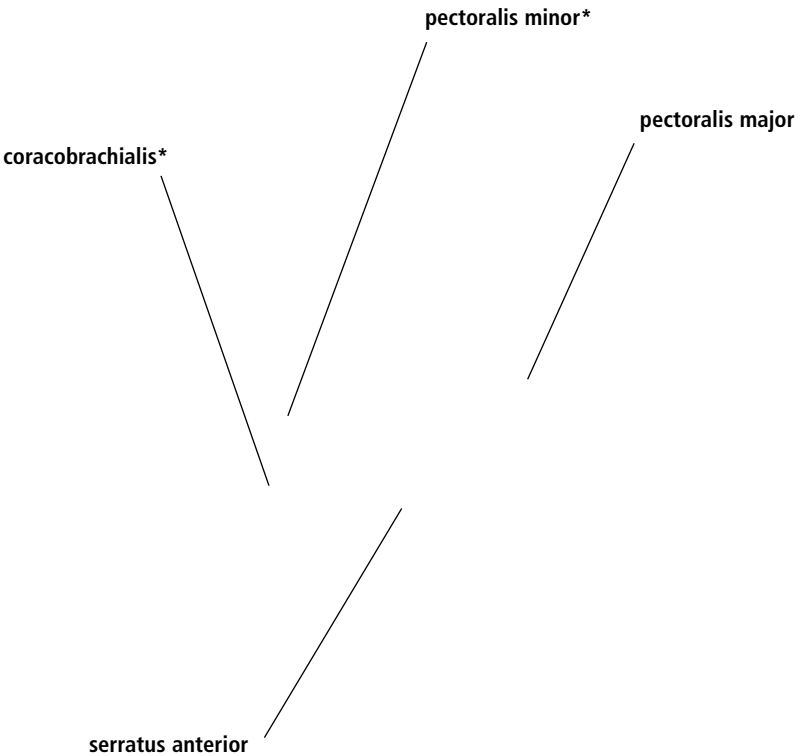
The muscles of the chest originate on the collarbone and sternum and insert on the upper arm. They are responsible for adduction, internal rotation, and forward flexion of the humerus. This muscle group is responsible for “pushing” movements and interacts synergistically with the deltoideus anterior of the shoulder and triceps brachii of the arm to accomplish this.

CHEST ANATOMY

CHEST



ANNOTATION KEY
* indicates deep muscles



ANNOTATION KEY
* indicates deep muscles

CHEST

PUSH-UPS

Starting Position: Lie flat on the floor, face down. Place your hands slightly outside of your shoulders and your fingertips parallel to your collarbone. Make sure your elbows are at 45-degree angles to your torso. Place both feet on your tiptoes.

1

Action: Raise your legs and hips off of the ground. Your lower back should arch slightly. Extend your arms, pushing into the ground. To return, lower your body (in a single plane) by bending your arms.

Movement Path: The plane of your body rotates upward in an arc. Use your feet as a lever.

LOOK FOR

- A single plane of movement, i.e., a straight line from head to ankle.

AVOID

- Segmental elevation, i.e., your shoulders rising before your hips or vice versa.
- Elevating your shoulders towards your ears.
- Moving your head forward.

STABILIZE BY

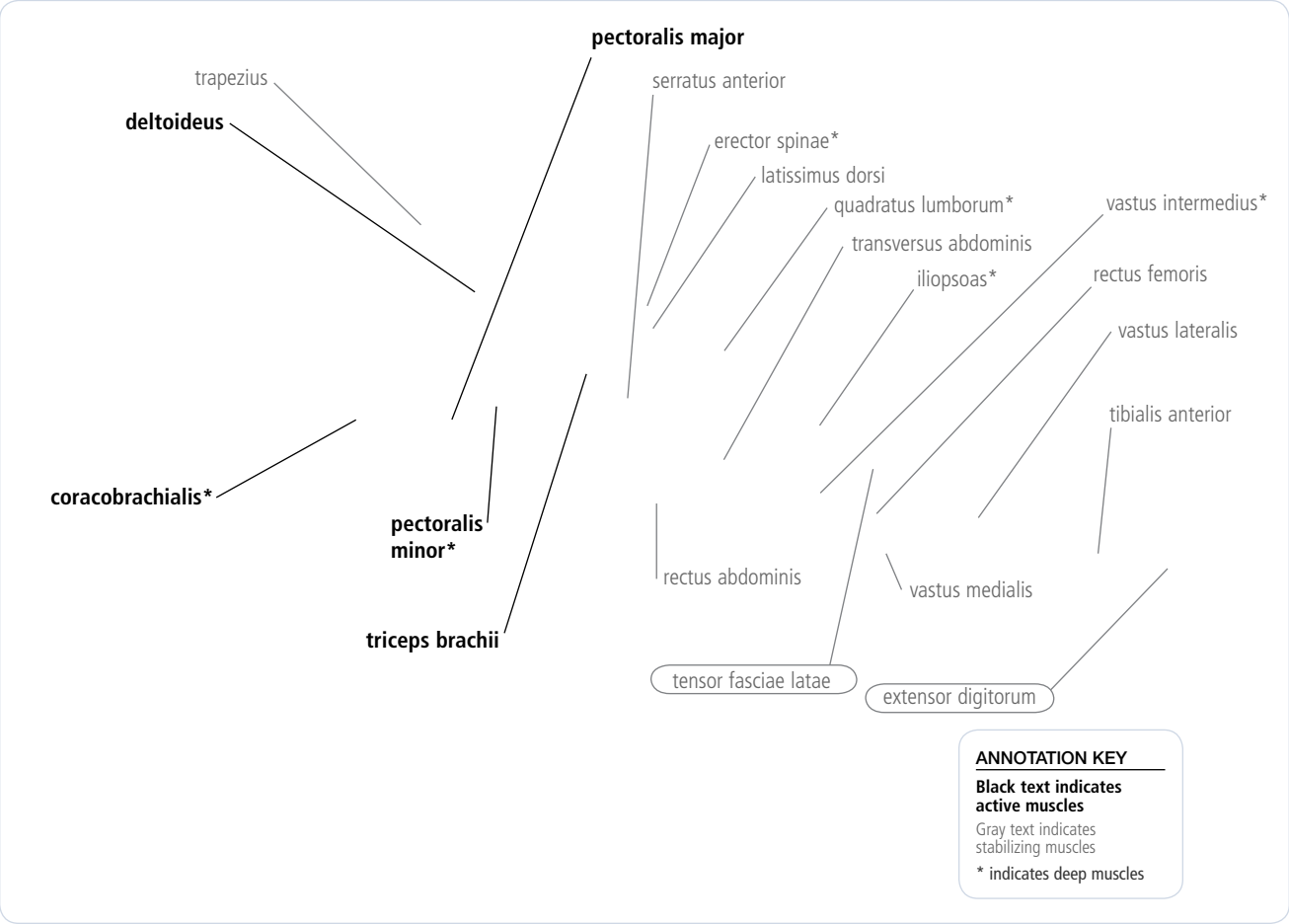
- Keeping your knees locked.
- Fixing your ankles in a stable position.
- Keeping your hips, abdominal muscles, and lower back rigid.

2

PUSH-UPS • CHEST

BEST FOR

- deltoideus
- coracobrachialis
- pectoralis major
- pectoralis minor
- triceps brachii



MODIFICATIONS

Easier: Shorten the lever by bending your knees to the floor. Maintain the same activation pattern and movement sequence.

More difficult: Raise the angle of elevation to 45 degrees, by placing your hands on a Swiss ball.

More difficult: Place your feet on a Swiss ball.

More difficult: Raise one leg and maintain the same activation pattern and movement sequence.

TOWEL FLY

Starting Position: Start from the top (i.e., with fully extended arms) of the push-up position, with your hands wider than shoulder width and placed on a towel so that the towel is taut between your hands and directly under your chest.

Action: Move your hands together while keeping your torso rigid and your arms extended. Return by spreading your hands to the starting position.

LOOK FOR

- Your arms to remain directly below your chest and perpendicular to your torso.

AVOID

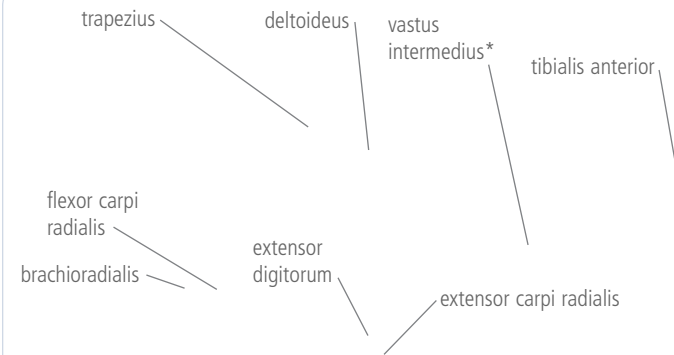
- Dropping your head forward or bending or extending your elbows.
- Any change in your spinal position.
- Elevating or widening your shoulder blades.

STABILIZE BY

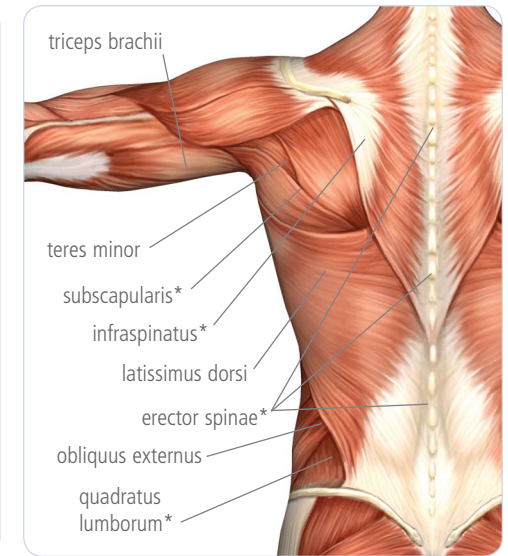
- Keeping your hips up and your knees and ankles locked.
- Keeping your shoulders retracted and depressed throughout the movement.

Movement Path: As your hands slide together, your torso (spine), hips, and legs elevate, using your toes as a lever. Your hand movement should be smooth and simultaneous.

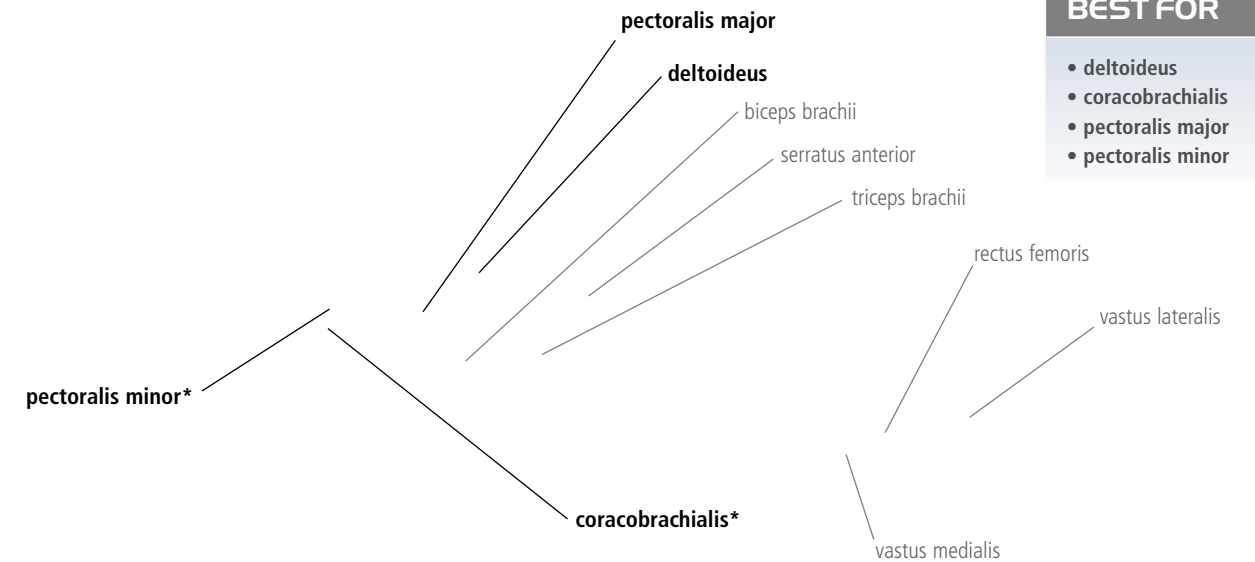
1



1



2



BEST FOR

- deltoideus
- coracobrachialis
- pectoralis major
- pectoralis minor

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

INCLINE DUMBBELL FLY

1

Starting Position: Sit on an incline bench with your shoulders higher than your hips at no greater than a 60-degree angle. Hold the dumbbells above your chest so that your palms face each other. Your elbows should be very slightly bent. Your shoulder blades should contact the bench. Keep your spine in a neutral position and your feet flat on the floor.

Action: Keeping your elbows bent, push your hands apart, and inhale until your hands drop to just below the height of your chest. Return to the starting position by squeezing your chest and bringing the weights back to the starting position along the same path as the descent, exhaling as you do so.

LOOK FOR

- Your chest and rib cage to rise as the weight descends.
- Your spine and shoulders to remain in the same position as you return to the starting position.

AVOID

- Moving your head or chin forward off the bench.
- Elevating your shoulders toward your ears.
- Bending your elbows excessively as the weight descends or flattening them as the weight ascends.

STABILIZE BY

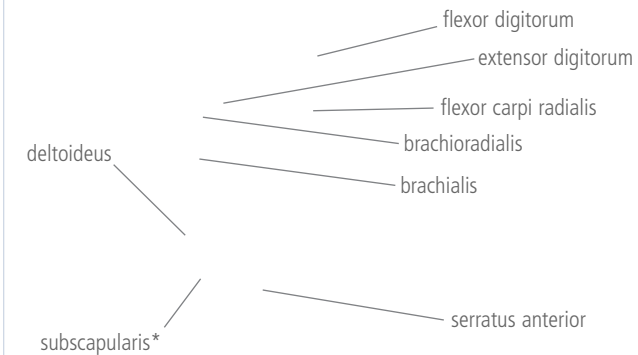
- Keeping your grip strong and your upper arms (both biceps and triceps) contracted.
- Ensuring that your shoulder blades remain in contact with the bench throughout the movement.
- Keeping your feet flat and your neck long.

2

Movement Path: As the dumbbells leave the starting position they come down and apart in an arc that begins vertically and drops to horizontal relative to the ground, not the bench.

INCLINE DUMBBELL FLY • CHEST

1

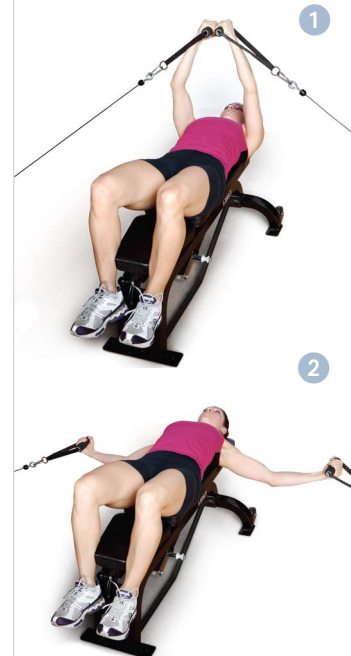


BEST FOR

- deltoideus
- coracobrachialis
- pectoralis major
- pectoralis minor

MODIFICATION

Similar difficulty: Place the bench equidistant between two low pulleys. Grasp the handles such that they are over your chest. Maintain the same activation pattern and movement sequence.



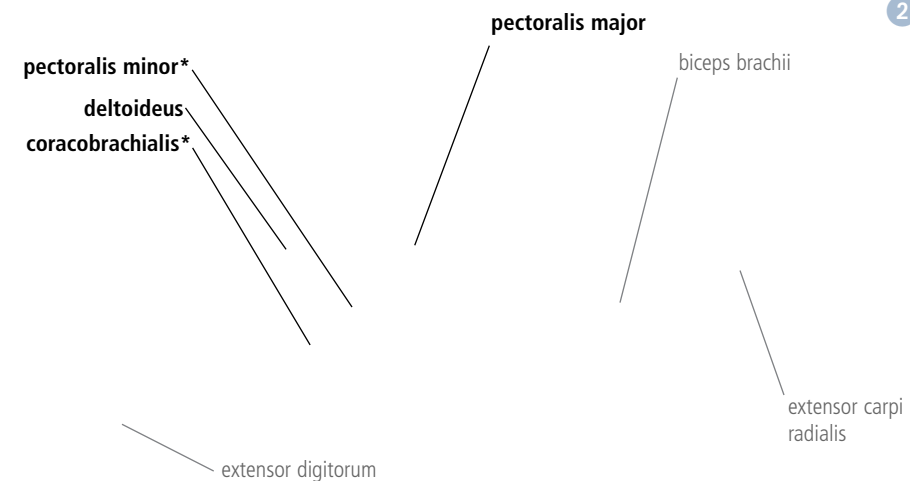
ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

2



BENCH PRESS

Starting Position: Lie on a bench, holding a barbell above your chest so that your arms are directly above your collarbone. Your arms should be extended, your shoulder blades and feet flat, and your spine in a neutral position. Grip the bar so that your hands are wider than your shoulder width in an overhand grip, with your thumb grasping the bar underneath.

Action: Inhale, and lower the barbell to your chest, finishing on or slightly above the nipple line. Exhale, extending your arms to push up to the ceiling until returned to the starting position.

1

- LOOK FOR**
- Your rib cage to remain open and rise during the descent phase.
 - Your shoulders to remain retracted and away from your ears during the ascent.

- AVOID**
- Dropping the weight quickly.
 - Bouncing the bar off your chest.
 - Changing your spinal position (either arching or flattening your back) during the movement.
 - Picking your feet up the floor.

- STABILIZE BY**
- Keeping your spine neutral and your forearms perpendicular to the bar (directly below it) throughout the movement.
 - Ensuring that your shoulders, head, and hips remain in contact with the bench at all times.

Movement Path: As the bar descends, it remains horizontal. There is a slight movement downward on the torso, from the clavicle toward the mid-lower chest, which returns on the upward movement.

2

BENCH PRESS • CHEST

1

BEST FOR

- deltoideus
- coracobrachialis
- pectoralis major
- pectoralis minor
- triceps brachii

2

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

MODIFICATION

More difficult: Grip the bar with your hands closer together. Maintain the same activation pattern and movement sequence.

1

2

INCLINE BARBELL PRESS

LOOK FOR

- Your rib cage to remain open and rise during the descent phase.
- Your shoulders to remain retracted and away from your ears during the ascent.

AVOID

- Dropping the weight quickly.
- Bouncing the bar off your chest.
- Changing your spinal position (either arching or flattening your back) during the movement.
- Picking your feet off the floor.

1

Starting Position: Lie on a bench at a 45-degree angle, holding a barbell above your chest so that in profile, your hands, arms, and shoulders are vertical. Your arms should be extended, your shoulder blades and feet flat, and your spine in a neutral position. Grip the bar so that your hands are wider than shoulder width in an overhand grip, with your thumb grasping the bar underneath.

Action: Inhale, and lower the barbell until it touches your upper chest, just below your collarbone. Exhale, extending your arms to push up to the ceiling until returned to the starting position.

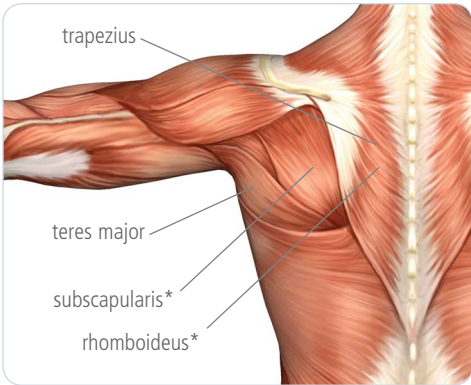
2

Movement Path: As the bar descends, it remains horizontal. Your elbows and forearms move directly downward in a vertical plane.

STABILIZE BY

- Keeping your spine neutral and your forearms perpendicular to the bar (directly below it) throughout the movement.
- Ensuring that your shoulders, head, and hips remain in contact with the bench.
- Keeping your feet in contact with the floor.

INCLINE BARBELL PRESS • CHEST



1

triceps brachii

pectoralis minor*

brachioradialis

deltoideus

pectoralis major

2

latissimus dorsi

serratus anterior

flexor digitorum

extensor carpi radialis

flexor carpi radialis

extensor carpi radialis

coracobrachialis*

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

BEST FOR

- deltoideus
- coracobrachialis
- pectoralis major
- pectoralis minor
- triceps brachii

DIP

Starting Position: With your feet on the assist bar, grasp the handles and extend your arms completely, with your hands adjacent to your hips and your torso leaning slightly forward. Make sure that your shoulders are down and away from your ears and that spine is in a neutral position.



Action: Lower your body until your upper arms are parallel to the floor. Your knees should be slightly behind your hips and your chest slightly in front of them. Return to the starting position by extending your elbows.

Movement Path: As your torso drops, there is a slight slide backward from your knees; your center of mass drops directly downward.

STABILIZE BY

- Arching your lower back slightly and keeping it solid throughout the movement.
- Placing your elbows directly over your hands.
- Keeping your chin up and your eyes looking directly forward.

LOOK FOR

- Your torso to lean forward as you descend.

AVOID

- Looking down.
- Shifting your elbows to a position that is not directly over your hands.

2

DIP • CHEST

BEST FOR

- deltoideus
- pectoralis major
- serratus anterior
- triceps brachii

MODIFICATION

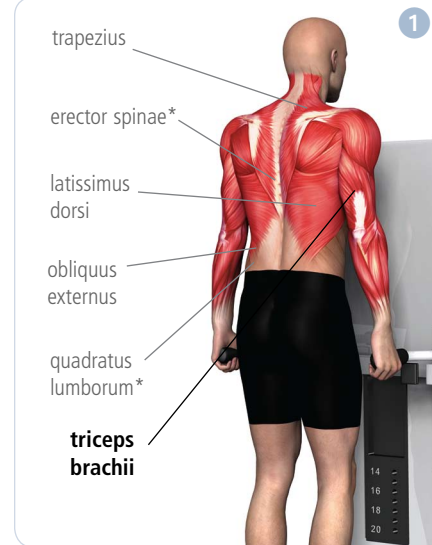
More difficult: Bend your knees at 90-degree angles, and cross one ankle over the other. Maintain the same activation pattern and movement sequence.

**ANNOTATION KEY**

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



deltoideus posterior

brachialis

levator scapulae*

trapezius

anconeus

extensor
carpi radialisflexor
digitorumobliquus
internus*deltoideus
anteriorpectoralis
majorserratus
anteriortransversus
abdominis*

iliopsoas*

MACHINE CHEST PRESS

Starting Position: Sit vertically with your feet flat, either on a footstool or the floor. Keeping your spine in a neutral position, grasp the handles so that your forearms are parallel to the floor and your elbows are directly behind the handles in the same plane. Keep your shoulders down, your chin back, and your head up.

Action: Inhale, and slowly allow the handles to come toward you by bending your elbows. Once your hands are parallel to your chest, pause, exhale, push the handles, and return to the starting position by extending your elbows.

Movement Path: The handles of the machine draw back in a horizontal plane toward your torso until your hands are parallel to your chest, and return to the starting position via the same path.

LOOK FOR

- Your chest to remain high, your forearms to remain parallel to the ground, and your elbows to remain directly behind your hands.

AVOID

- Flattening your lower back, overextending your shoulders forward, or allowing your elbows to either elevate or drop during any portion of the exercise.
- Picking your feet up.

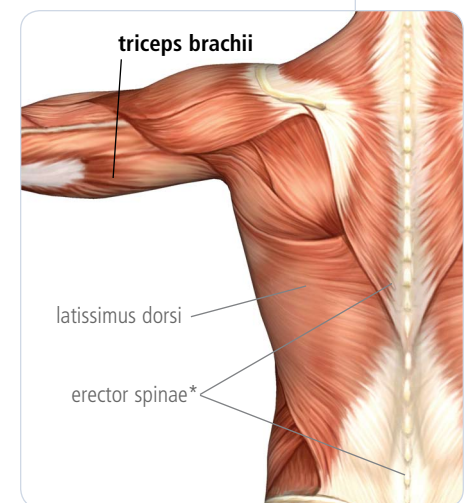
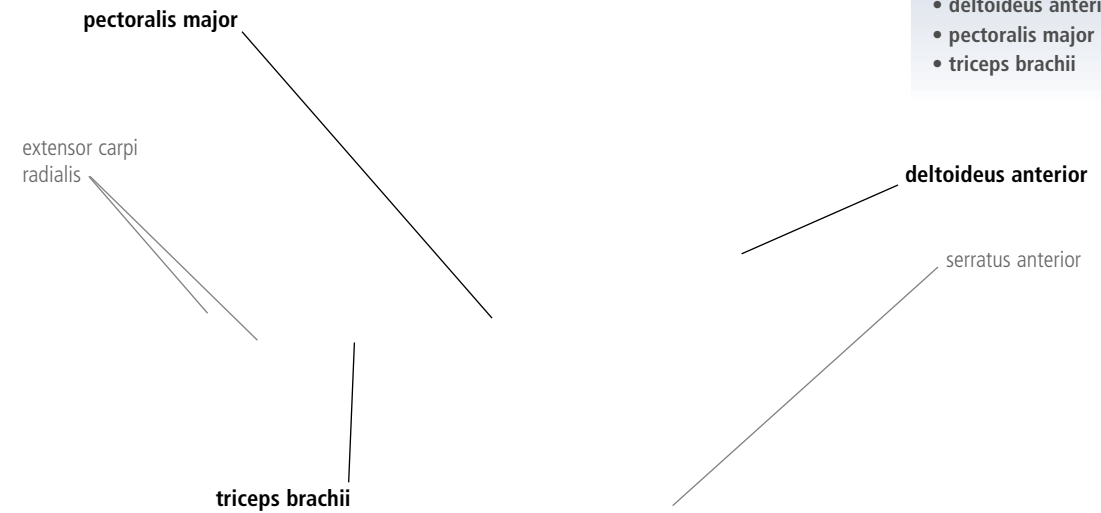
STABILIZE BY

- Keeping your tail bone, upper back, and head in contact with the bench (your lower back should be slightly off it) throughout the movement, with your shoulders down and away from your ears

MACHINE CHEST PRESS • CHEST

BEST FOR

- deltoideus anterior
- pectoralis major
- triceps brachii



ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

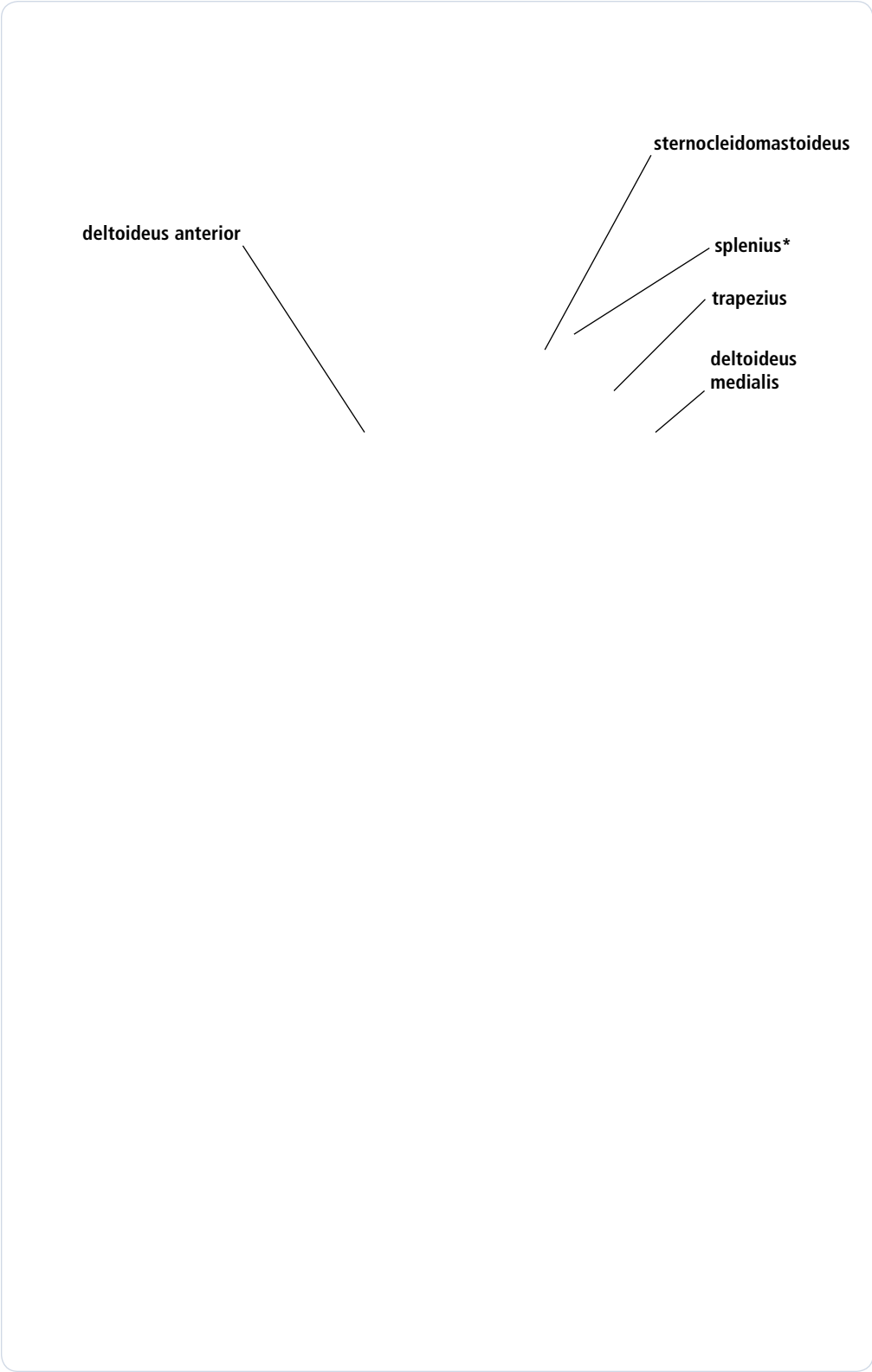
SHOULDERS

Because the structure of the shoulder joint is inherently mobile and less stable than the hip or knee joints, there are many muscles that traverse it from multiple directions in order to maintain stability. They are responsible for the depression and elevation of the scapula, and take part in moving the upper arm in every direction and stabilizing it.

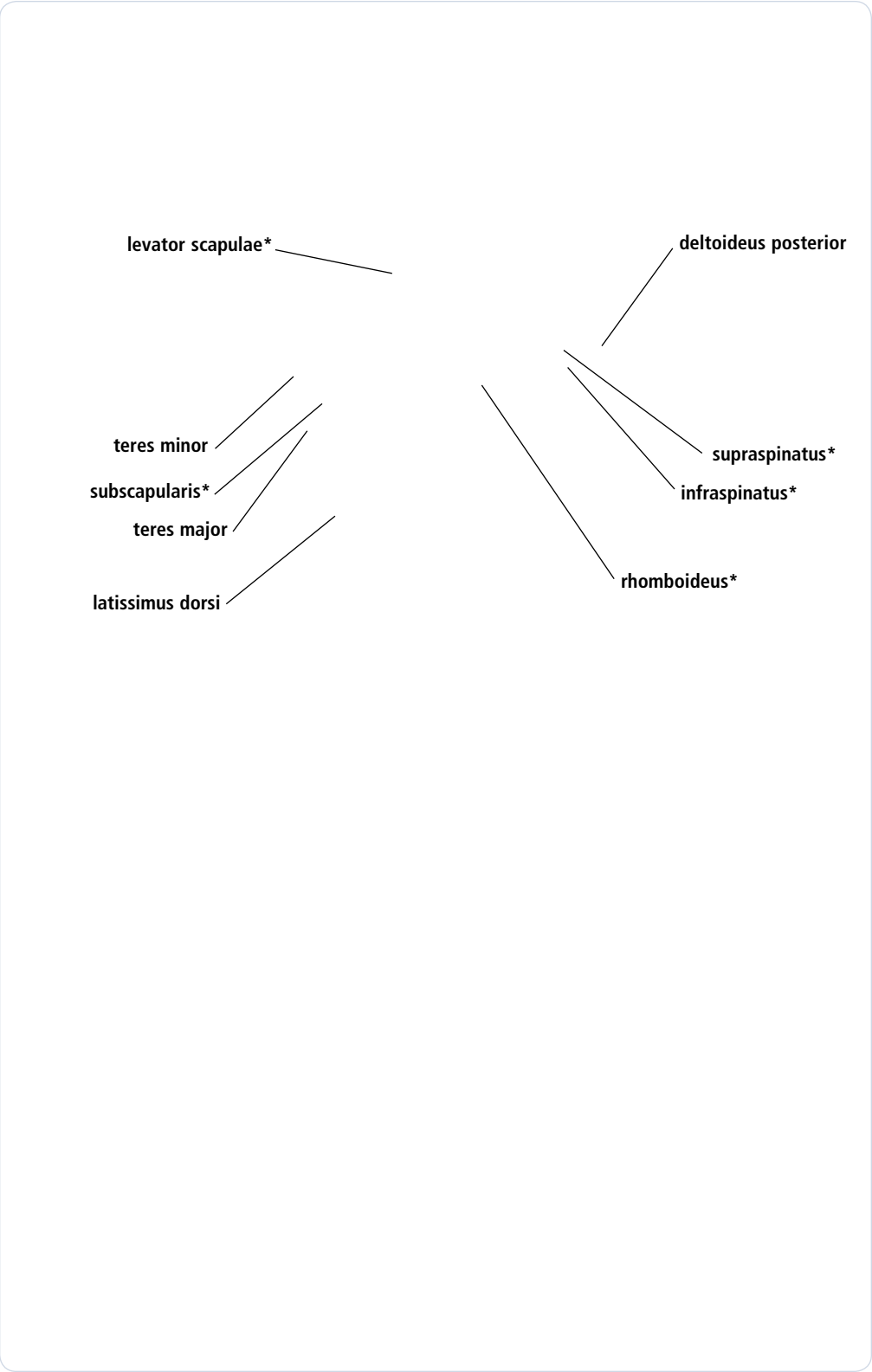
In any exercise or activity that involves movement of the arms, the muscles of the shoulder play a part. Because these muscles interact directly with the torso and arm, the muscles of the chest, back, core, and arms are integral to their development and function.

SHOULDER ANATOMY

SHOULDERS



ANNOTATION KEY
* indicates deep muscles



ANNOTATION KEY
* indicates deep muscles

SHOULDERS

SHOULDER EXTERNAL ROTATION

Starting Position: Lie on your side, supporting your head with your hand and keeping your bottom arm bent. Bend your knees slightly, with your spine in a neutral position and your upper arm in an exact line with your spine. Bend your elbow at 90 degrees, with the palm of the hand holding the dumbbell facing your bellybutton.

1

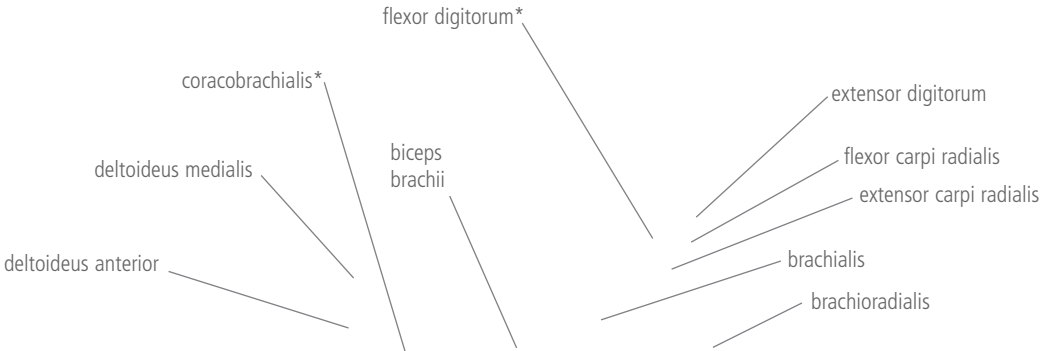
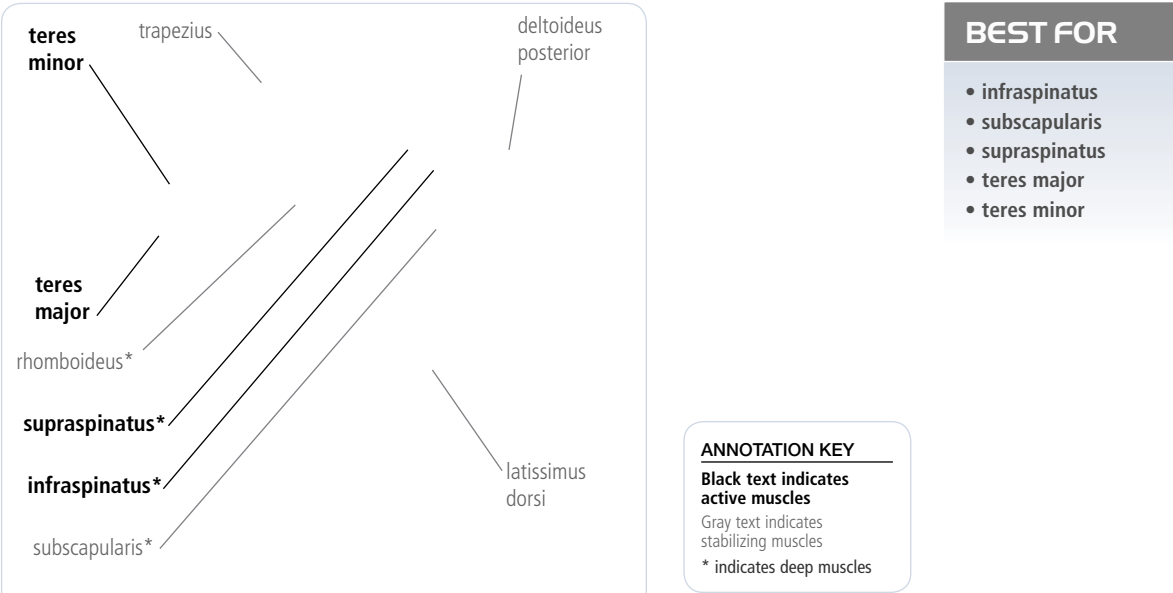
Action: Holding your elbow by your side, rotate the back of your hand and forearm toward the ceiling, until just below perpendicular or as high as possible without changing your shoulder or upper-arm position.

Movement Path: Your forearm should remain at a 90-degree angle to your elbow and upper arm. Your hand moves directly upward.

STABILIZE BY

- Keeping your shoulder in one position and your upper arm held tightly to your torso.

SHOULDER EXTERNAL ROTATION • SHOULDERS



LOOK FOR

- A rotational movement around your torso, in a line parallel to your waistline.

AVOID

- Raising your upper arm or elbow away from your torso.
- Letting your shoulder joint slide either forward toward your chest or backward by pinching your shoulder blades.

2

REAR RAISE

Starting Position: Bend your knees slightly, and drop your torso forward, so that your spine is just above 90 degrees from your hips, almost parallel to the floor. Keep your spine in a neutral position and your chin slightly elevated. Your shoulders should be down. Grasp the dumbbells with your hands directly under your chest toward the floor, bending your elbows slightly.

1

STABILIZE BY

- Keeping your torso stationary.
- Keeping your shoulders down and your neck long, with your head up.
- Pulling your abdomen up and in.
- Contracting your gluteals and hamstrings, with your spinal muscles active.

2

Action: Your arms should remain at a 90-degree angle to your torso, your knees should remain bent, and your elbows should remain in their extended position. Bring your hands to just above a horizontal line, contracting and depressing your shoulder blades.

Movement Path: The dumbbells should trace a semicircle in a vertical plane.

LOOK FOR

- Your spine to remain in one position.
- Your hands to move not only back, but also away (or out) from your torso in a slow and controlled fashion.
- A small pause at the top of the movement.

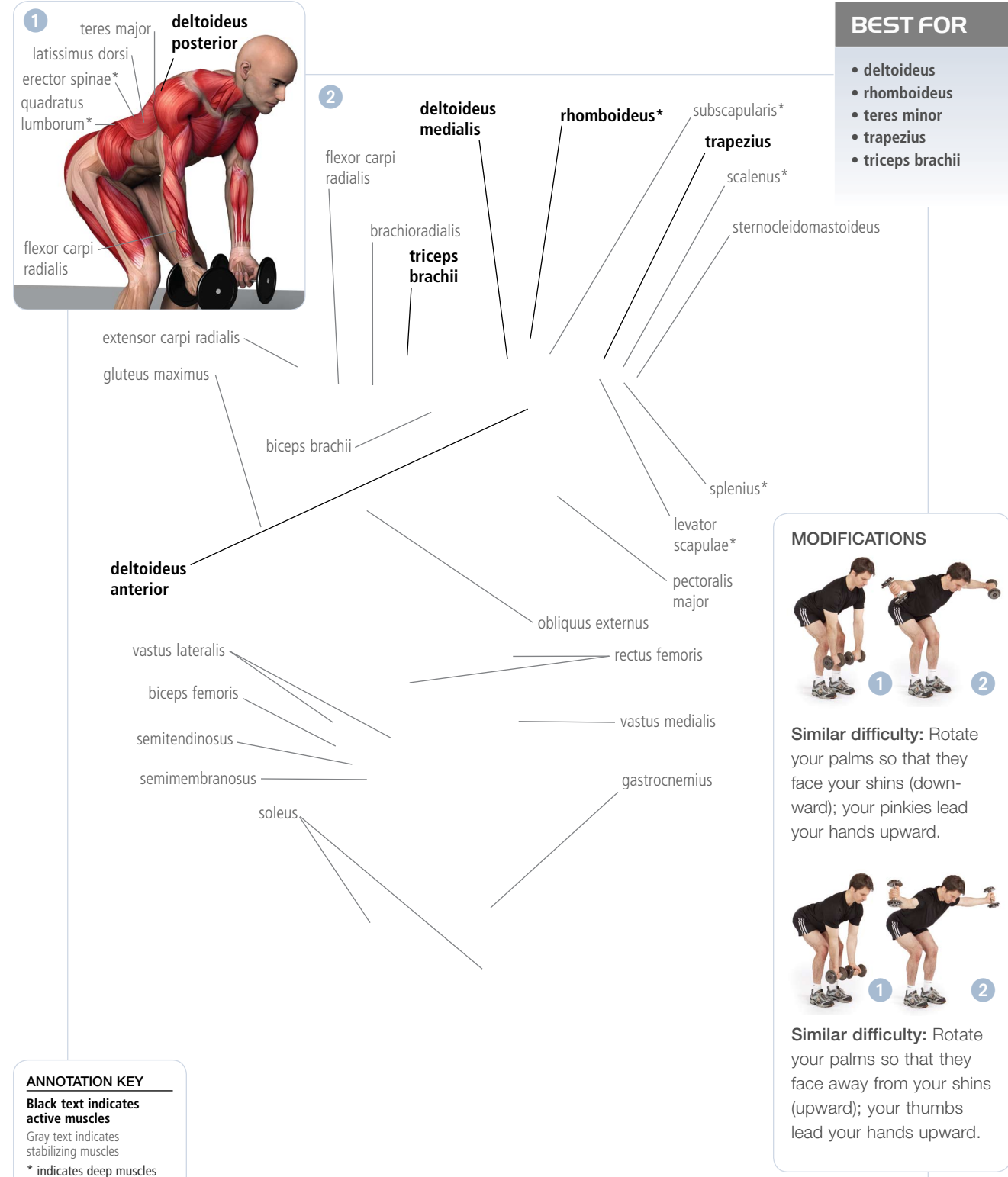
AVOID

- Any spinal, hip, knee, or head movement.
- Elevating your shoulders toward your ears or extending your arms forward, toward the ground, in the bottom position.
- Moving your shoulder blades from their down and flat position as your arms descend toward the floor.

REAR RAISE • SHOULDERS

BEST FOR

- **deltoideus**
- **rhomboideus**
- **teres minor**
- **trapezius**
- **triceps brachii**



MODIFICATIONS



Similar difficulty: Rotate your palms so that they face your shins (downward); your pinkies lead your hands upward.



Similar difficulty: Rotate your palms so that they face away from your shins (upward); your thumbs lead your hands upward.

UPRIGHT ROW

Starting Position: Stand with your feet shoulder-width apart, grasping the dumbbells so that your palms face your body and your knuckles point away. Keep your knees slightly bent and your chest up.



LOOK FOR

- Your arms to rise simultaneously; your hands should never be higher than your elbows.
- An upright posture to be maintained throughout the movement.

AVOID

- An excessive shrug or rounding your shoulders.
- Keeping your elbows close to your body or forward of the lateral plane of your body (from the shoulder joint).

STABILIZE BY

- Keeping your chest and rib cage high.
- Keeping your spine solid and in a neutral position.
- Bending your hips and knees slightly.
- Keeping your eyes and head forward.

Action: Pull your hands up until your elbows and forearms are virtually parallel to the ground. The dumbbells should end in front of your collarbones.

Movement Path: Pull your arms directly up, parallel to the line of your torso, with a slight retraction of your elbows and shoulders at the top of the movement.



UPRIGHT ROW • SHOULDERS

BEST FOR

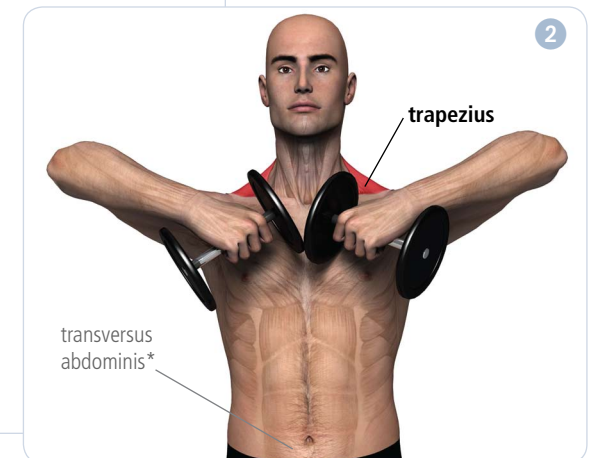
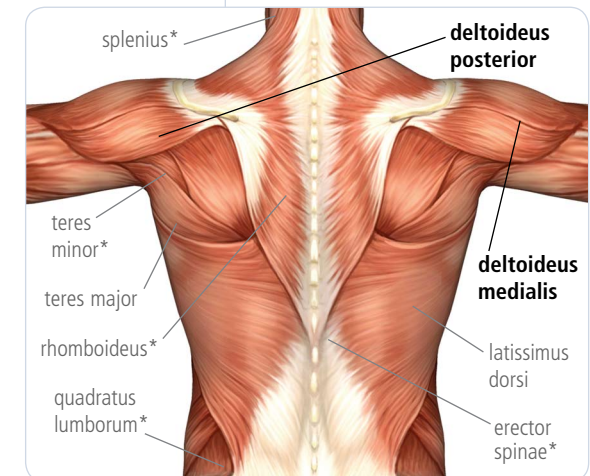
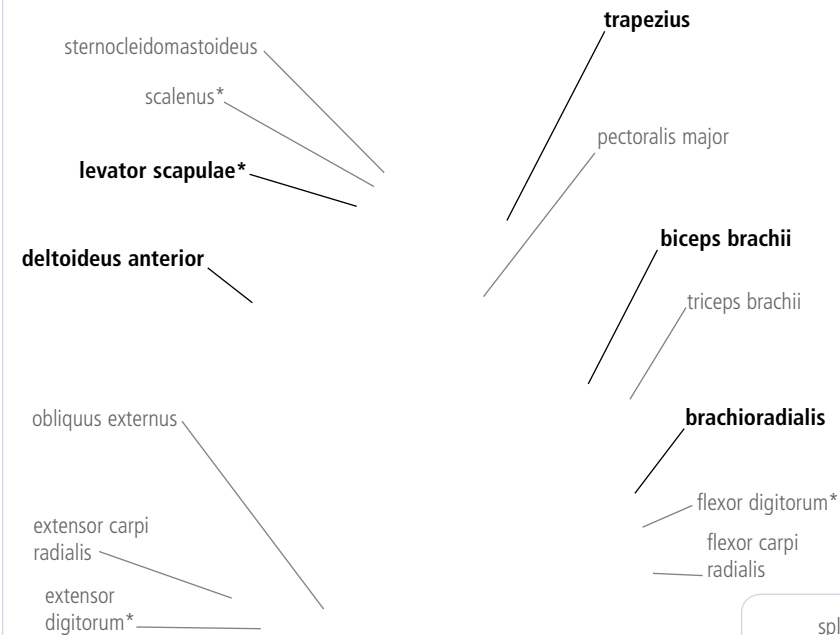
- deltoideus
- trapezius
- biceps brachii
- brachioradialis

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



DUMBBELL SHRUG

Starting Position: Stand straight up, gripping dumbbells at your side.

Action: Elevate your shoulders toward your ears, keeping your head in a neutral position and exhaling as you contract and elevate.



Movement Path: Your shoulders move up and in toward your ears. Your elbows remain straight and your head remains in one position.

STABILIZE BY

- Keeping your head up and your chest high.
- Keeping your spine in a neutral position and your hips and knees slightly bent.

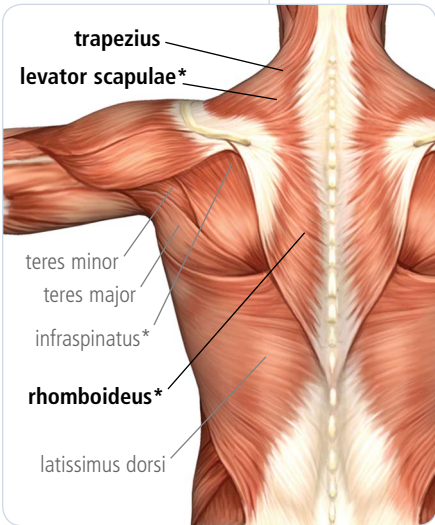
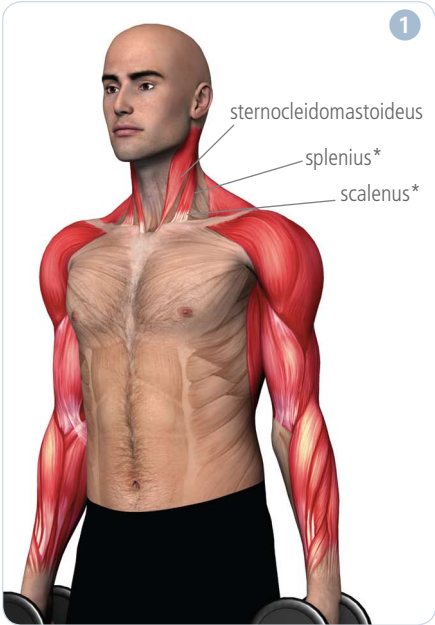
- LOOK FOR**
- Movement up and in from your shoulder joints.

- AVOID**
- Tilting your head in any direction, bending your elbows, or jutting your chin forward.

DUMBBELL SHRUG • SHOULDERS

BEST FOR

- levator scapulae
- rhomboideus
- trapezius



flexor carpi radialis

flexor digitorum*

- pectoralis major
- deltoideus anterior
- deltoideus medialis
- biceps brachii
- brachialis
- triceps brachii
- brachioradialis
- extensor carpi radialis
- extensor digitorum

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

OVERHEAD PRESS

1



Starting Position: Seated on a Swiss ball, begin with your hands parallel with your collarbone and slightly wider than shoulder-width apart, your elbows pointing directly down at the floor, and your palms away from you. Keep your spine in a neutral position and look directly forward. Draw your abdomen in and up, and keep your chest and ribs high.

Action: Exhale, pushing your palms directly up toward the ceiling. Keep your shoulder blades down.

Movement Path: Your hands should remain in the same plane as your torso.

Your arms should finish with your hands directly above your shoulder joints and your shoulders down and away from your ears.

Arc slightly through the push, making sure your bone structure is supporting the weight.

STABILIZE BY

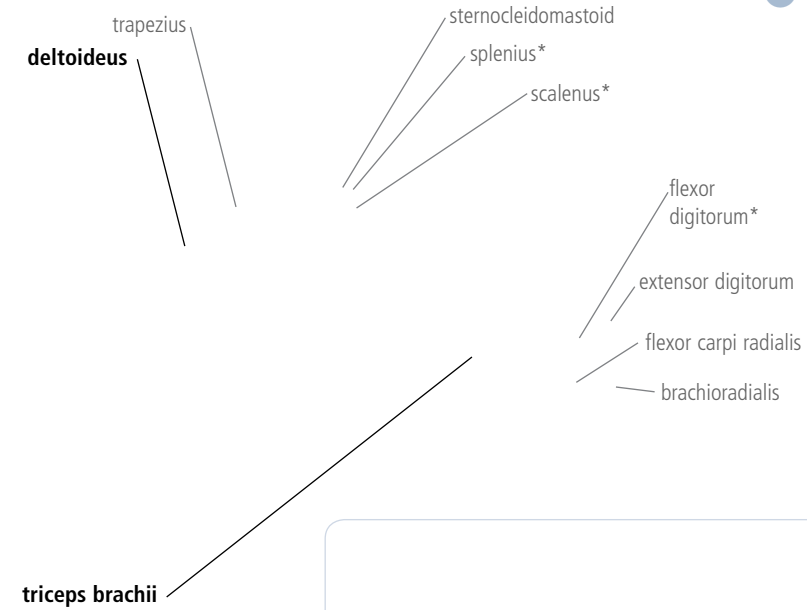
- Pulling your abdominal muscles up and in.
- Maintaining a neutral spinal position
- Keeping your knees directly over your feet and your feet pressed firmly into the floor.
- Keeping your shoulder blades down and your rib cage high.



2

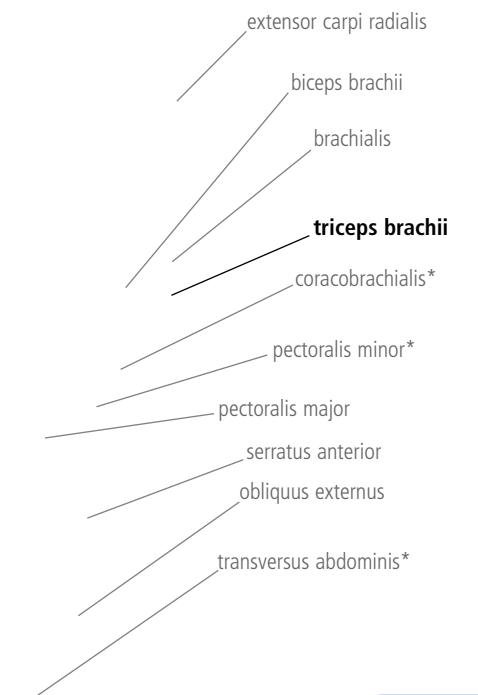
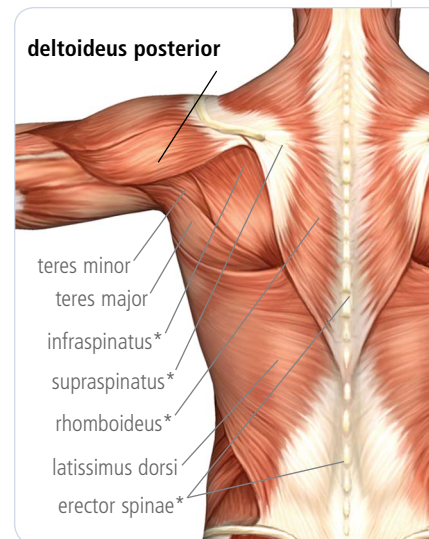
OVERHEAD PRESS • SHOULDERS

1

**BEST FOR**

- deltoideus
- triceps brachii

2

**ANNOTATION KEY**

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

LOOK FOR

- A smooth transition from the bottom to the top of the movement.
- The identical speed and movement path with each hand.

AVOID

- Extending your chin forward.
- Elevating your shoulders.
- Rounding your back.
- Accelerating the weight by altering the speed with which you move toward the ceiling.
- Any end position other than holding the weight directly over your shoulders.

FORWARD RAISE

Starting Position:

Begin with your hands by your sides, facing your thighs, with your elbows straight.

Action:

Raise your hands forward to a horizontal position with your thumbs up.

**LOOK FOR**

- A sharp 90-degree angle between your arms and your torso in the finish position.

AVOID

- Extending your shoulders forward from your shoulder blade; keep them back and down.
- Shrugging or allowing your back to round.

Movement Path: Your arms should be parallel; raise them in a frontal plane so that your hands end up directly in front of your shoulder joints.

Keep your elbows locked, your shoulders down, and your chest high.

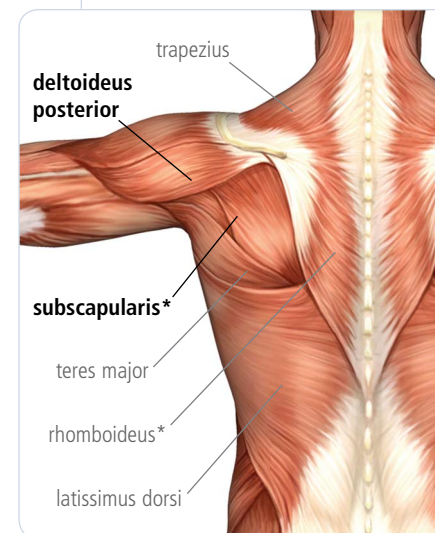
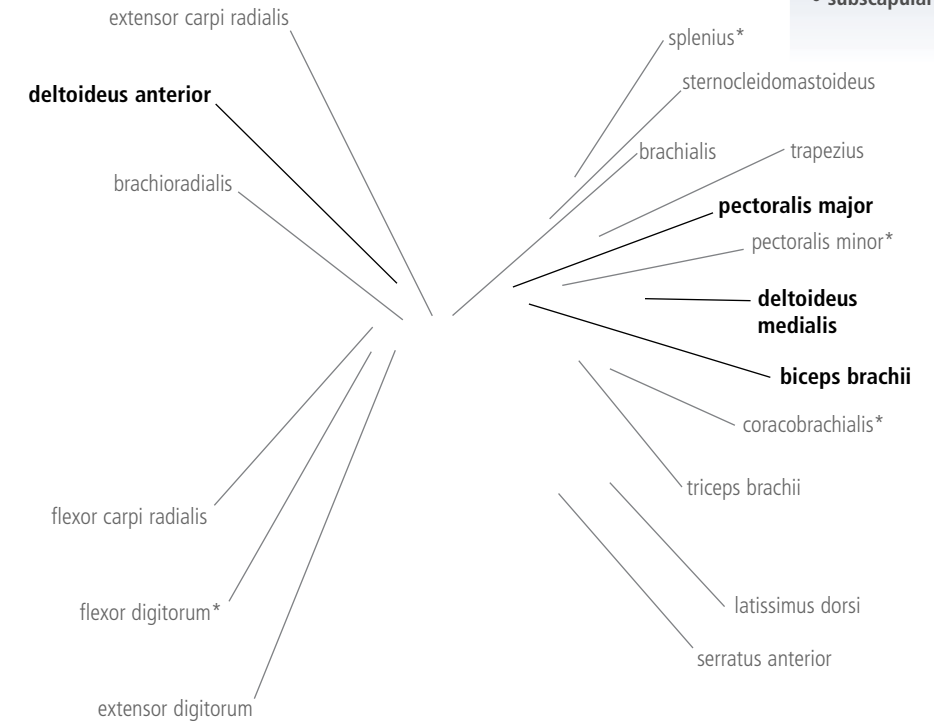
STABILIZE BY

- Keeping your chest up and your shoulders retracted and depressed throughout the movement.
- Pulling your abdominal muscles up and in.
- Keeping your legs and gluteals firm and motionless.

FORWARD RAISE • SHOULDERS

BEST FOR

- deltoideus
- pectoralis major
- subscapularis

**ANNOTATION KEY**

Black text indicates active muscles

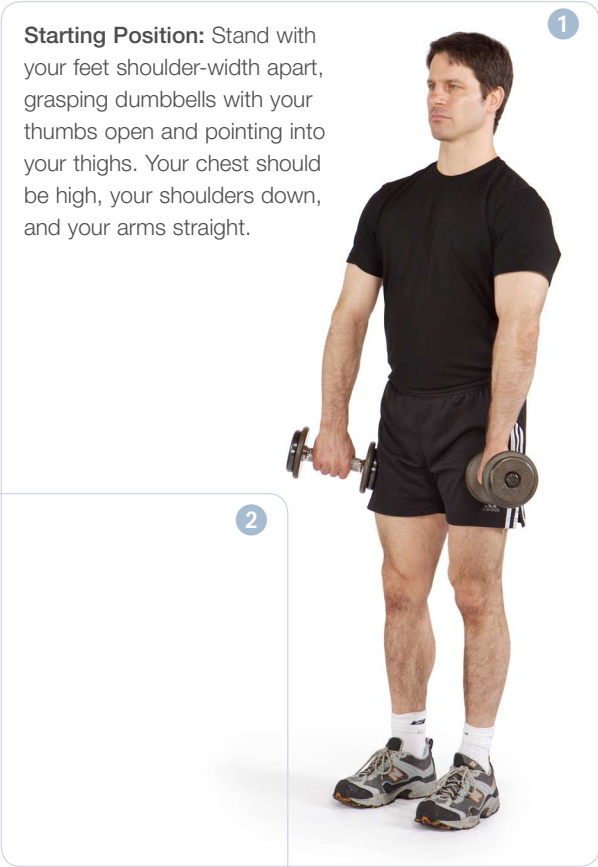
Gray text indicates stabilizing muscles

* indicates deep muscles

EMPTY CANS

SHOULDERS

- LOOK FOR**
- Your thumbs to remain pointed down toward the floor and your pinkies up toward the ceiling.
 - Your shoulders to remain down and back and your chest to remain high.
- AVOID**
- Reaching forward with your hands,
 - Rounding your upper back.
 - Bending your elbows.
 - Allowing your hands or thumbs to turn up away from the floor.

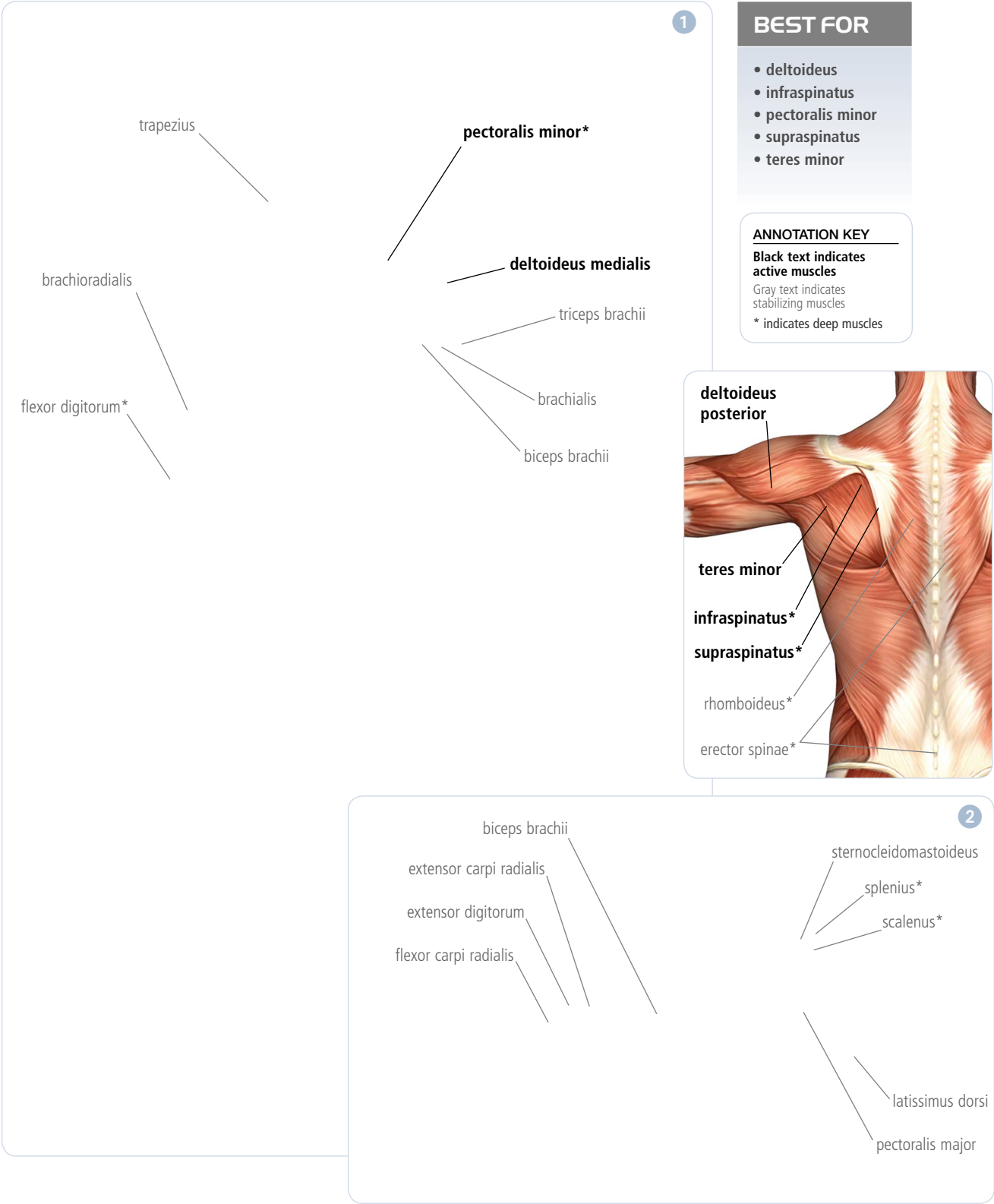


Action: Lift your hands up and away from your body, pointing your thumbs down to the ground and keeping your arms at 45-degree angles to your midline.

Movement Path: The movement is a 45-degree angle raise. Lift your hands until your arms are parallel to the ground.

- STABILIZE BY**
- Keeping your chest up and your shoulders retracted and depressed throughout the movement.
 - Pulling your abdominal muscles up and in.
 - Keeping your legs and gluteals firm and motionless.

EMPTY CANS • SHOULDERS



LATERAL RAISE

Starting Position:

Begin with your hands at your sides and your elbows and knees slightly bent. Grasp the dumbbells with your palms facing your sides.



Action: Elevate your hands to your sides until in a horizontal position. Your palms should face the ground, and your arms should be slightly bent.

Movement Path: Raise your arms laterally from the sides of your body, keeping your elbows in a fixed position.

STABILIZE BY

- Keeping your spine in a neutral position and your hips and knees slightly bent.

LOOK FOR

- Your wrists to remain in the same position as your arms are elevated.
- The action to come from your shoulder joints, not your neck.

AVOID

- Shrugging.
- Flexing or extending your elbow as your arm is raised.
- Rotating your hands in either direction.
- Extending your head or chin forward.
- Leaning backward with your upper body.

LATERAL RAISE • SHOULDERS

BEST FOR

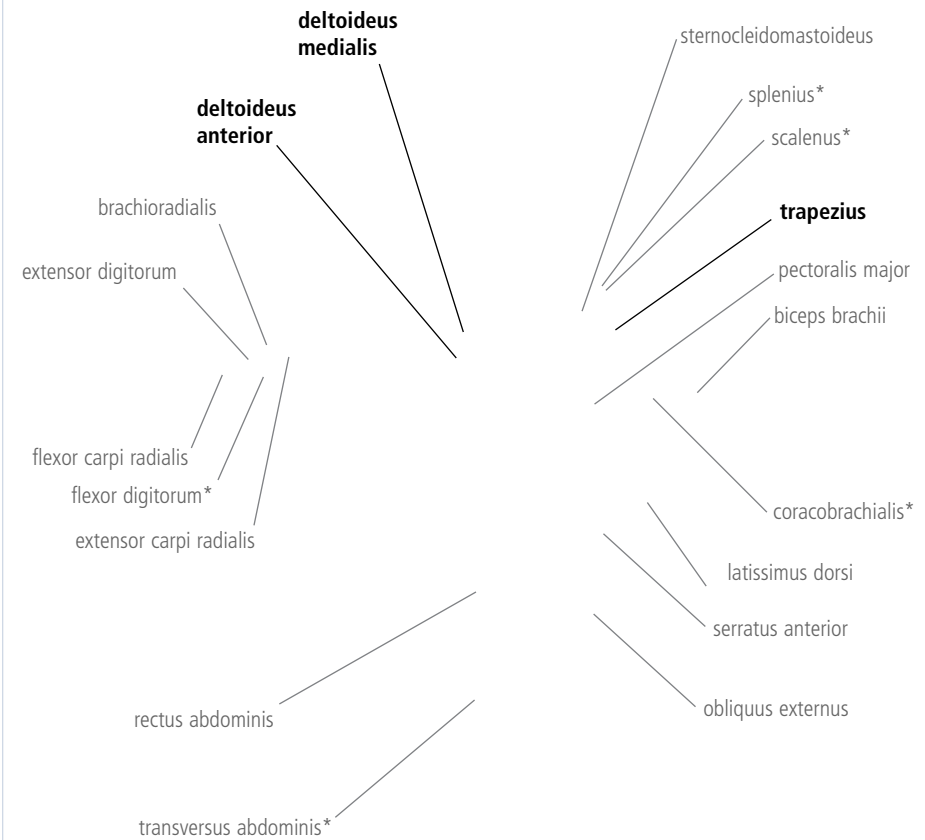
- **deltoideus**
- **trapezius**

MODIFICATIONS

Similar difficulty: Stand equidistant between two cables, grabbing the left cable with your right hand, and vice versa. Maintain the same activation pattern and movement sequence.



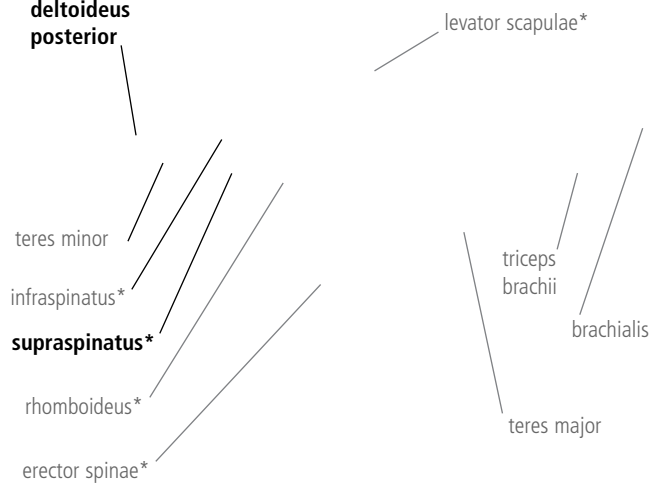
More difficult: As above, but use only a single arm, and stand only on the foot closest to the cable.

**ANNOTATION KEY**

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

deltoideus posterior

ARMS

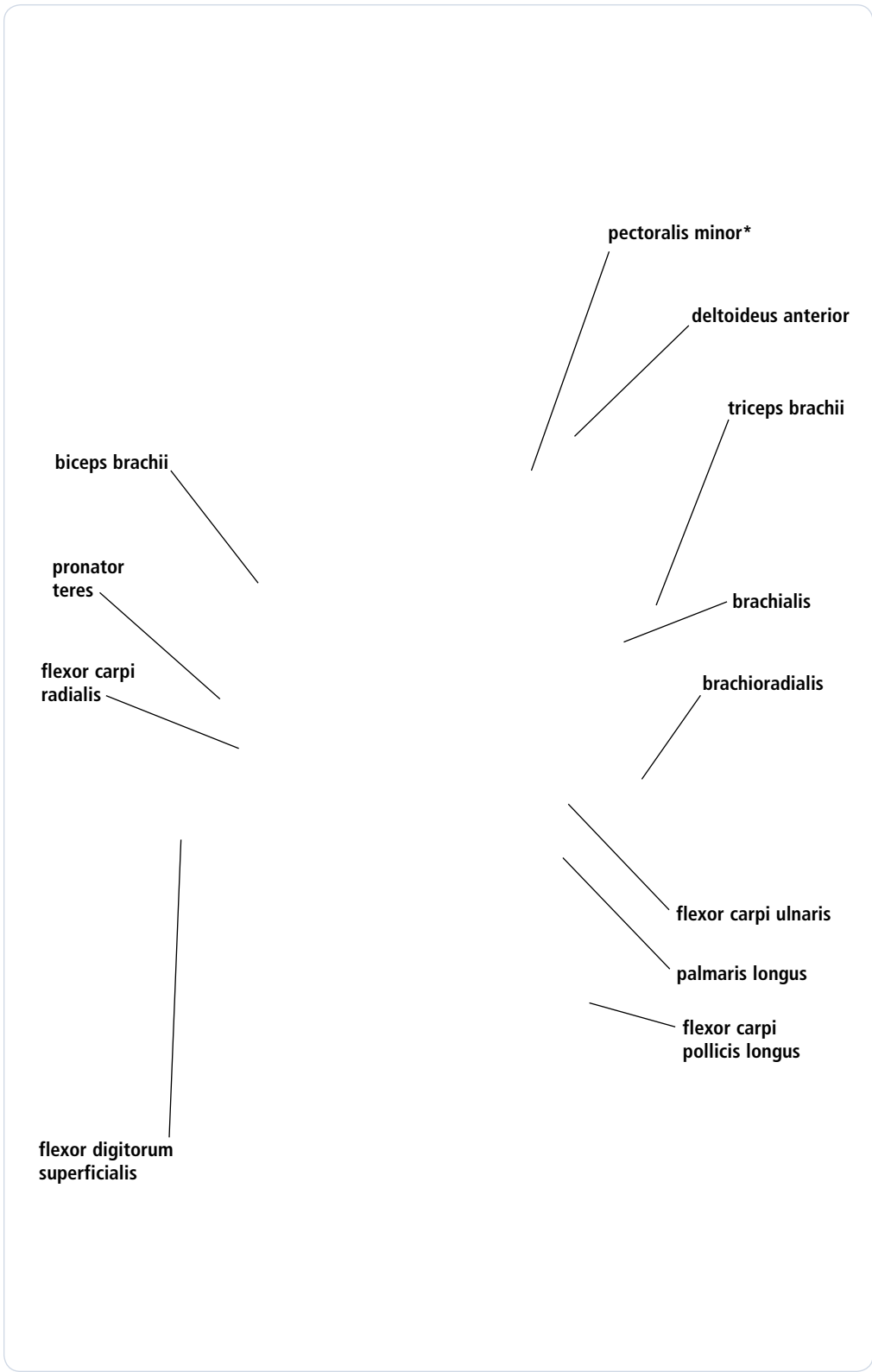
The muscles of the arm originate from the scapula and the humerus.

Their function is to extend and flex the elbow, and they are responsible for movement and stabilization of the wrist and hand in all directions.

The upper muscles of the arm are also involved in forward flexion and backward extension from the shoulder joint. These muscles act in conjunction with the muscles of the shoulders, chest, and back in performing all pushing and pulling movements.

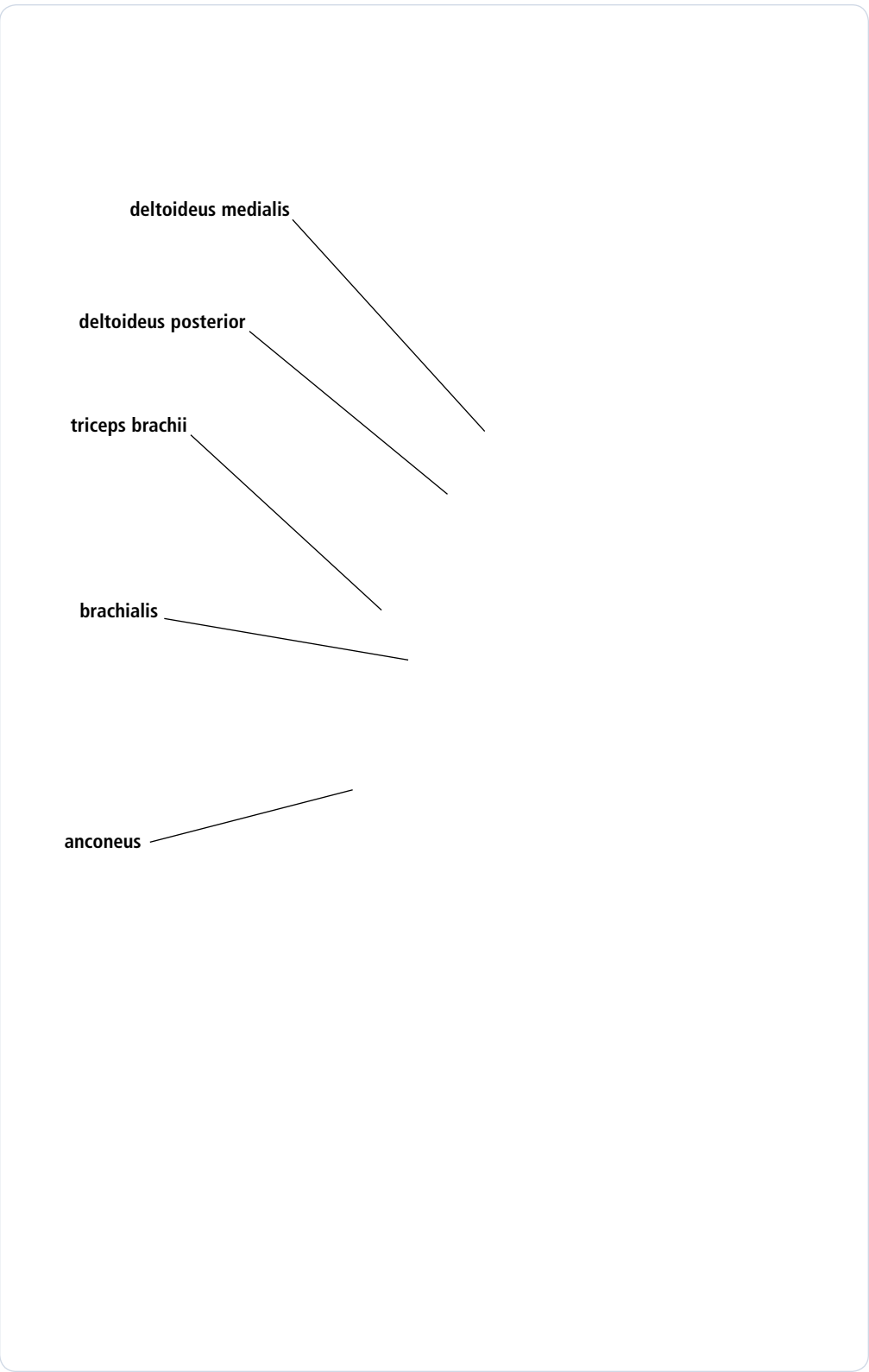
ARM ANATOMY

ARMS



ANNOTATION KEY
* indicates deep muscles

ARMS



ANNOTATION KEY
* indicates deep muscles

TRICEPS EXTENSION

Starting Position: Lie on your back on a bench, extending your arms straight above your chest. Your hands should be shoulder-width apart, with your palms facing up toward the ceiling as you grip the barbell. Keep your shoulders retracted, your knees bent, and your spine neutral. Exercise extreme caution when lowering the barbell over your head.

1

LOOK FOR

- Your upper arms to remain motionless and your wrists to remain solid throughout the movement.

AVOID

- Extending your shoulders toward the ceiling or elevating your shoulders toward your ears.
- An excessive arch in your back.

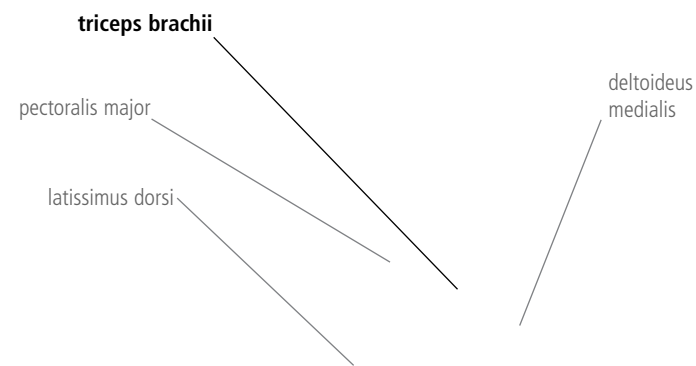
Action: Lower the barbell by bending your elbows toward your head. Stop just above your forehead, and extend your elbows to return to the starting position. Your upper arms remain perpendicular to the torso in a vertical position throughout the movement.

Movement Path: The barbell curves in a linear arc, from perpendicular to approximately 35 degrees from your elbow joints. Fix your upper arms in a perpendicular position.

2

STABILIZE BY

- Keeping your chest active and your shoulders down, in contact with the bench.
- Keeping your upper arms parallel.
- Maintaining a neutral spinal position with your feet flat.
- Keeping your neck long and your chin up.



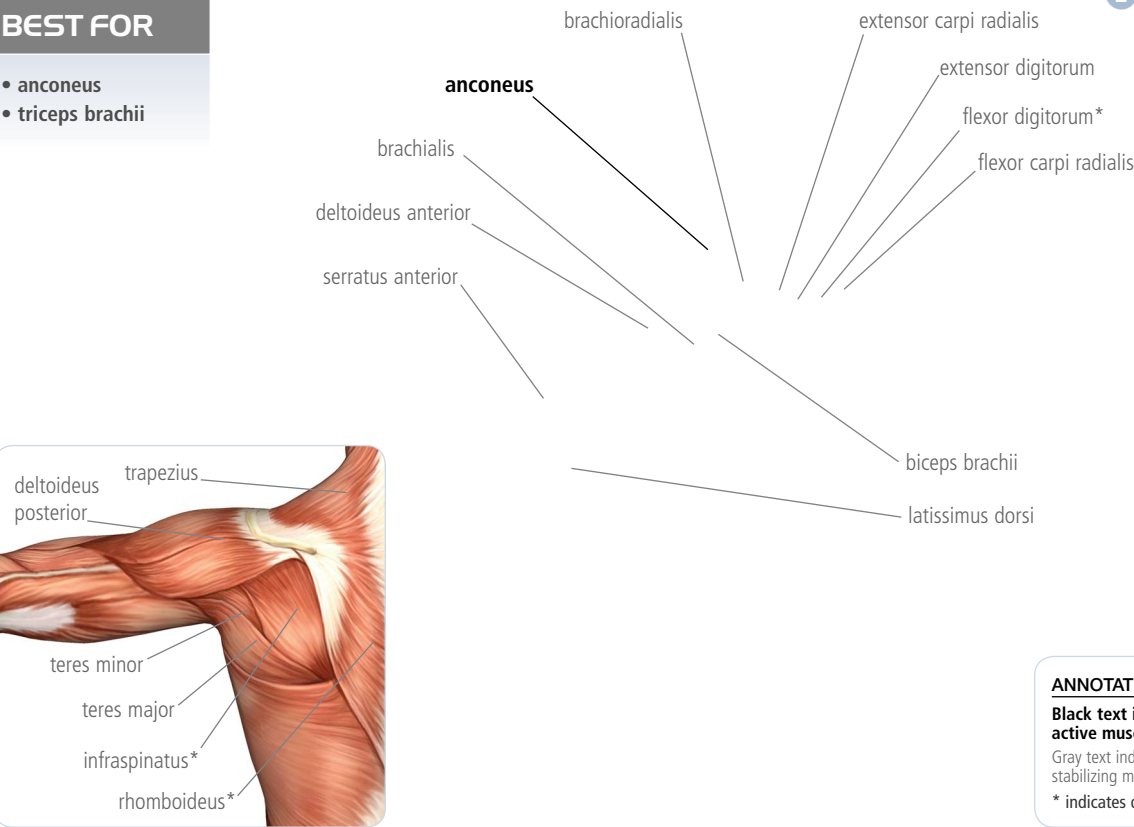
1

MODIFICATION
Similar difficulty: Standing, clasp a single dumbbell above your head with straight elbows; then lower the dumbbell behind your head by bending your elbows and keeping your upper arms straight up.



BEST FOR

- anconeus
- triceps brachii



2

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

BENCH DIP

1

Starting Position: Sit on a bench, with the bench perpendicular to your torso. Place your hands beside your hips on the bench and step forward so that your knees are bent. Your fingertips should be off the bench, so that only the heel of each hand contacts the bench. Your hips and torso should be at a 90-degree angle.

LOOK FOR

- Your body to remain close to the bench; your weight should primarily be translated through your hands, not your feet.

AVOID

- Rounding your back, sliding your hips away from the bench.
- Pushing with only your legs and elevating your shoulders at the top of the movement.

STABILIZE BY

- Pulling your abdomen in.
- Making sure your body remains close to the bench.
- Placing the palms of your hands securely on the bench.

2

Action: Lower your tailbone to the ground, bending your knees, elbows, and shoulders. At the bottom of the movement, extend your elbows and shoulders, keeping your head up. Pushing down into the bench, return to the start position.

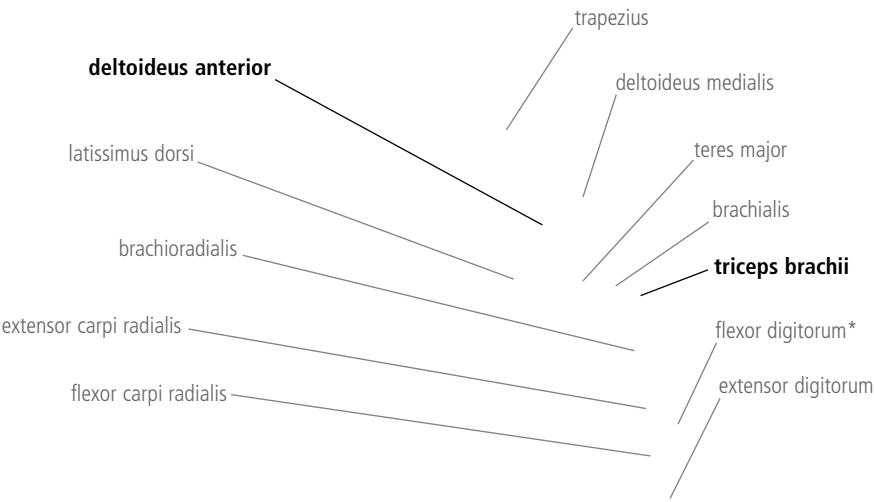
Movement Path: Your spine should descend in a straight line, parallel to the bench.

BENCH DIP • ARMS

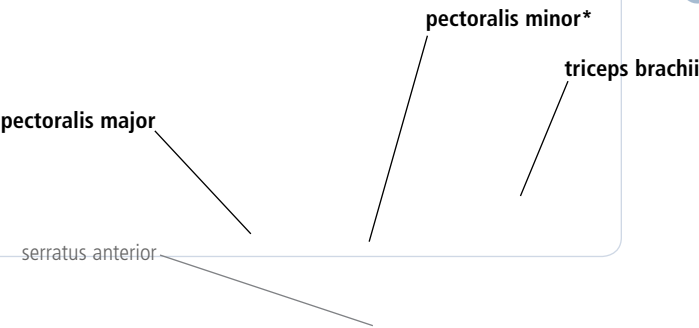
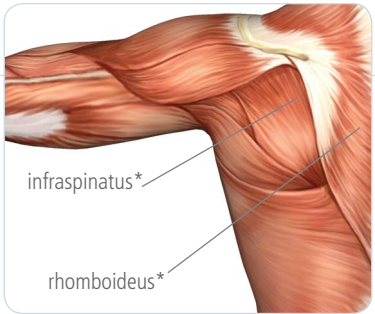
1

BEST FOR

- **deltoideus**
- **pectoralis major**
- **pectoralis minor**
- **triceps brachii**



2



MODIFICATION

Easier: Use assistance from a lower bench (extend your legs, placing your feet directly in front of your torso and planted firmly on the assist bench).



ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

BICEPS CURL

Starting Position:
Stand straight up with your knees slightly bent. Keep your chest up and shoulders down. Grasp dumbbells with your palms facing your thighs, directly at your sides.



Action: Exhale and bring the weights up, rotating your hands so that your thumbs turn outward as your hands rise, keeping your elbows tightly at your sides. Finish with your palms facing directly toward your shoulder joints. Inhale, and return the weights in a slow, controlled manner.

Movement Path: Your torso, hips, and legs are motionless as your lower arms are drawn upward in an arc from a vertical position in which they pointing down to one in which your hands are adjacent to your shoulder joints.

STABILIZE BY

- Retracting your shoulder blades throughout the entire movement.
- Pulling your abdomen up and in.

- LOOK FOR**
 - Your elbows to stay close to your body.
 - Your head to stay upright and your shoulder blades to stay down.
 - Your palms to rotate gradually as the weights are lifted and lowered.

- AVOID**
 - Extending your back during the curl; your spine should remain neutral.
 - Elevating your shoulders.
 - Moving your elbows too far away from your body.
 - Moving your head forward.
 - Letting the dumbbells down in an uncontrolled manner during the eccentric phase.
 - Any torso rotation.
 - Swinging the weights up.

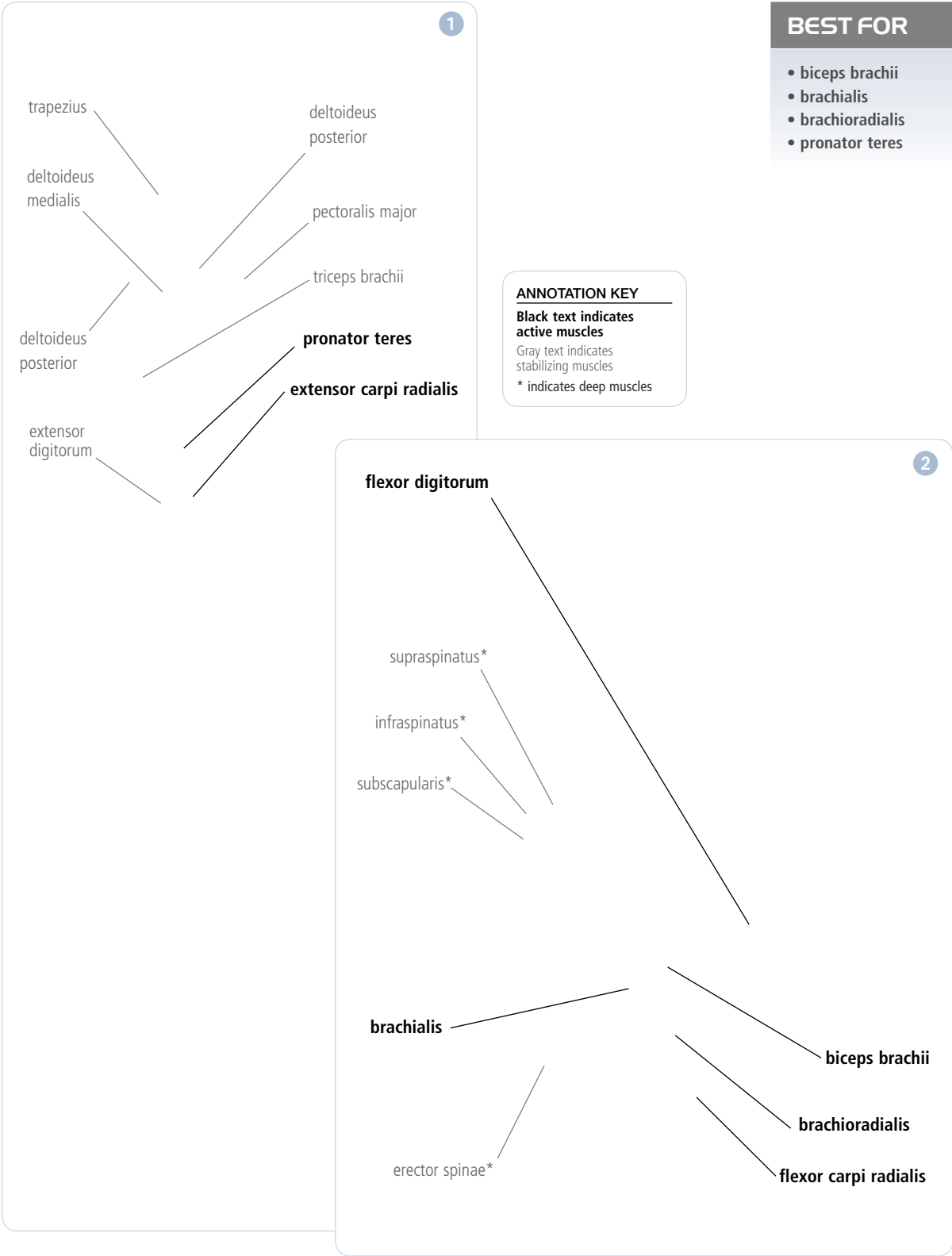
BICEPS CURL • ARMS

BEST FOR

- biceps brachii
- brachialis
- brachioradialis
- pronator teres

ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles



HAMMER CURL

Starting Position: Stand with your knees slightly bent and your arms extended at your sides. Hold the dumbbells with your palms facing in, toward your body. Your spine should be in a neutral position, your head and chest should be up, and your abdomen should be pulled up and in.



LOOK FOR

- Your elbows to stay close to your body.
- Your head to stay upright and your shoulder blades to stay down.

AVOID

- Extending your back during the curl; your spine should remain neutral.
- Elevating your shoulders.
- Moving your elbows too far away from your body.
- Moving your head forward.
- Letting the dumbbells down in an uncontrolled manner during the eccentric phase.
- Any torso rotation.
- Swinging the weights up.

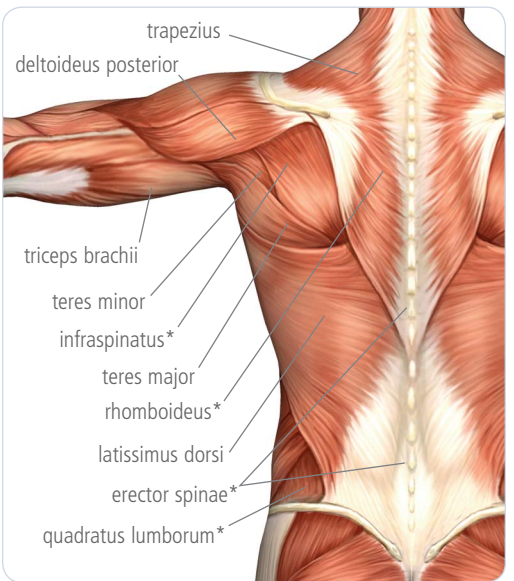
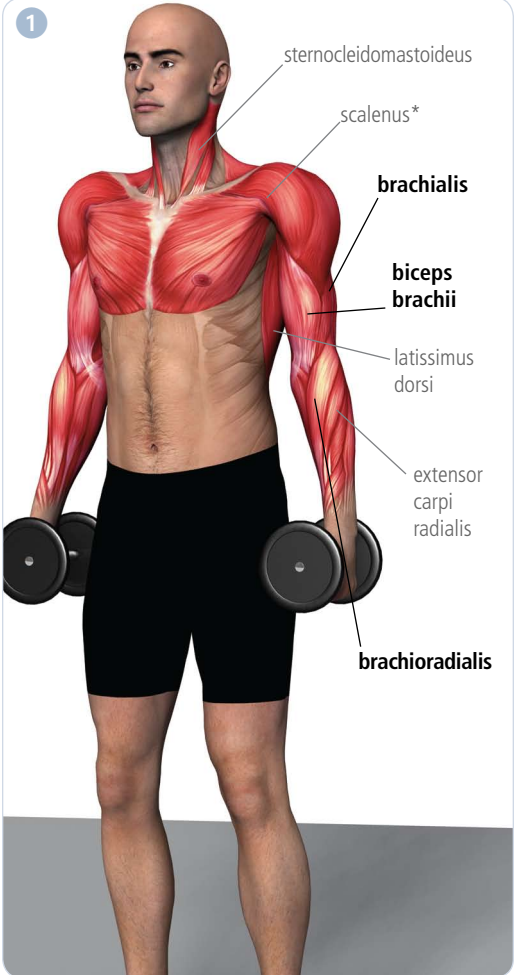
Action: Exhale, and flex your arms at the elbow, bringing the weights upward while your elbows remain at your sides. At the top of the movement, your thumbs should point towards the ceiling. Return along the same path by extending your arms at the elbows.

Movement Path: Your torso, hips, and legs are motionless as your lower arms are drawn upwards in an arc, from a vertical position in which they point down, to one in which your hands are adjacent to your shoulder joints.

STABILIZE BY

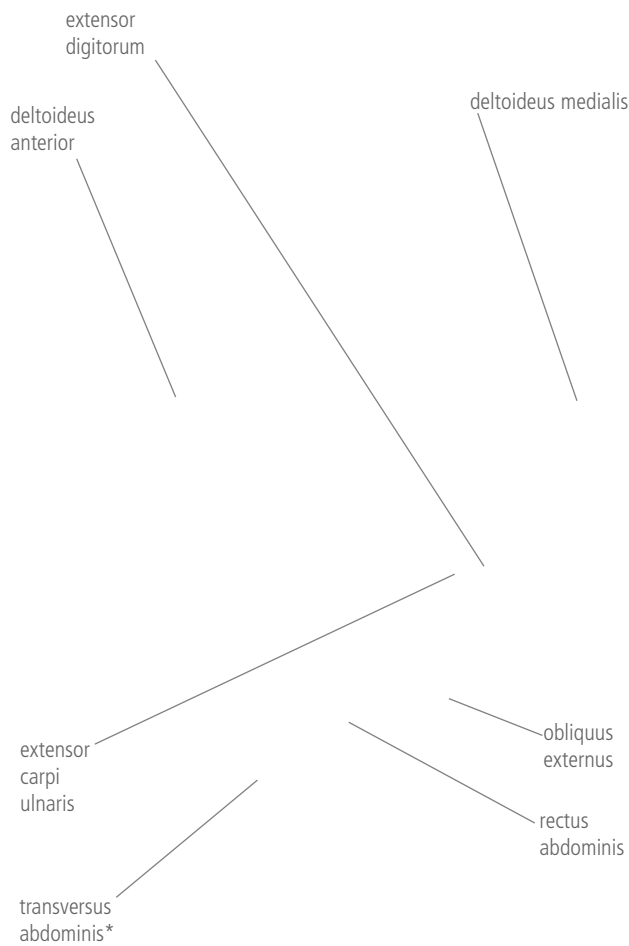
- Keeping your shoulder blades retracted throughout the entire movement.
- Pulling your abdomen up and in.

2



HAMMER CURL • ARMS

2



MODIFICATION

Easier: Raise and lower only one dumbbell at a time with alternating arms.



BEST FOR

- biceps brachii
- brachialis
- brachioradialis

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

WRIST CURL

Starting Position:

Sit on a bench and lean your torso forward, so that your forearms and elbows rest on your upper leg and your palms face the ceiling as you grip a barbell.



STABILIZE BY

- Firmly pushing your elbows into your legs (or the bench, if you chose to kneel on the floor and rest your forearms across a bench).
- Keeping your forearms parallel.

Action: Allow your wrists to open, letting the barbell roll down to your fingertips. Slowly contract your fingers, rolling your wrists toward the ceiling. Your wrists remain on your knee.

Movement Path: The weight should drop directly parallel to your shin bones (tibia), from your knee toward your foot. Curve your wrists in toward your body at the top of the movement.

LOOK FOR

- A full range of motion: your hands should fully extend.
- Your elbows to remain in contact with your upper thighs and your forearms to remain in contact with your lower thighs as the barbell moves through the range of the motion.

AVOID

- Excessive speed, movement in your upper torso, or movement of your forearms from your legs.

WRIST CURL • ARMS

BEST FOR

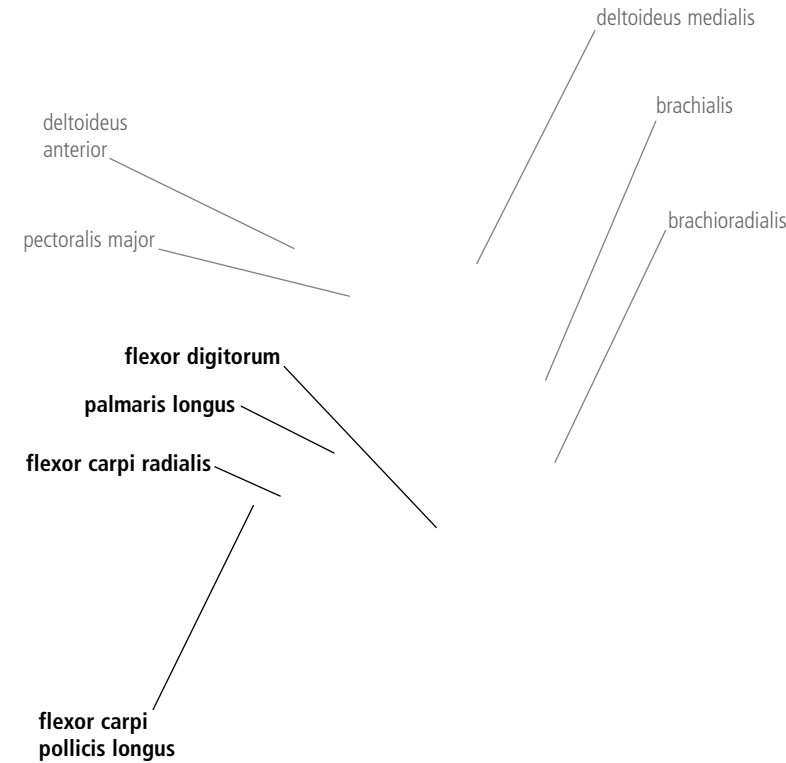
- **flexor carpi pollicis longus**
- **flexor carpi radialis**
- **flexor carpi ulnaris**
- **flexor digitorum**
- **palmaris longus**

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



MODIFICATION

More difficult: Turn your palms down, and roll the barbell up. Your knuckles drop to contact your tibia, then move up toward the ceiling.



DUMBBELL KICKBACK

Starting Position: Place one knee and one hand on a bench, with your spine horizontal and parallel to the bench. Your knee should be directly below your hip and your hand directly below your shoulder. Place your opposite foot on the ground, bending your knee slightly. Your feet should be slightly wider than shoulder-width apart. Your working arm should be bent, with the elbow at your side, parallel to your spine. Keep your shoulder retracted, with your hand directly in front of your shoulder joint.

Action: Keeping your entire body in a fixed position, push your hand backward, from your shoulder toward your hip, and straighten your elbow so that your arm is at least parallel with your torso and the ground.

Movement Path: The dumbbell moves from your shoulder to your hip in an arc.

- STABILIZE BY**
- Keeping your spine neutral.
 - Keeping your shoulders parallel and contracted (down and flat)
 - Keeping your hips even.
 - Distributing your weight evenly among the three contact points.

- LOOK FOR**
- Your elbow to remain at your side.
 - Your shoulder to remain in the same plane as your spine and the opposite shoulder.

- AVOID**
- Allowing your elbow to drop toward the floor or migrate away from your side.
 - Allowing your shoulder to drop forward or your torso to rotate.
 - Elevating your shoulder toward your ear.

1



2

MODIFICATION
More difficult: Stand with your torso bent to 80 degrees, with your shoulders retracted, your elbows slightly behind the plane of your spine, your hands in front of your shoulder joints, and dumbbells held in front of your stomach. Extend your elbows back, behind the plane of your hip, keeping your upper arms stable, your shoulders down, and your spine long.

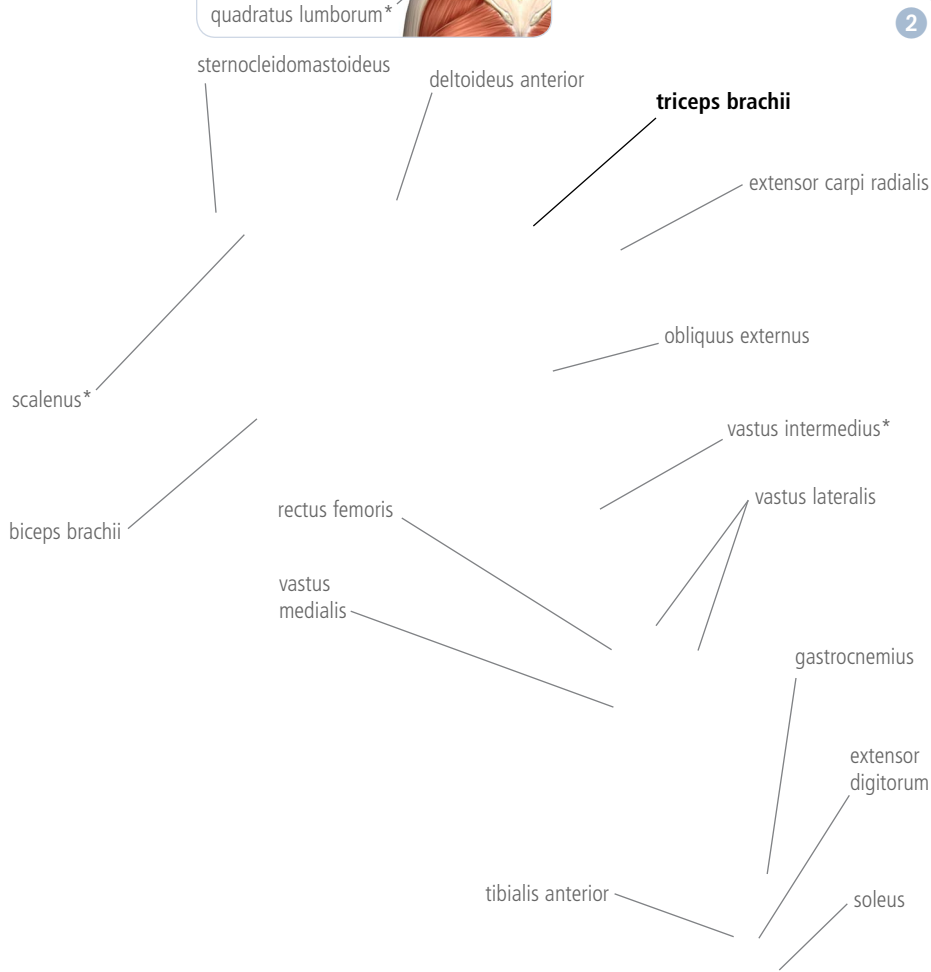
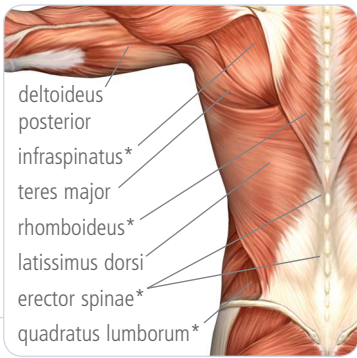


ANNOTATION KEY
 Black text indicates active muscles
 Gray text indicates stabilizing muscles
 * indicates deep muscles

DUMBBELL KICKBACK • ARMS

BEST FOR

- anconeus
- triceps brachii



2

TRICEPS PUSHDOWN

Starting Position: Stand facing the cable stack with your knees and hips slightly bent and your torso leaning slightly forward. Grasp the bar with your palms facing down. Your elbows should be at your sides in line with your spine and your chest and head should be up.



1

- LOOK FOR**
- Your body to remain completely stationary.
 - The angle of your spine to be parallel with the path of the cable.

- AVOID**
- Allowing your elbows to leave your sides.
 - Shrugging or rounding your shoulder.
 - Leaning backward.

Action: Exhale, and push the weight down toward the floor and into your lap, straightening your arms completely. Pause, and then inhale as the weight returns in the opposite manner.

Movement Path: Your torso, hips, and legs are stationary while your lower arms travel outward and downward in an arc to your legs.

- STABILIZE BY**
- Keeping your elbows in by your sides and pulled back, parallel with your spine.
 - Pulling your abdomen up and in.
 - Keeping your spine neutral and your shoulders down and back.

2

TRICEPS PUSHDOWN • ARMS

BEST FOR

- anconeus
- triceps brachii

MODIFICATION

Similar difficulty: Use a rope with a neutral (parallel) grip.



1



2



ANNOTATION KEY

Black text indicates active muscles
 Gray text indicates stabilizing muscles
 * indicates deep muscles

BARBELL CURL

Starting Position: Grasp the barbell with your palms facing forward (away from your body) and with your hands slightly wider than your hips. Your torso should be upright and your hips and knees should be slightly bent. Keep your abdomen pulled up and in, your chest high, and your chin up.



LOOK FOR

- Your torso to remain stationary.
- The bar to move at one rate through the full range of the motion.
- Your shoulders to stay down, your chest to stay up, and your abdomen to stay pulled in.

AVOID

- Swinging the bar.
- Leaning backward.
- Allowing your elbows to slide backward during the upward movement of the bar.
- Moving your head forward.

Action: Exhale, and bring the weight up, keeping your elbows tightly at your sides. Finish with your palms facing directly backward. Inhale, and return the barbell to the starting position in a slow, controlled manner.

Movement Path: Your torso, hips, and legs are motionless as your lower arms are drawn upward in an arc, from a vertical position in which they point down, to one in which your hands are adjacent to your shoulder joints.

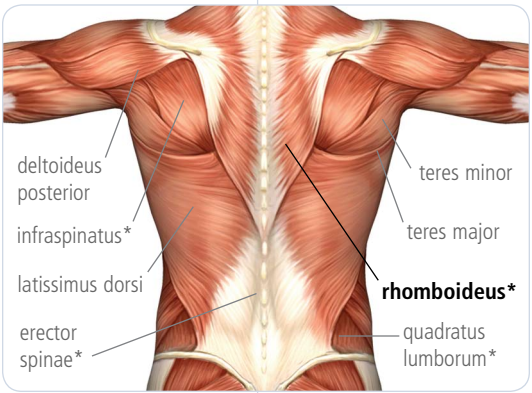
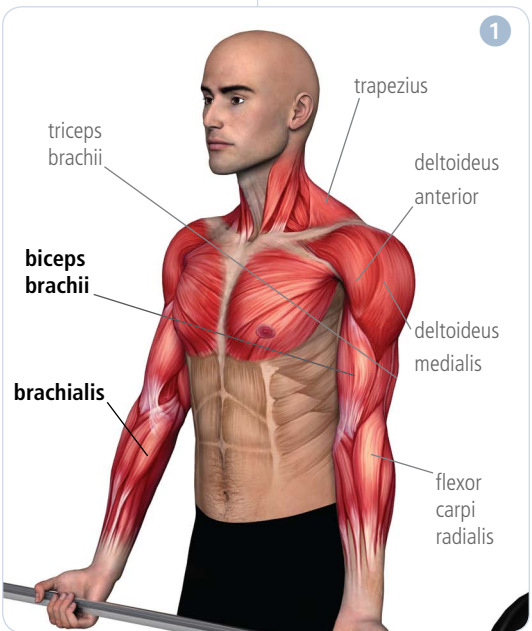
STABILIZE BY

- Keeping your elbows in by your sides and pulled back, parallel with your spine.
- Pulling your abdomen up and in.
- Keeping your spine neutral and your shoulders down and back.

2

BEST FOR

- brachioradialis
- biceps brachii



flexor digitorum*

extensor digitorum

sternocleidomastoideus

levator scapulae*

splenius*

trapezius

extensor carpi radialis

anconeus

pectoralis major

obliquus externus

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

CORE

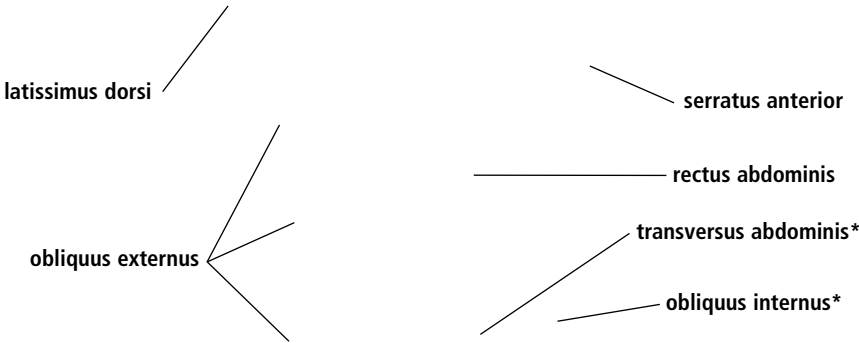
The muscles of the core originate on either the pelvis or spinal column and are responsible for forward flexion, extension, lateral flexion, and rotation of the spine. These muscles are intimately connected to the muscles of the hip and back and are critical to the proper functioning of both.

Along with hip and back muscles, the is made up of both superficial and deep layers, some of which include muscles (for example, the transversus abdominis) that are responsible only for stabilization, not movement. Core training has typically consisted of movements that are primarily forward flexion and rotation biased; however, understanding that each plane of movement is necessary for proper function and, subsequently, injury prevention, is imperative.

All athletic activities require the core muscles to stabilize and translate forces from the lower to upper body and vice versa.

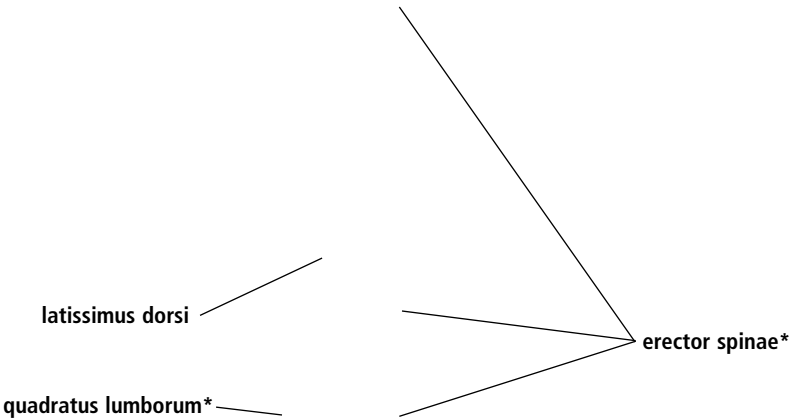
CORE ANATOMY

CORE



ANNOTATION KEY
* indicates deep muscles

CORE



ANNOTATION KEY
* indicates deep muscles

TRANSVERSE ABDOMINALS

Starting Position: Sit on a Swiss ball with your hands on your hips. Your shoulders should be down and your head should face directly forward.

1

- LOOK FOR**
- Your chest to remain high and your ribs to remain up and apart.
- AVOID**
- Torso flexion (bending forward) or any hip movement.

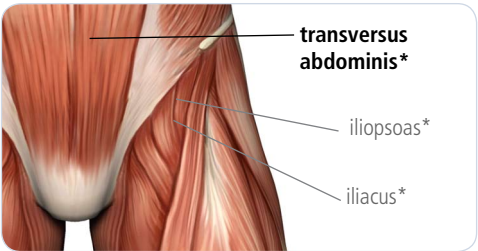
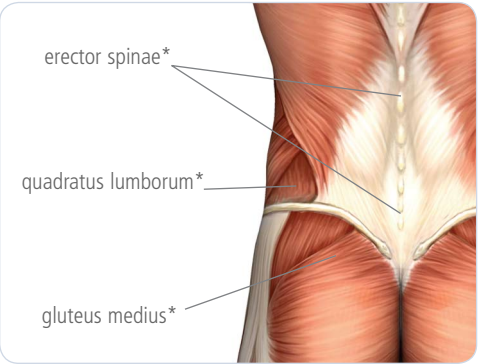
Action: Inhale through your lower abdomen, pushing your belly button out. Exhale, and drag your belly button in and up, elevating your rib cage and cinching your waist.

Movement Path: Your abdomen contracts backward toward your spine.

- STABILIZE BY**
- Keeping your chest up and your shoulders down and relaxed.
 - Maintaining a neutral spinal position.

2

TRANSVERSE ABDOMINALS • CORE



ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

BEST FOR

- transversus abdominis

infraspinaus*

obliquus externus

CRUNCH

Starting Position: Place your hands behind your head while lying flat on the floor. Bend your knees so that your feet are flat on the floor, keeping your spine long.

1

LOOK FOR

- A smooth movement throughout the entire length of your spine.
- Your abdominal muscles to contract and pull in
- Your hips to remain stable.

AVOID

- Pulling with your hands.
- Bringing your chin toward your chest or collarbone.
- Arching your back.
- Elevating your feet.

Action: Push your lower back into the floor, keeping your spine long. Contract your abdominal muscles, and lift your upper back off the floor and slightly forward, exhaling as you come up.

Movement Path: Your torso curves from your mid-low back to the top of your head, in straight line up and toward the knees.

2

STABILIZE BY

- Keeping your shoulders down with your elbows widely spread.
- Keeping your hips even and your feet flat.

CRUNCH • CORE

BEST FOR

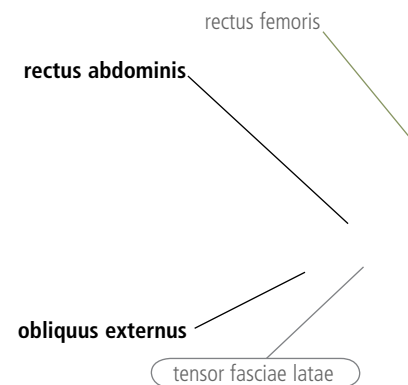
- obliquus externus
- rectus abdominis

ANNOTATION KEY

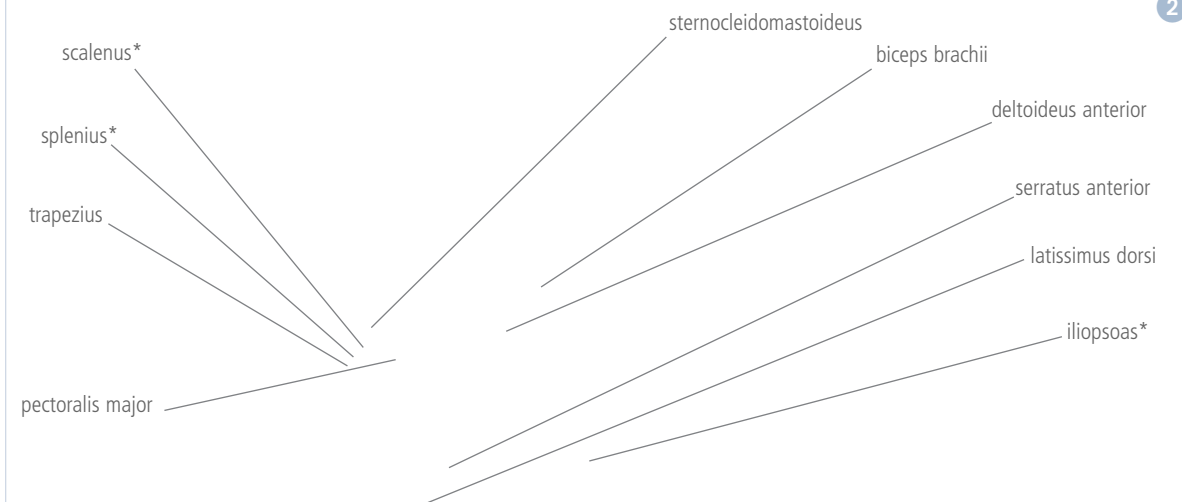
Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



1



2

MODIFICATIONS



Similar difficulty:

Place your feet on a Swiss ball while you do a crunch. Do not let the ball roll.



Similar difficulty:

Do a regular crunch while sitting on a Swiss ball. Do not let the ball roll.



More difficult:

Raise your shins up to 45-degree angles.



More difficult:

Lift a medicine ball from your chest straight up. Keep your arms straight as you do a crunch.



More difficult:

As before, but raise one leg and extend the ball to your outstretched foot as you do a crunch.

CROSSOVER CRUNCH

CORE

LOOK FOR

- A smooth movement throughout the entire length of your spine.
- Your abdominal muscles to contract and pull in.
- Your neck to remain long and your chin to remain away from your chest.
- Both hips to remain stable and on the ground.

AVOID

- Pulling with your hands,
- Bringing your chin toward your chest or collarbone.
- Arching your back.
- Elevating your foot.
- Raising the pivot elbow off the ground.
- Moving the elbow on the active arm faster than your shoulder.

STABILIZE BY

- Keeping your shoulders down with your elbows widely spread.
- Keeping your hips even and your feet flat.

Starting Position: Place your hands behind your head while lying flat on the floor. Cross one ankle over the opposite knee, keeping one foot on the floor and your knees bent. The knee that is crossed should be in the same plane as the shoulder on that side (*not* wider).

Action: Bring your active arm and shoulder up and simultaneously across your body, bringing the elbow toward the crossed-over knee, flexing and rotating your torso. Pivot on your opposite elbow; your opposite arm always remains in contact with the floor. Pivoting on the elbow on the same side as the elevated leg, turn your body so that the opposite elbow comes across and up, toward your bent knee. Exhale as you rise.

Movement Path: Your torso flexes and simultaneously rotates downward and inward.

CROSSOVER CRUNCH • CORE

serratus anterior
pectoralis major
gluteus maximus

iliopsoas*
pectineus*
adductor longus
adductor magnus
gracilis*

gluteus medius*
rectus abdominis
tensor fasciae latae
obliquus internus*
obliquus externus
latissimus dorsi
sternocleidomastoideus
scalenus*
levator scapulae*
deltoideus anterior

MODIFICATION

Similar difficulty: Sit on a Swiss ball, and do a sit-up, twisting as you rise so that your torso is turned to almost 90 degrees to your legs at the top of the movement.



ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

BEST FOR

- obliquus externus
- obliquus internus
- rectus abdominis

OBLIQUES

Starting Position: With one hip resting on a Swiss ball, spread your feet widely apart on the floor, and place your hands behind your head.

Action: Exhale, and bend upward from your hips, curving your spine laterally. Keep your hands behind your head, your elbows wide, and your shoulders down.

Movement Path: Your hips and legs remain stationary on the ball. Your torso bends from one side to a vertical position.

- STABILIZE BY**
- Keeping the edges of your feet rigid and in contact with the floor.
 - Distributing your weight evenly between your feet.
 - Keeping your legs straight, and your hips contracted.

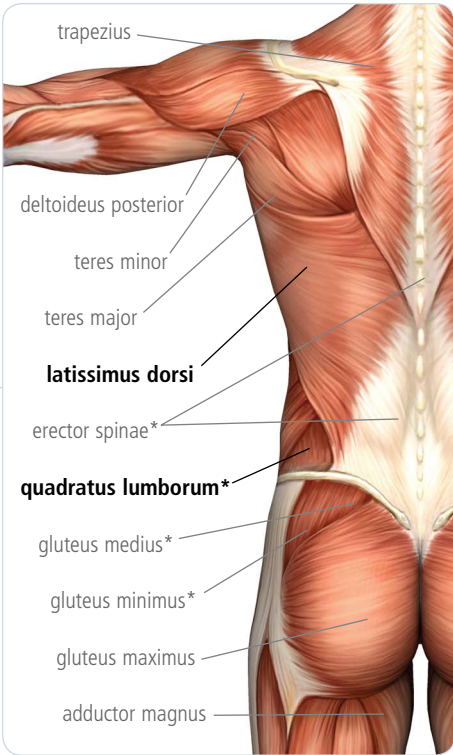
- LOOK FOR**
- Your torso to move from a 45-degree angle to a 90-degree angle.
 - Your shoulders, hips, and spine to remain in alignment.

- AVOID**
- Rotation of any kind
 - Moving your hips on the ball.
 - Bending your knees.

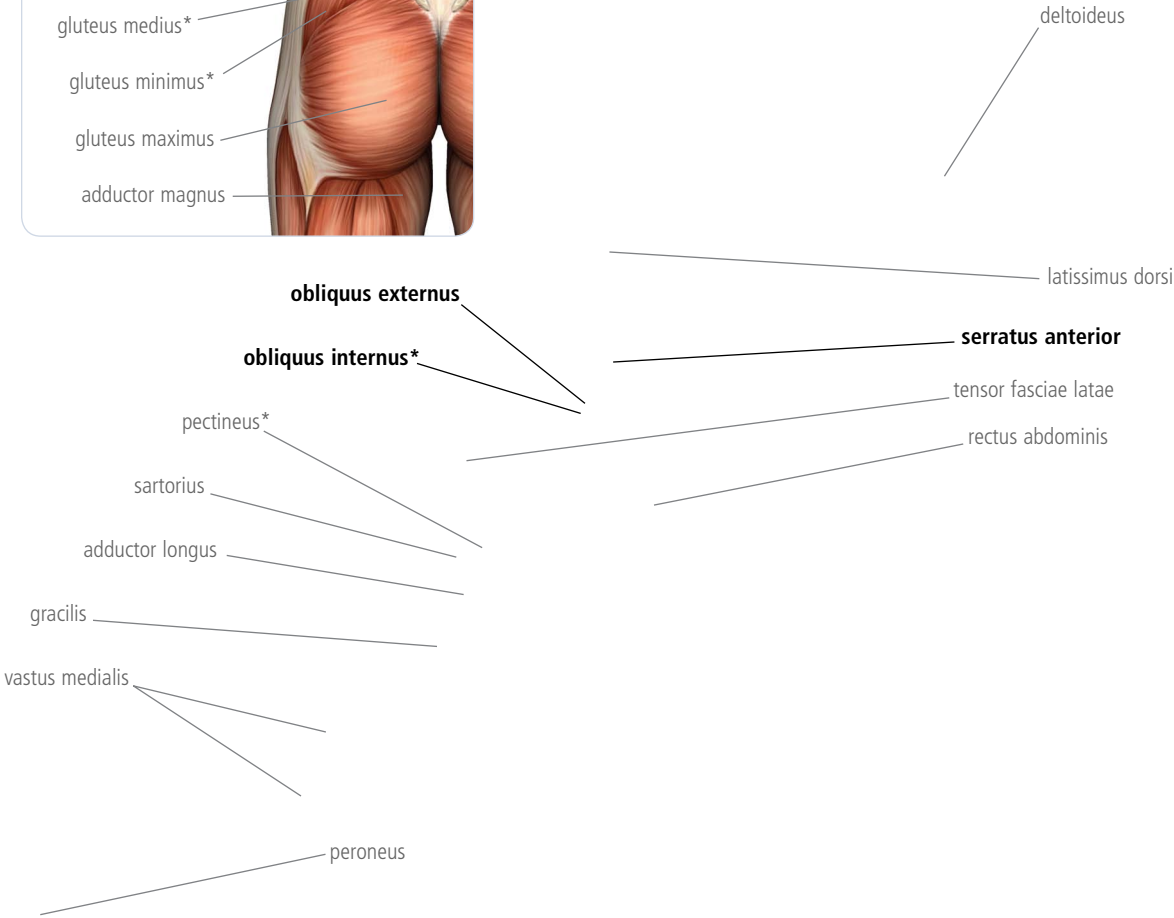
OBLIQUES • CORE

BEST FOR

- obliquus externus
- obliquus internus
- quadratus lumborum



ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles



AB WHEEL

Starting Position: On your knees, bend your torso forward at a 45-degree angle with your spine in a neutral position. Extend your arms forward at a 45- to 90-degree angle to your torso, with your elbows straight and your hands grasping the wheel.

1

Action: Inhale, and extend your arms forward, allowing your torso to drop until your chest is almost parallel to the floor, rolling the wheel in a straight line away from you. Your hips move forward, following your torso, but your knees remain stationary. Exhale, and draw your arms and hips back simultaneously; your torso elevates and returns to the starting position.

Movement Path: Your center of mass is translated forward and downward as your arms and hips extend into a linear position, with your knees as the fulcrum.

2

STABILIZE BY

- Pulling your abdomen up and in.
- Keeping your shoulders down and back throughout the movement.
- Keeping your arms extended and your wrists solid.
- Maintaining a neutral spinal position throughout the movement.

LOOK FOR

- All joints to move at the same time.
- Your head and spine to remain aligned.

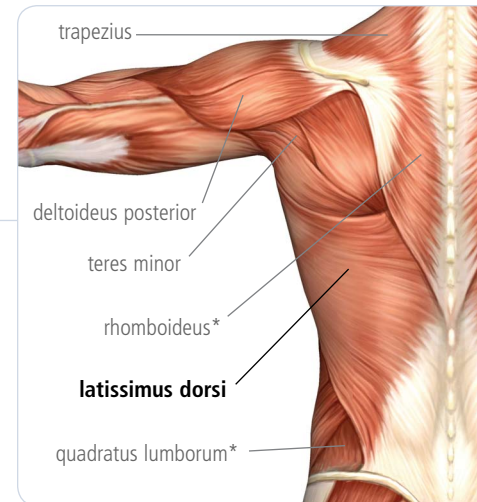
AVOID

- Rounding or arching your spine.
- Allowing your joints to move sequentially.
- Moving quickly in either direction.

AB WHEEL • CORE

BEST FOR

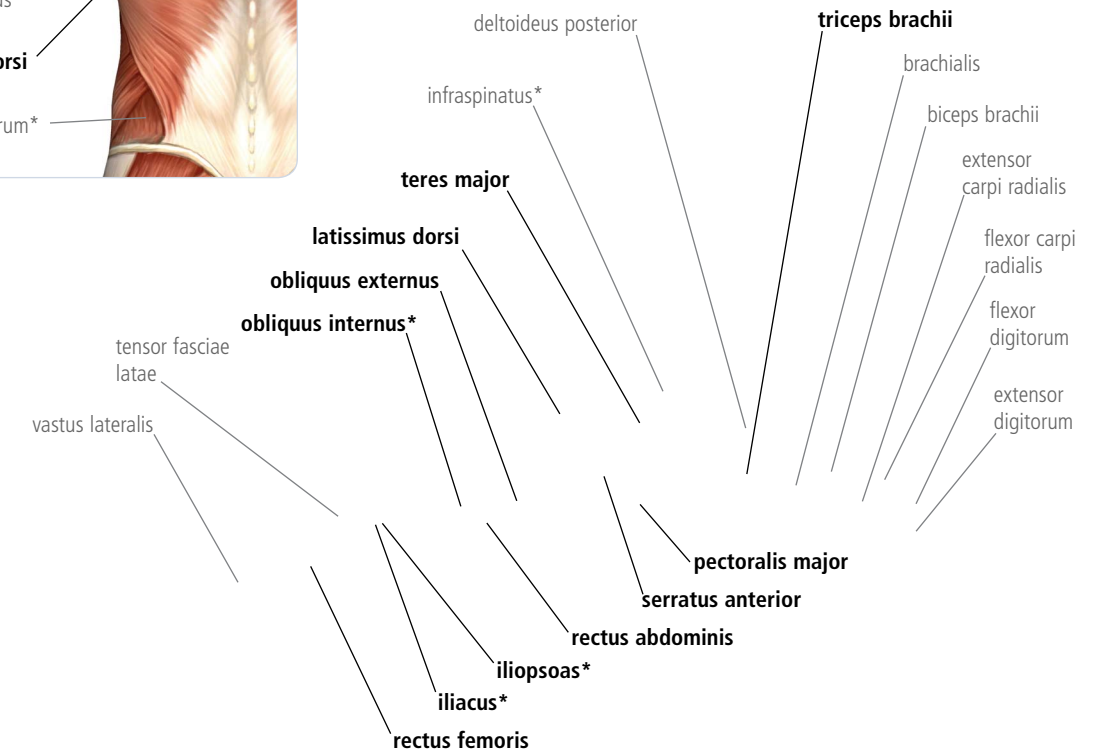
- | | |
|---------------------|---------------------|
| • iliacus | • rectus abdominis |
| • iliopsoas | • rectus femoris |
| • latissimus dorsi | • serratus anterior |
| • obliquus externus | • teres major |
| • obliquus internus | • triceps brachii |
| • pectoralis major | |

**ANNOTATION KEY**

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

**MODIFICATION****Similar difficulty:**

Replace the wheel with a Swiss ball; your hands begin higher up.



2

HANGING LEG RAISE

CORE

LOOK FOR

- Your knees to bend as your upper legs are raised (your lower legs remain vertical).
- Your legs to move upward together.

AVOID

- Swinging.
- Extending your arms upward to more than 5 degrees above horizontal.
- Moving your hips backward.



Starting Position: Hang with your upper arms in stirrups, with your elbows bent at 90-degree angles, pointing forward just above shoulder height. Grasp the stirrups with your hands and make sure your torso, legs, and hips are straight.

Action: Pull your upper arms and elbows downward and your upper legs and knees upward toward your elbows, flexing at the hips. Tuck your hips forward and bring your chest forward slightly. Return to the starting position in a slow and controlled manner. Exhale as you rise, and inhale as you return to the starting position.

Movement Path: Your torso rounds slightly as your hips flex upward and your upper arms are pulled downward. Your center of mass makes no appreciable movement.

STABILIZE BY

- Keeping your upper arms parallel and your shoulders down.
- Gripping the stirrups firmly.
- Keeping your legs parallel.

HANGING LEG RAISE • CORE

BEST FOR

- iliacus
- iliopsoas
- rectus abdominis
- rectus femoris
- tensor fasciae latae

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



MODIFICATION

More difficult:

Clasp a medicine ball between your knees, and maintain the same activation pattern and movement sequence.



BRIDGE

Starting Position: Bend your knees, with your feet flat on the floor. Bend your elbows to 90-degree angles, with your hands facing the ceiling and a slight arch in your lower back, which should not contact the floor.

1

LOOK FOR

- Your hip to be the hinge; there should be no movement in your spine from the hip to the shoulder joint.

AVOID

- Sequential lifting, a pelvic tuck, or any rotational elevation (one hip rising faster than the other).

Action: Using your upper arms, shoulders, and feet, push your hips and ribs simultaneously up toward the ceiling. Return to the starting position.

Movement Path: Curvilinear; the spinal movement is straight up from the floor.

2

STABILIZE BY

- Keeping your upper arms and elbows pulled down and into the floor.
- Distributing your weight evenly, keeping your feet flat.
- Keeping your hips, knees, and feet in a single line.

BRIDGE • CORE

BEST FOR

- biceps femoris
- gluteus maximus
- rhomboideus
- quadratus lumborum
- semimembranosus
- semitendinosus

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



MODIFICATIONS



More difficult: Raise one leg, and maintain the same activation pattern and movement sequence.

More difficult: Lay your arms flat and balance your feet on a Swiss ball, keeping your hips elevated and the plane of your body flat.

More difficult: Same as previous, but bend one leg toward your body.

More difficult: With your legs on a Swiss ball, raise one leg straight up in the top position.

ARM-LEG EXTENSION

CORE

Starting Position: Lie flat on the ground, with one arm bent, your elbow on the floor, your palm down, and the hand under your chin. Extend your other arm, holding your thumb up toward the ceiling.

1

Action: Simultaneously lift your extended arm, torso, and the opposite leg.

Movement Path: Your arm and leg rise straight up to form an arc with your torso.

2

LOOK FOR

- A simultaneous movement of the raised arm and leg.
- An equivalent range or distance from the floor for the raised arm and leg.
- Your hip bones to remain in contact with the ground.

AVOID

- Any rotation of your torso or hips from the floor.
- Elevating your shoulder or bending your knee or elbow.

STABILIZE BY

- Keeping your shoulder blades down and back.
- Keeping your hips even and down.
- Keeping your legs and arms straight.

MODIFICATIONS

1



1



1



2



2



2



More difficult: Raise both arms and legs simultaneously.

More difficult: On your hands and knees, place a medicine ball on your back. Extend one arm and the opposite leg (stabilizers: all spinal and hip musculature).

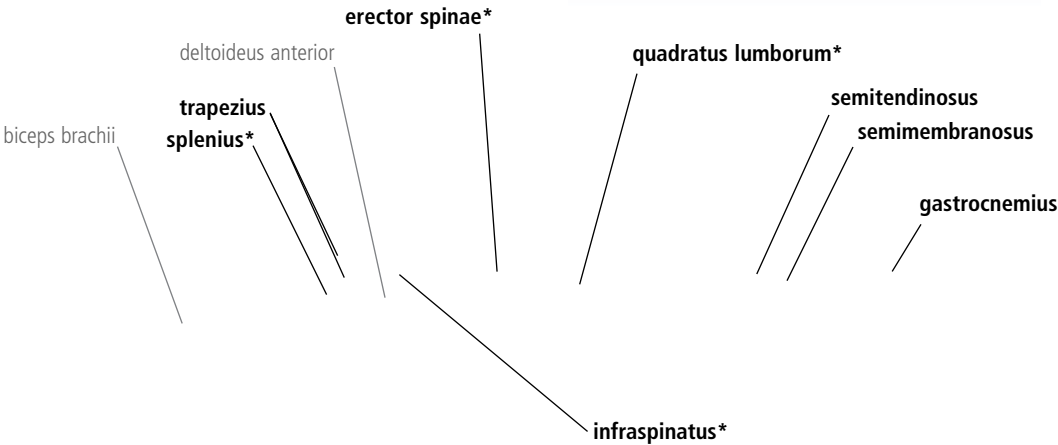
More difficult: In a front plank position, raise one arm and the opposite leg (stabilize with all core muscles).

ARM-LEG EXTENSION • CORE

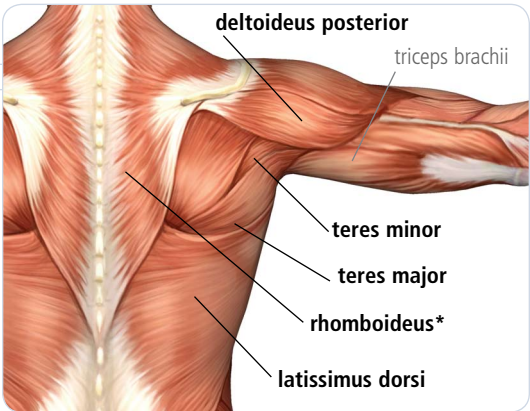
BEST FOR

- erector spinae
- gluteus maximus
- infraspinatus
- quadratus lumborum
- rhomboideus
- splenius
- teres minor
- trapezius

1

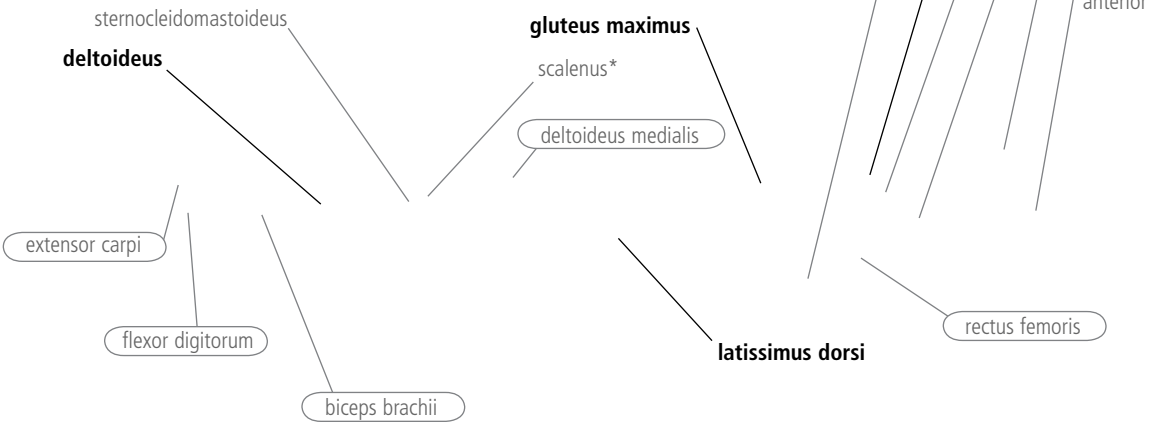


2



ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles



FRONT PLANK

Starting Position: Lie face down on the floor, and fold your hands directly beneath your chin with your elbows by your sides and your feet on your toes.

Action: Raise the length of your torso to a horizontal position with a slight arch in your lower back. Your shoulder blades should be flat and your spine long.

Movement Path: None.

STABILIZE BY

- Keeping your spine neutral.
- Keeping your shoulders down and your head up.
- Maintaining the contraction of your gluteals and legs.
- Keeping your legs straight and your ankles bent at 90-degree angles, with your toes pointing directly into the ground.

MODIFICATIONS

Easier: Raise your forelegs and rest your weight on your knees to shorten the lever.

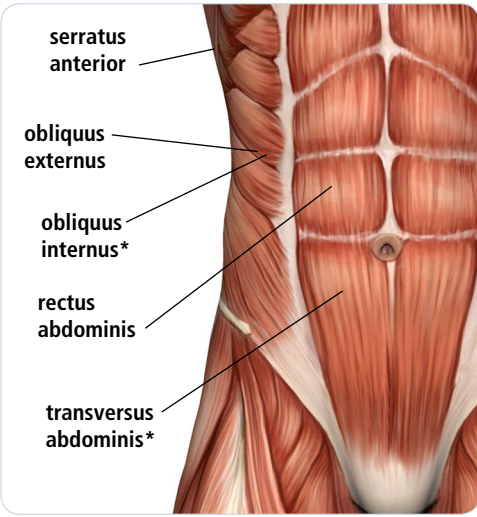
LOOK FOR

- A neutral spinal position.
- Locked knees, with your ankles at 90-degree angles and your elbows directly under your shoulder joints.

AVOID

- Rounding your spine.
- Dropping your hips.
- Elevating your shoulders toward your ears.

FRONT PLANK • CORE



BEST FOR

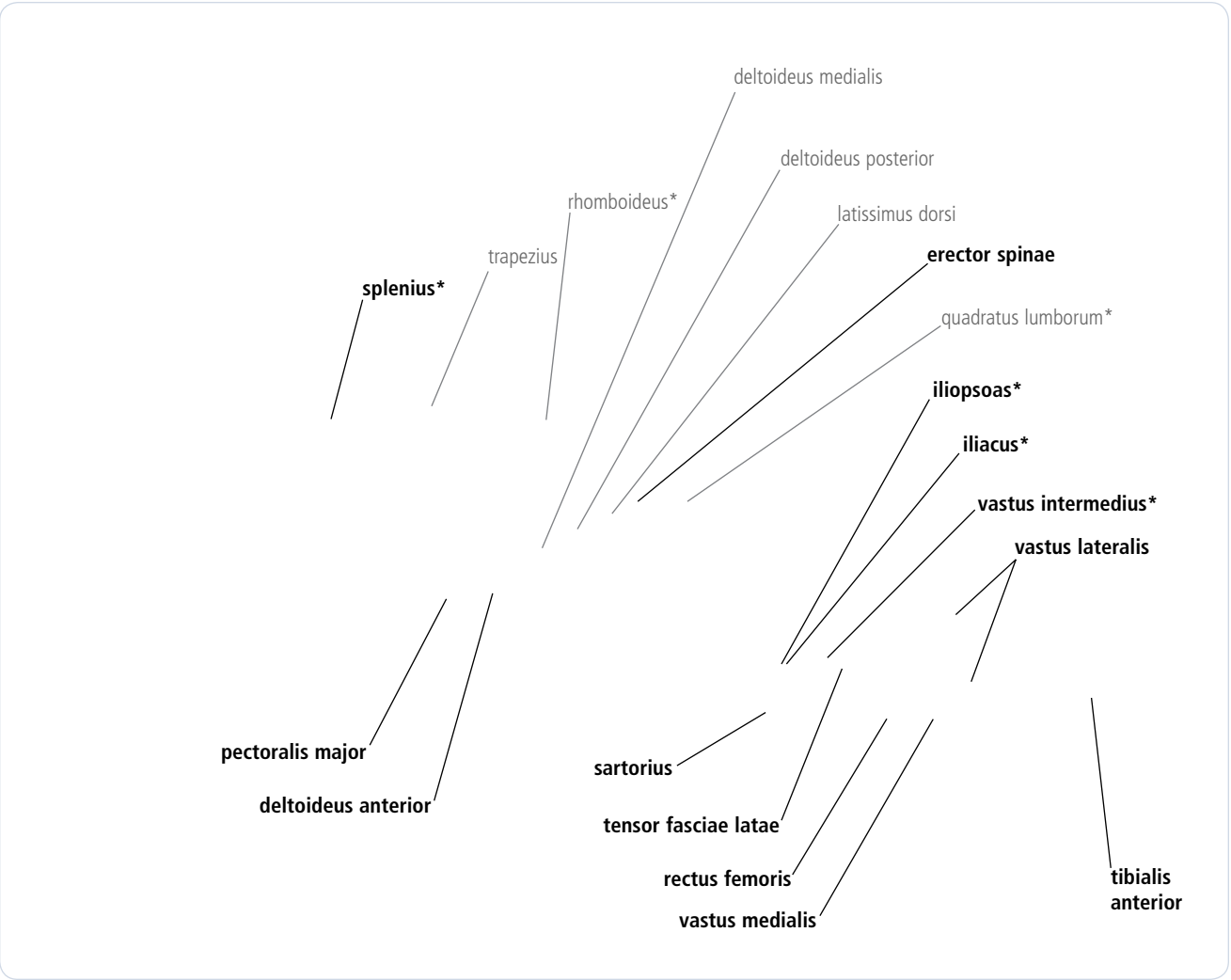
• erector spinae	• serratus anterior
• iliacus	• splenius
• iliopsoas	• tibialis anterior
• obliquus internus	• transversus abdominis
• rectus abdominis	
• rectus femoris	• vastus intermedius

ANNOTATION KEY

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles



SIDE PLANK

Starting Position: Lie on your side with the lower arm bent at the elbow. The lower elbow should be underneath the shoulder joint and the upper hand should be on your hip. Align your ankles, hips, shoulders, and head.

Action: Push the length of your body toward the ceiling, balancing on the edge of your bottom shoe with one foot directly over the other.

Movement Path: Your center of mass rises directly upward.

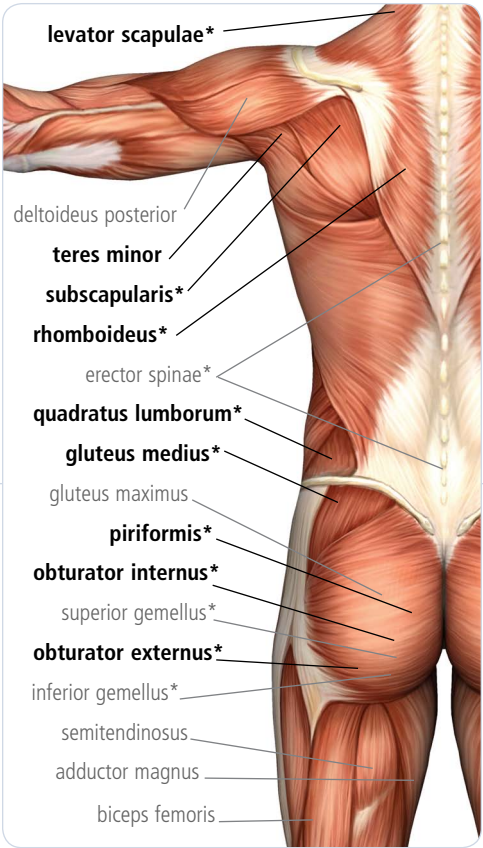
- STABILIZE BY**
- Pulling your abdomen up and in, keeping your spine neutral.
 - Keeping your shoulder blades away from your spine and flat on your rib cage.
 - Straightening and contracting your legs.
 - Balancing on the edge of your bottom foot or shoe.

- LOOK FOR**
- Your spine to be in a straight position, similar to standing.
 - Your elbow to be directly under your shoulder joint, with a slight arch in your lower back, your knees straight, and your ankles at 90-degree angles.

- AVOID**
- Rotating your hips.
 - Allowing your shoulder blades to slide toward your spine.
 - Moving your head forward.

1

2



ANNOTATION KEY
Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles

SIDE PLANK • CORE

BEST FOR

- | | |
|----------------------|------------------------|
| • gluteus medius | • subscapularis |
| • obliquus externus | • tensor fasciae latae |
| • obliquus internus | • teres minor |
| • peroneus | • trapezius |
| • quadratus lumborum | • vastus lateralis |
| • rhomboideus | |

MODIFICATIONS



Easier: Bend your lower leg so that, keeping your knees parallel, your weight is balanced on your bent knee.



Easier: Bend both legs, crossing your top leg in front so that both knees are on the floor.



Push your hip and torso toward the ceiling, extending the top leg and knee up and out.



More difficult: Raise your top arm and balance with both arms straightened.

SANTANA PUSH-UP

Starting Position: Lie flat on the floor with your hands slightly wider than shoulder width, grasping dumbbells, so that the dumbbell handles are parallel to your spine. Point your elbows directly at the ceiling. Your feet should be slightly wider than shoulder width, and your spine should be neutral.

1

Action: Push up toward the ceiling; once your arms are fully extended, rotate your hips and feet, lifting one arm in an arc toward the ceiling so that your arms are aligned in a straight line and your feet are split apart, with your weight on the edges of your shoes. Your torso, hips, and legs are rigid.

2

Movement Path: Your entire body moves up and away from the floor, and then rotates around your spine 180 degrees.

3

STABILIZE BY

- Pulling your abdomen up and in.
- Keeping your shoulder blades down and flat.
- Keeping your knees straight and your legs contracted.
- Maintaining a neutral spinal position throughout the movement.

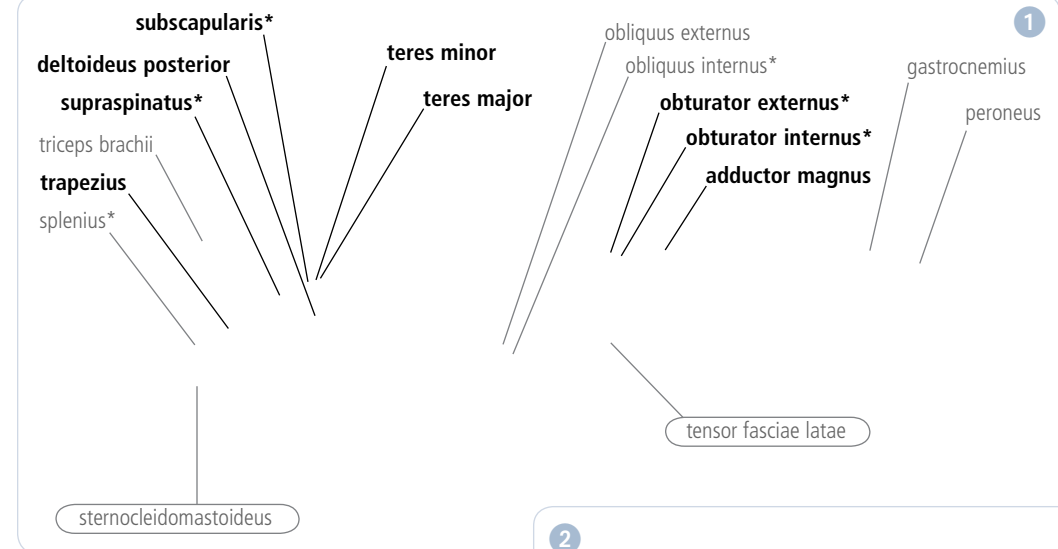
LOOK FOR

- Your shoulders to remain depressed.
- Your neck to remain long.
- Your hips to remain elevated.
- Your shoulder, hip, and feet to remain in the same plane from the floor.

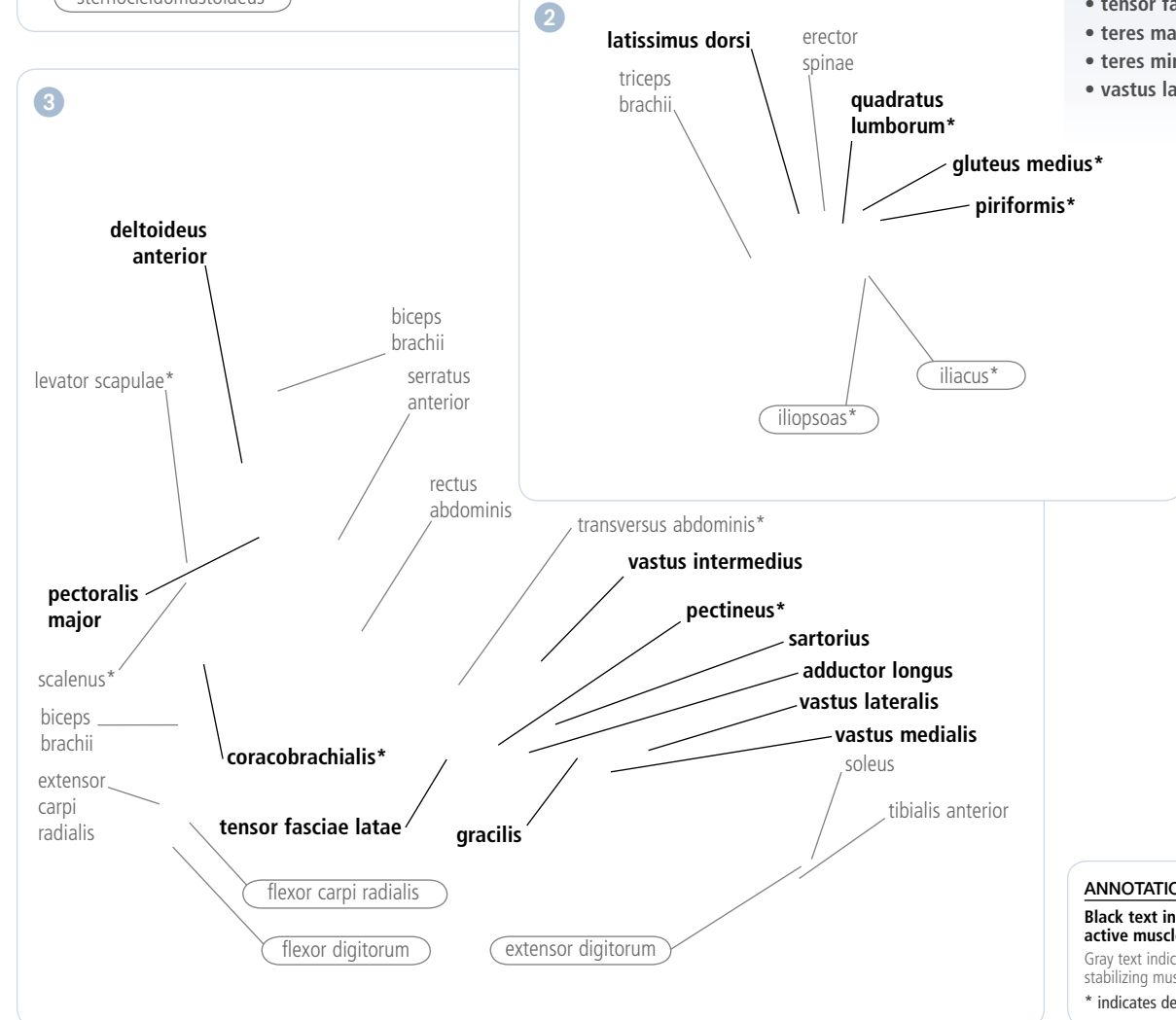
AVOID

- Bending your knees.
- Dropping your hips.
- Excessive rotation in shoulder and hip.

SANTANA PUSH-UP • CORE

**BEST FOR**

- adductor longus
- adductor magnus
- deltoideus
- coracobrachialis
- gluteus medius
- gracilis
- obturator externus
- obturator internus
- pectoralis major
- piriformis
- quadratus lumborum
- sartorius
- subscapularis
- supraspinatus
- tensor fasciae latae
- teres major
- teres minor
- vastus lateralis

**ANNOTATION KEY**

Black text indicates active muscles

Gray text indicates stabilizing muscles

* indicates deep muscles

PNF RAISE



Starting Position: Stand on one foot, bending the raised knee, and grasp a medicine ball just below and to the outside of the knee on the standing leg.

Action: Stand, extending your leg, while bringing the ball across your body to above and outside the opposite shoulder.

Movement Path: Your upper body rotates as your center of mass shifts upward. The ball moves in an arc across your body.

STABILIZE BY

- Pulling your abdomen up and in.
- Distributing your weight evenly across your foot.
- Using all muscles and joints in a coordinated, relaxed manner.

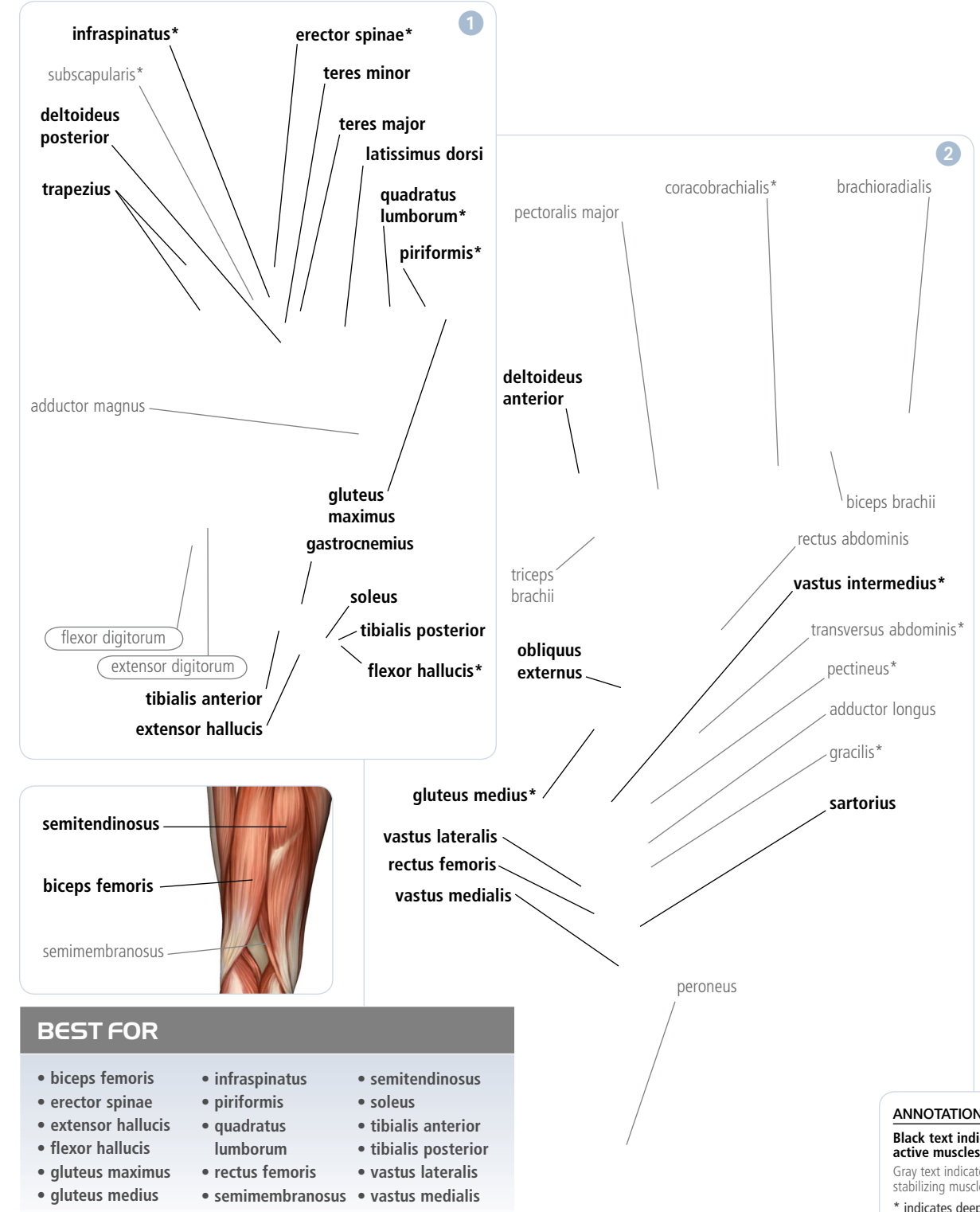
LOOK FOR

- Your knee and hip to extend and rise at the same time.
- The ball to remain equidistant from your torso throughout the movement.
- Your elbows to remain extended.

AVOID

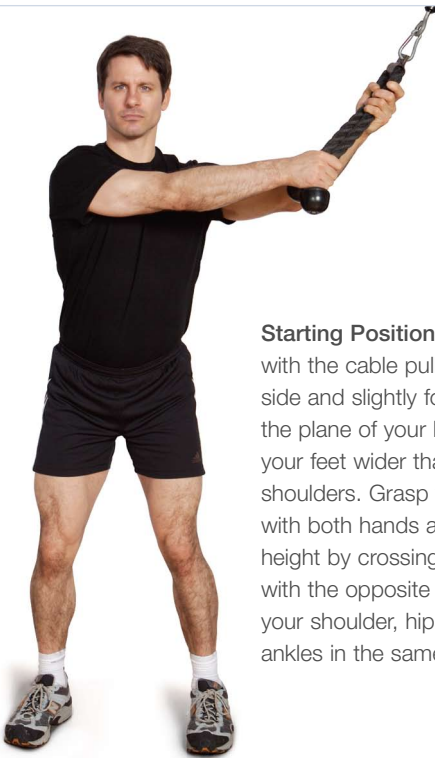
- Excessive flexion of your torso and spine.
- Bringing the ball close to your body or lifting any part of your foot from the floor.

PNF RAISE • CORE



WOODCHOPPER

CORE



Starting Position: Stand with the cable pulley to one side and slightly forward of the plane of your body, with your feet wider than your shoulders. Grasp the handle with both hands at shoulder height by crossing your torso with the opposite arm. Align your shoulder, hips, and ankles in the same plane.

LOOK FOR

- Your torso and hips to remain stationary.
- Your arms to be fully extended at the bottom of the movement.
- Elevating your chest at the bottom of the movement.

AVOID

- Rotating your hips.
- Bending the elbow of the arm crossing your body.
- Shifting your weight from side to side during the movement.

STABILIZE BY

- Keeping your shoulders down and back.
- Pulling your abdomen up and in and keeping your chest high.
- Distributing your weight evenly across your feet.
- Keeping your hips and knees slightly bent and solid.

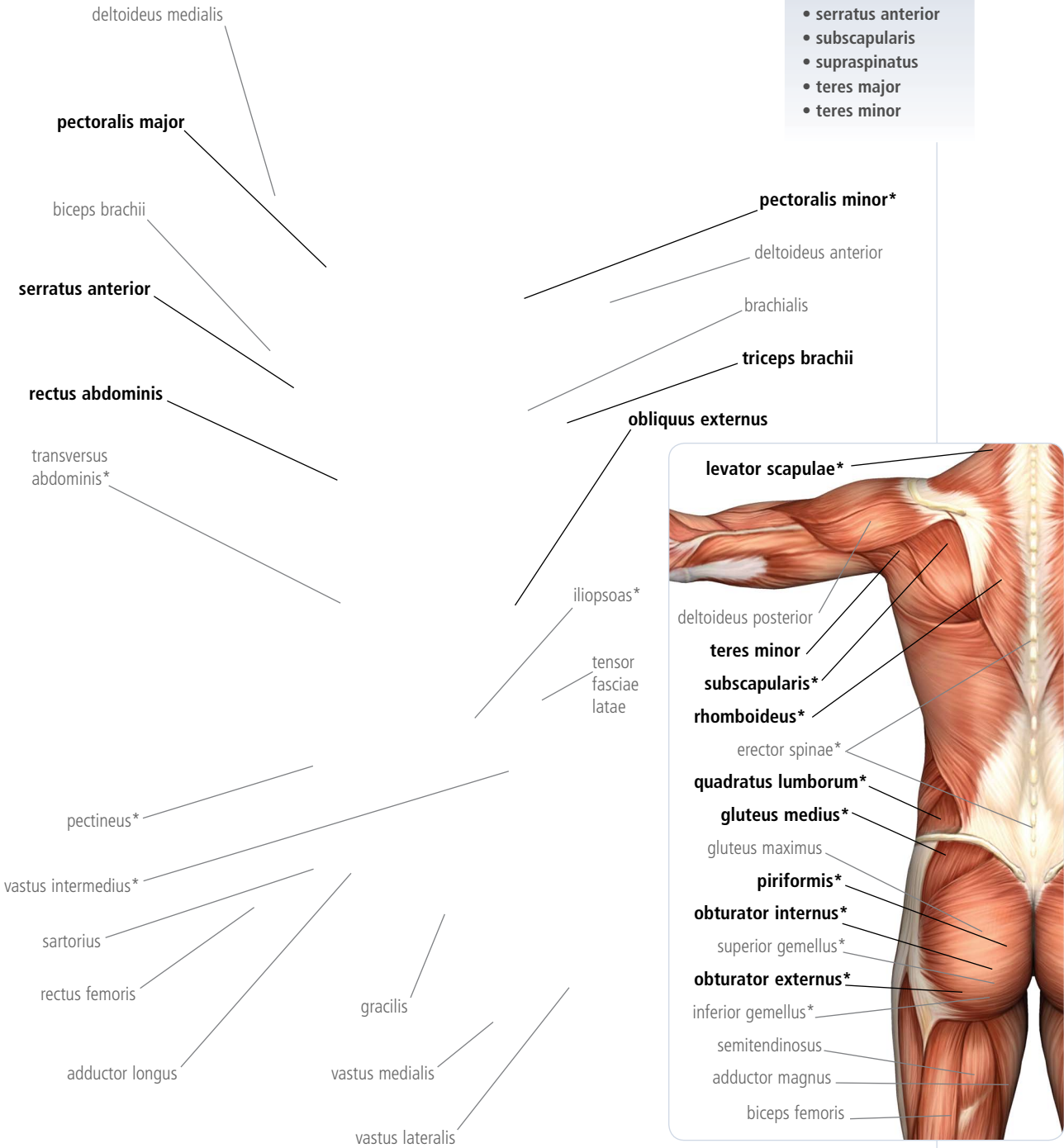
Action: Pull the handle downward and inward with straight arms in a 90-degree arc, stopping in front of the opposite leg; return along the same path. Exhale as you pull the weight down, and inhale as you return to the starting position.

Movement Path: Your torso and center of mass remain stationary as your arms move in a 90-degree arc from the side of your body downward and inward until your hands cross your midline.

2

ANNOTATION KEY

Black text indicates active muscles
Gray text indicates stabilizing muscles
* indicates deep muscles



WOODCHOPPER • CORE

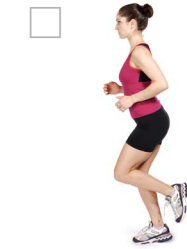
BEST FOR

- obliquus externus
- pectoralis minor
- serratus anterior
- subscapularis
- supraspinatus
- teres major
- teres minor

INDEX

This checklist provides a quick and easy way to keep track of the exercises described in this book. Once you have mastered the techniques described for these exercises and their variations, you can use this list to devise different workouts to keep your workout sessions fresh and productive.

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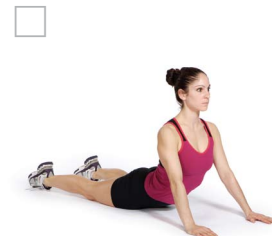
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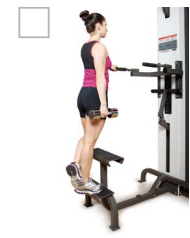
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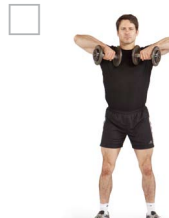
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GLOSSARY

abduction. Movement away from the body.

Active (also Active Isolated Stretching). Actively contracting a given muscle group (agonist) in order to stretch the opposing (antagonist) muscle group. Developed by Aaron Mattes.

adduction. Movement toward the body.

alternating grip. One hand grasping with the palm facing toward the body and the other facing away.

anterior. Located in the front.

dynamic. Continuously moving.

extension. The act of straightening.

curvilinear (movement path). Moving in a curved path.

flexion. The bending of a joint.

isometric. Muscles contracting against an equal resistance, resulting in no movement.

iliotibial band (ITB). A thick band of fibrous tissue that runs down the outside of the leg, beginning at the hip and extending to the outer side of the tibia just below the knee joint. The band functions in coordination with several of the thigh muscles to provide stability to the outside of the knee joint. See page 13.

neutral position (spine). a spinal position resembling an “S” shape, consisting of a *lordosis* in the lower back, when viewed in profile.

PNF. Proprioceptive Neuromuscular Facilitation; refers to a neuromuscular pattern of contraction which utilizes the greatest efficiency regarding positional awareness.

Pose Technique. Running methodology developed by Dr. Nicholas Romanov.

posterior. Located behind.

medial. Located on, or extending toward, the middle.

lateral. Located on, or extending toward, the outside.

lordosis. Forward curvature of the spine and lumbar region.

scapula. The protrusion of bone on the mid to upper back, also known as the shoulder blade.

static. No movement; holding a given position.

LATIN GLOSSARY

The following glossary explains the Latin terminology used to describe the body’s musculature. Certain words are derived from Greek, which has been indicated in each instance.

CHEST

coracobrachialis. Greek *korakoeidés*, “ravenlike,” and *brachium*, “arm”

pectoralis (major and minor). *pectus*, “breast”

ABDOMEN

obliquus externus. *obliquus*, “slanting,” and *externus*, “outward”

obliquus internus. *obliquus*, “slanting,” and *internus*, “within”

rectus abdominis. *rego*, “straight, upright,” and *abdomen*, “belly”

serratus anterior. *serra*, “saw,” and *ante*, “before”

transversus abdominis. *transversus*, “athwart,” and *abdomen*, “belly”

NECK

scalenus. Greek *skalénós*, “unequal”

semispinalis. *semi*, “half,” and *spinae*, “spine”

splenius. Greek *splénion*, “plaster, patch”

sternocleidomastoideus. Greek *stérnon*, “chest,” Greek *kleís*, “key,” and Greek *mastoeidés*, “breastlike”

BACK

erector spinae. *erectus*, “straight,” and *spina*, “thorn”

latissimus dorsi. *latus*, “wide,” and *dorsum*, “back”

multifidus spinae. *multifid*, “to cut into divisions,” and *spinae*, “spine”

quadratus lumborum. *quadratus*, “square, rectangular,” and *lumbus*, “loin”

rhomboideus. Greek *rhembesthai*, “to spin”

trapezius. Greek *trapezion*, “small table”

SHOULDERS

deltoideus (anterior, medial, and posterior). Greek *deltoeidés*, “delta-shaped”

infraspinatus. *infra*, “under,” and *spina*, “thorn”

levator scapulae. *levare*, “to raise,” and *scapulae*, “shoulder [blades]”

subscapularis. *sub*, “below,” and *scapulae*, “shoulder [blades]”

supraspinatus. *supra*, “above,” and *spina*, “thorn”

teres (major and minor). *teres*, “rounded”

UPPER ARM

biceps brachii. *biceps*, “two-headed,” and *brachium*, “arm”

brachialis. *brachium*, “arm”

triceps brachii. *triceps*, “three-headed,” and *brachium*, “arm”

LOWER ARM

anconeus. Greek *anconad*, “elbow”

brachioradialis. *brachium*, “arm,” and *radius*, “spoke”

extensor carpi radialis. *extendere*, “to extend,” Greek *karpós*, “wrist,” and *radius*, “spoke”

extensor digitorum. *extendere*, “to extend,” and *digitus*, “finger, toe”

flexor carpi pollicis longus. *flectere*, “to bend,” Greek *karpós*, “wrist,” *pollicis*, “thumb,” and *longus*, “long”

flexor carpi radialis. *flectere*, “to bend,” Greek *karpós*, “wrist,” and *radius*, “spoke”

flexor carpi ulnaris. *flectere*, “to bend,” Greek *karpós*, “wrist,” and *ulnaris*, “forearm”

flexor digitorum. *flectere*, “to bend,” and *digitus*, “finger, toe”

palmaris longus. *palmaris*, “palm,” and *longus*, “long”

pronator teres. *pronate*, “to rotate,” and *teres*, “rounded”

HIPS

gemellus (inferior and superior). *geminus*, “twin”

gluteus maximus. Greek *gloutós*, “rump,” and *maximus*, “largest”

gluteus medius. Greek *gloutós*, “rump,” and *medialis*, “middle”

gluteus minimus. Greek *gloutós*, “rump,” and *minimus*, “smallest”

iliopsoas. *ilium*, “groin,” and Greek *psoa*, “groin muscle”

iliacus. *ilium*, “groin”

obturator externus. *obturare*, “to block,” and *externus*, “outward”

obturator internus. *obturare*, “to block,” and *internus*, “within”

pectineus. *pectin*, “comb”

piriformis. *pirum*, “pear,” and *forma*, “shape”

quadratus femoris. *quadratus*, “square, rectangular,” and *femur*, “thigh”

UPPER LEG

adductor longus. *adducere*, “to contract,” and *longus*, “long”

adductor magnus. *adducere*, “to contract,” and *magnus*, “major”

biceps femoris. *biceps*, “two-headed,” and *femur*, “thigh”

gracilis. *gracilis*, “slim, slender”

rectus femoris. *rego*, “straight, upright,” and *femur*, “thigh”

sartorius. *sarcio*, “to patch” or “to repair”

semimembranosus. *semi*, “half,” and *membrum*, “limb”

semitendinosus. *semi*, “half,” and *tendo*, “tendon”

tensor fasciae latae. *tenere*, “to stretch,” *fasciae*, “band,” and *latae*, “laid down”

vastus intermedius. *vastus*, “immense, huge,” and *intermedius*, “between”

vastus lateralis. *vastus*, “immense, huge,” and *lateralis*, “side”

vastus medialis. *vastus*, “immense, huge,” and *medialis*, “middle”

LOWER LEG

adductor hallucis. *adducere*, “to contract,” and *hallex*, “big toe”

adductor digiti minimi. *adducere*, “to contract,” *digitus*, “finger, toe,” and *minimum* “smallest”

extensor hallucis. *extendere*, “to extend,” and *hallex*, “big toe”

extensor digitorum. *extendere*, “to extend,” and *digitus*, “finger, toe”

flexor digitorum. *flectere*, “to bend,” and *digitus*, “finger, toe”

flexor hallucis. *flectere*, “to bend,” and *hallex*, “big toe”

gastrocnemius. Greek *gastroknémía*, “calf [of the leg]”

peroneus. *peronei*, “of the fibula”

plantaris. *planta*, “the sole”

LATIN GLOSSARY

soleus. *solea*, “sandal”

tibialis anterior. *tibia*, “reed pipe,” and *ante*, “before”

tibialis posterior. *tibia*, “reed pipe,” and *posterus*, “coming after”

trochlea tali. *trochleae*, “a pulley-shaped structure,” and *talus*, “lower portion of ankle joint”

ACKNOWLEDGMENTS

In my daily life for the past 20 years, several times a day I have been instructing people using the exercises in this book. I have worked with clients as young as 12 and as old as 92, from every different walk of life imaginable. From recreational athletes to professional ones, cardiac patients to pregnant and postpartum women, for entertainment or business, regardless of where they came from two things have remained constant for me: I have enjoyed every minute of it and felt privileged to be able to do it, and I am constantly reminded that exercise is relevant to everyone. Period.

As with any book, there are many people who have contributed to it. Their contributions also come in many different forms. For their help I would like to personally thank Sean, Karen, Lisa, and Amber at Moseley Road for originally bringing this project to me and seeing it through. All of my staff here at La Palestra, Teodoro Chavez, Greg Cimino, Elisama Colon, Alex Evans, Gloria Robles, Shirley Gauthiers, Luis Gonzalez, Engenio Guerrero, Carrie Lane, Adam Palmer, Thierno Diallo, Jermaine Phanord, Ann Reginald, Molly Morgan, SebaJ Adele, Kofi Sekyiamah, Paul Knapic, Craig Maltese, Doug Dickinson, Melissa Mora, and Grace Eichinger deserve mention; and particularly for the contributions in helping with the text, Gillian Mounsey, Greg Peters, Marissa O'Neil, Garth Wakeford, and Sydney Foster, who along with Mark Tenore so skillfully and flawlessly modeled all of the exercises in this book. Dr. Rob DeStefano, Dr. Jennifer Solomon, and Dr. Kyler Brown were invaluable with their anatomical expertise, as was my assistant Shannon Plumstead, who typed this entire work and is masterful at reading my handwriting. Thanks go to my good friends and colleagues Jim and Phil Wharton, who introduced me to the technique of Active Isolated Stretching, and Nicholas Romanov, whose "Pose" method of running has had a profound influence on my thinking about all movement.

Lastly and most importantly I thank my family: my wife, Deborah, who is my advisor on all things and whose opinion is the one I count on most when making any decision; my daughter, Milena; and my son, Tony (who was born while I was writing this book), who make everything I do have a reason; and for my mom, who I thank every day for giving me the life that I have.